



**BEN-GURION UNIVERSITY OF THE
NEGEV**

Master's Thesis

**Design, Review and Improvement of the
Adhesive Gripper**

Author:

Yoav Mann Mannhaim

Supervisor:

Prof. Amir Shapiro

A thesis submitted in fulfillment of the requirements
for a Master of Science
in the

**DEPARTMENT OF MECHATRONICS
FACULTY OF ENGINEERING SCIENCES**

September 30, 2026



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Chairman of Graduate Studies Committee: Date:

Abstract

Adhesive end-effectors offer a versatile alternative to mechanical, vacuum, and magnetic grippers for robotic pick-and-place, but unlike a mechanical clamp, the holding capacity of an adhesive bond is a dynamic function of the forces the manipulator's own motion applies to it. This thesis develops and evaluates a manipulation system built around this constraint, combining empirical material characterization, force-aware trajectory planning, and real-time grasp validation.

The mechanical properties of the adhesive used throughout this work (3M™ Value Duct Tape 1900) were measured directly rather than taken from a datasheet: thickness, tensile modulus (150 MPa), shear modulus, and surface energy on both uncoated (288.2 J/m²) and coated (471.4 J/m²) cardboard substrates. These measured values parameterize a Force-Aware Rapidly-exploring Random Tree Star (RRT*) planner, which rejects candidate motions whose required acceleration or energy expenditure would exceed the adhesive's characterized limit surface. The planner was evaluated on a bank of fifty successful simulated trajectories with the payload raised to 58-96% of the adhesive's static tensile capacity, at which the force-aware constraint actively rejects roughly a quarter of candidate motions and is demonstrably governing the search.

Because a plan is only as good as the assumptions it was built on, this thesis also develops a grasp validation system that monitors the bond during execution rather than trusting the plan alone. Two candidate sensing modalities, infrared reflectance and photoresistor occlusion, were compared quantitatively across three objects and under both nominal and direct-illumination lighting; the photoresistor array was selected on the basis of a substantially larger effect size against an empty-gripper baseline (Cohen's d up to 1.70, against negligible values for infrared) and its use of 87.7% of the available sensor dynamic range against 16.6% for infrared. A neural classifier trained on this sensor array distinguishes four grasp-quality classes with 84.9% and 79.9% test accuracy on two test objects, and detects complete loss of grasp with perfect recall on both. A two-phase protocol built on this classifier assesses grasp quality before motion begins and applies a graduated response, from trajectory de-rating to an immediate abort, to degradation detected during execution, with thresholds derived from the statistics of the recorded sensor data rather than assumed.

These components are integrated into a single system architecture in which the offline planner and the online monitor share one physical basis: the same measured adhesive

limit surface that constrains planned motion also grounds the monitor's alarm thresholds. The planner and the monitor have each been characterized and validated independently, the planner in simulation, the monitor against recorded sensor data -- but their combined operation on physical hardware has not yet been demonstrated, and this thesis states that limitation explicitly rather than implying a result that was not obtained. The open questions these leaves, along with the specific next steps required to close them, are set out in the concluding chapter.

The complete implementation described in this thesis -- the Force-Aware RRT* planner and the grasp-validation system, including installation instructions -- is available at:

<https://github.com/YoavMannhaim/Thesis-documentation>

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Nomenclature

The following table summarizes the principal symbols used throughout this thesis. Symbols with strictly local meaning, confined to a single derivation, are defined at their point of use and are not repeated here.

Latin Symbols

Symbol	Description	Units
a	Length of the adherent/joint surface (peel model); also joint acceleration context-dependent	m
b	Width of the adhesive film / joint	m
t	Adhesive layer / film thickness	m
E	Young's (elastic) modulus	Pa
G	Shear modulus	Pa
P	Applied peel force	N
P_{cr} / P_T	Critical peel force / tensile peel force component	N
U_T, U_s, U_p, U_E, U_B	Total, surface, potential, elastic, and bend strain energy terms	J
q	Robot joint configuration (vector)	rad
q_{init}, q_{start}	Initial / start configuration	rad
q_{goal}	Goal configuration	rad
$q_{rand}, q_{near}, q_{new}$	Sampled, nearest-neighbor, and newly steered configurations (RRT / RRT*)	rad
n, N	Number of planning iterations / maximum iterations	—
$r, r(n)$	Near-node connection (rewiring) radius	rad
T	Tree data structure used by RRT / RRT*	—
F, C, C_{obs}	Free space, configuration space, obstacle space	—
d	Distance metric between two	—

	configurations	
J	Total path cost (weighted RRT* cost function)	—
$w_{time}, w_{dist}, w_{advance}$	Cost-function weighting coefficients used in RRTStar.py (time, distance, goal-advance)	—
S_i	Safety penalty term used in thesis Eq. 6.5 cost formulation	—
L_{cm}	Center-of-mass offset length of the gripped object	m
a_{req}, a_{limit}	Required / maximum allowable joint linear acceleration	m/s^2
S_a, S_v	Safety factors applied to acceleration and velocity limits	—

Greek Symbols

Symbol	Description	Units
θ	Peel angle / tilt angle	rad / deg
Γ	Surface energy per unit area (Ch.3); also cost-function safety weight in Eq. 6.5 — see note	$J/m^2 / —$
ν	Poisson's ratio of the adhesive	—
σ	Normal (cleavage) stress	Pa
τ	Shear stress (mechanics chapters) — also used elsewhere for the planned path τ ; see note	$Pa / —$
Δ	RRT / RRT* steer step size	rad
$\alpha_{req}, \alpha_{limit}$	Required / maximum allowable angular acceleration	rad/s^2
ω	Angular velocity	rad/s

Abbreviations

Abbreviation	Meaning
RRT	Rapidly-exploring Random Tree
RRT*	Rapidly-exploring Random Tree Star (asymptotically optimal variant)
DOF	Degrees of Freedom
FK	Forward Kinematics
LSTM	Long Short-Term Memory (recurrent neural network)
IR	Infrared (sensor)
LDR	Light-Dependent Resistor
ML	Machine Learning
POC	Proof of Concept

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1. Introduction

Automated pick-and-place is one of the most common tasks in warehouse and logistics robotics, and one of the most sensitive to how an object is held. Conventional end-effectors, parallel-jaw and multi-fingered mechanical grippers, vacuum cups, magnetic pads, each impose their own constraints on object geometry, surface material, and porosity. Thin, flat, irregularly shaped, or delicate items are frequently difficult to grasp reliably with any of these approaches: mechanical jaws require accessible parallel surfaces, vacuum cups fail on porous or highly textured materials, and magnetic grippers are limited to ferrous objects. Adhesive-based end-effectors, which bond directly to an object's surface using a pressure-sensitive adhesive film, offer a complementary approach: they require no enclosing geometry, exert no clamping force on the object, and can in principle grip a wide range of surface types. This versatility comes at a cost that mechanical grippers do not share. A mechanical jaw's hold on an object is, to first order, a fixed geometric constraint, it does not change as the arm accelerates, decelerates, or reorients. An adhesive bond is not: its ability to resist a given combination of normal, shear, and peel loading is a property of the adhesive's mechanics, and that capacity is consumed by the very motion the manipulator is trying to execute. A trajectory that is perfectly safe for a mechanical gripper, fast, direct, involving sharp reorientations, can be exactly the kind of motion that peels an adhesive bond loose. Any manipulator built around an adhesive end-effector must therefore treat the adhesive's mechanical limits as a first-class constraint on motion, not an afterthought, and must be able to detect when a bond is degrading during execution rather than discovering the failure only when the object is dropped. This thesis addresses that problem directly, and does so in four parts that mirror the four related-work areas surveyed in Chapter 2: the mechanics of the gripper and adhesive interface, the motion planning that must respect them, the sensing required to validate a grasp in real time, and the integration of these pieces into one system. The first part (Chapter 5) is experimental: this thesis empirically characterizes the mechanical properties of the adhesive used throughout, 3M™ Value Duct Tape 1900, across thickness, tensile modulus, shear modulus, and surface energy, on both coated and uncoated cardboard substrates. These are not textbook constants; they are measured directly, using purpose-built jigs, and the resulting values (for example a tensile modulus of 150 MPa and a surface energy of 288 J/m² on uncoated cardboard) are the inputs that everything downstream in this thesis depends on. The second part (Chapter 6) is algorithmic: a Force-Aware variant of the Rapidly-exploring Random Tree Star (RRT*) planner, which uses the measured adhesive properties of Chapter 5 to reject candidate motions that would overload the bond, before they are ever executed. The planner was evaluated across a bank of fifty successful simulated scenarios with the object mass raised to 58-96% of the adhesive's static load capacity, so that the force-aware constraint is demonstrably the limiting factor rather than a check that is never exercised. The third part (Chapter 7) is a sensing and validation system that monitors the grasp while the manipulator is moving, rather than assuming the plan was correct. Two candidate sensing modalities, infrared reflectance and photoresistor occlusion, were compared directly and quantitatively; the photoresistor array was selected on the basis of that comparison, and a neural classifier trained on it distinguishes grasp quality with the results reported in Chapter 7. A two-phase protocol built on this classifier

assesses the grasp before motion begins and monitors it continuously during execution, with a graduated response to degradation rather than a single pass/fail check.

The fourth part (Chapter 8) draws these pieces together into a single system architecture, in which the same measured adhesive limit surface that constrains the offline planner also grounds the online monitor's alarm thresholds. That chapter is also explicit about what has and has not been demonstrated: the planner and the validator have each been tested and characterized on their own, but their joint operation on physical hardware remains future work, and Chapter 8 states this directly rather than implying an end-to-end result that was not obtained.

The remainder of this thesis is organized accordingly. Chapter 2 reviews related work in gripper design, trajectory planning, adhesive mechanics, and grasp validation. Chapter 3 develops the theoretical background -- peel and shear mechanics, the RRT/RRT* planning framework, and the neural architectures used later for grasp classification. Chapter 4 formalizes the problem this thesis solves. Chapter 5 presents the material characterization experiments. Chapter 6 develops and evaluates the Force-Aware RRT* planner. Chapter 7 presents the sensor selection, the trained grasp-quality classifier, and the monitoring protocol. Chapter 8 integrates these results into a system architecture and states its validation status plainly. Chapter 9 discusses the work remaining before this system could be deployed.

The thesis contributes new knowledge in the following subjects:

- An empirical mechanical characterization of the adhesive interface (3M™ Value Duct Tape 1900), thickness, tensile modulus, shear modulus, and surface energy on coated and uncoated cardboard, used directly, rather than assumed from a datasheet, throughout the rest of this thesis.
- A Force-Aware RRT* planner that combines a hard feasibility gate, derived from the measured adhesive limit surface, with a soft cost function that selects an efficient path among the motions the gate admits. The planner was evaluated on a bank of fifty successful simulated trajectories with the payload raised to 58-96% of the adhesive's static capacity, so that the force-aware constraint is demonstrably governing the search rather than a check that is never exercised.
- A quantitative comparison of two candidate grasp-sensing modalities (infrared reflectance and photoresistor occlusion), leading to a data-justified sensor selection, and a neural classifier trained on the selected array that distinguishes grasp-quality classes with measured test accuracy and perfect recall on complete grasp loss.
- A two-phase on-line validation protocol that assesses the grasp before motion begins and applies a graduated response to degradation detected during execution, with thresholds derived from the statistics of recorded sensor data.
- An integrated system architecture (Chapter 8) in which the offline planner and the on-line validator share one physical basis, the same measured adhesive limit surface, and an explicit statement of what has and has not yet been demonstrated end-to-end.

2. Related Work

This work is related to four main areas in the field of Robotic Object Manipulation: Robotic Grippers, Robotic Manipulator's Trajectory Planning, Modeling of Adhesives in Robotics, and Grasp Validation

In this chapter, we will review all related work fields

2.1 Review of Robot Grippers

Robotic Manipulators are versatile machines. Used in an increasing number of applications, from process automation, decreasing the difficulty or risk of tasks for human workers, or performing tasks impossible for human workers. The end effector attached to the manipulator will define the tasks that can be performed. The most common end-effectors used are grippers. There are various grippers used for different applications. Selecting the correct gripper for the task at hand is a critical part of the design process. The definition of manipulation and grasping changes with the application it is being used for. Generally, mechanical manipulation can be defined as an act of exerting force or torque on an object, causing motion or deformation, whereas holding an object can be termed as grasping [1].

One of the standard procedures performed by manipulators is grasping. For performing a grasping task, a gripper must be used as the end effector of the manipulator. Manipulation speed, object characteristics, and other variables are important factors when choosing a gripper type. Some grippers are designed for general purposes and changing objects. As can be seen in [2, 3, 4], versatile grippers are proposed for grasping objects

with different geometries. Minsch et al. in [5] present a review on composite grippers, which benefits from a substantial weight reduction compared to other grippers. In [6] and [7], a detailed review of soft robot grippers is presented. Agnus et al. [8] provides an overview of micro grippers.

2.1.1 Conventional Grippers

Conventional grippers are the most commonly used types of grippers in industrial settings. The three primary categories are multi-finger, vacuum grippers, and magnetic grippers.

Multi-Finger Grippers Conventional mechanical grippers employ multiple rigid or non-rigid "fingers" that clamp onto an object. The "fingers" rely on contact force to apply a wrench, force & torque to the object, enabling stable grasping against external disturbances such as gravity or inertial forces. Rigid-finger grippers are well-suited for grasping and manipulating heavy, rigid objects due to their ability to apply large forces [9]. Grasp stability relies on force closure and contact friction, typically requiring at least two contact points [10]. Golan et al. in [11] present a gripper's fingertip design that can switch between frictionless to frictional modes.

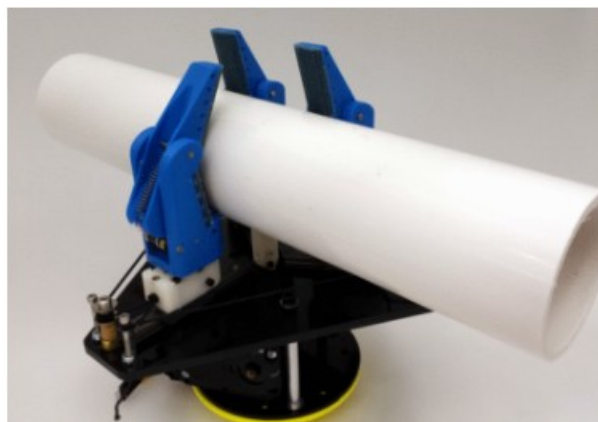


Fig 2.1 – Three-Fingered Gripper [4]

Multi-Finger grippers excel in applications with well-defined object geometries and consistent material properties. They are widely deployed in manufacturing environments for tasks such as machine tending, assembly operations, and materials handling [12]. The deterministic nature of mechanical grasping enables reliable operation with minimal sensing requirements, contributing to their dominance in industrial automation.

However, in multi-finger grippers, the placement of the fingers is critical; as such, not all objects are suitable for grasping by any rigid-finger mechanical gripper. [13] presents a variable-structure general-purpose robotic hand that can use the environment to change the configuration of its fingers.

The rigid nature of these grippers presents difficulties when handling non-rigid, soft, deformable, or delicate objects. Since high forces cannot be applied to delicate items, the grasping force magnitudes are limited, making rigid grippers unsuitable for such

applications. Furthermore, the reliance on a known, stable geometric configuration means they struggle with objects whose shape changes with the application of force.

Vacuum Grippers Pneumatic-based gripper, utilizing a suction cup, usually made of soft materials, connected to a vacuum pump to adhere to an object using vacuum force [14]. The applied force is directly proportional to the area of the suction cup and the internal pressure differential [14]. The force applied by the gripper is not affected by the size of the object being lifted, therefore, a single vacuum gripper can be used to lift and displace different objects. Vacuum grippers are commonly used in "pick and place" assignments [15,16], such as in the package delivery industry [12].

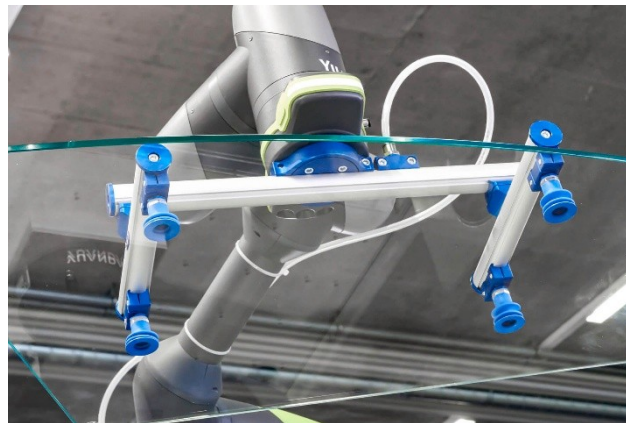


Fig 2.2 – Vacuum Gripper [87]

Vacuum grippers are extensively used in industries requiring hygienic handling, such as food processing and pharmaceutical manufacturing, because the suction cups can be easily cleaned and sanitized [17]. They also provide gentle handling characteristics, as the distributed pressure across the suction cup surface reduces stress concentrations compared to point contacts in mechanical grippers [17].

Vacuum grippers have significant disadvantages related to contact surface requirements and force limits. They require a smooth, relatively flat contact area; grasping porous or textured objects is highly inefficient. Furthermore, localized exposure of even a small portion of the suction cup to atmospheric conditions can severely decrease vacuum quality, rendering the gripper practically useless. The maximal grasping force is limited by the product of contact area and atmospheric pressure (in standard conditions),

making them unsuitable for lifting heavy objects. Crucially, they cannot be effectively used in low-ambient-pressure environments, such as aerospace applications.

Magnetic Grippers employ magnetic fields to generate attractive forces on ferromagnetic materials, enabling contactless grasping of steel, iron, and other magnetically susceptible objects [18]. These grippers can utilize either permanent magnets or electromagnets, with the latter offering controllable magnetic fields through electrical current modulation [18].



Fig 2.3 – Magnetic Gripper [88]

The primary advantage of magnetic grippers is their simplicity and speed. Attachment occurs immediately upon bringing the gripper into proximity with the object, with no actuation delay or complex control requirements. Electromagnet-based systems provide controllable attachment and release by energizing and de-energizing the magnetic field [18]. Magnetic grippers require minimal positioning accuracy compared to mechanical grippers, as the magnetic attraction provides some tolerance for misalignment.

These characteristics make magnetic grippers well-suited for rapid material handling in manufacturing environments, particularly in metal fabrication, automotive assembly, and machining operations where ferromagnetic workpieces are common [19]. They can handle parts with complex geometries that would be challenging for mechanical or vacuum grippers and can grasp through thin non-ferromagnetic coatings or layers.

However, the application domain of magnetic grippers is fundamentally restricted to ferromagnetic materials. They cannot manipulate non-magnetic materials such as aluminum, plastics, wood, or glass [19]. Residual magnetism after gripper release can be problematic, potentially interfering with sensitive electronic components or causing unwanted attraction of ferromagnetic debris to the handled object [19]. The magnetic field may also affect nearby objects, limiting use in densely packed environments or near magnetically sensitive equipment [19].

2.1.2 Non-Conventional Grippers

The limitations of conventional grippers have driven extensive research into alternative gripping technologies that can provide greater adaptability, gentler handling, or capabilities suited to specific application domains.

Soft Pneumatic Grippers represent a paradigm shift from rigid mechanisms, utilizing compliant elastomer structures that deform under pneumatic pressure to conform around objects [20,21]. These grippers employ pneumatic networks (PneuNets) or similar fluidic actuator designs where internal chambers inflate when pressurized, causing controlled bending or expansion of the gripper fingers. In [22], a detailed review of soft grippers is presented.

In **Fig 2.4**, a soft-bodied gripper with integrated curvature sensors for shape control is presented.

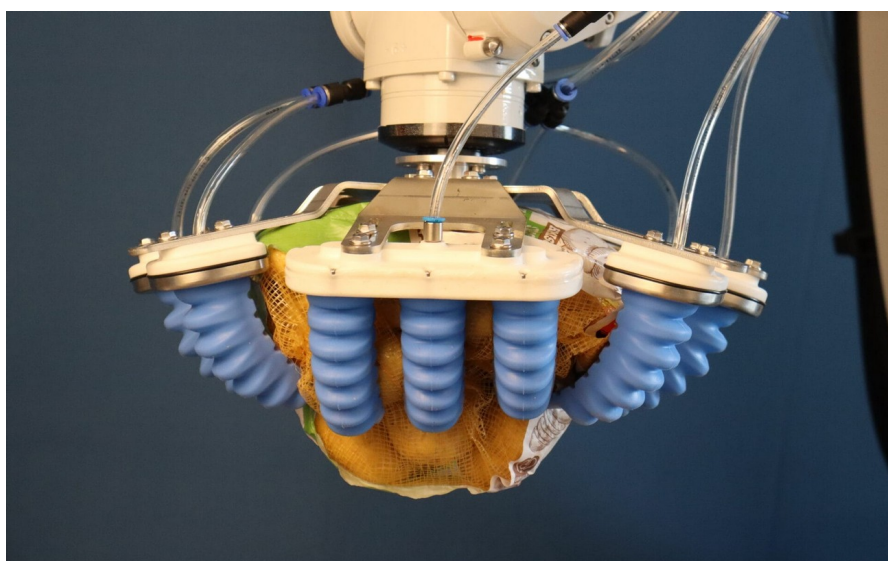


Fig 2.4 – Soft Pneumatic Gripper with Integrated Curvature Sensors [22]

The fundamental advantage of soft grippers is their intrinsic compliance and adaptability. The elastic deformation allows a single gripper design to accommodate objects with widely varying shapes and sizes without requiring precise positioning or custom tooling [23]. Compliant contact distributes forces over larger surface areas, enabling gentle handling of delicate objects such as biological materials, food products, and fragile components that would be damaged by rigid grippers.

Despite these advantages, soft pneumatic grippers typically provide lower gripping forces compared to rigid systems of similar size, limiting their application to relatively lightweight objects. The response time is slower due to the time required to pressurize and depressurize the pneumatic chambers, making them less suitable for high-speed pick-and-place operations. Precise force control is challenging because the relationship between pneumatic pressure and gripping force is nonlinear and depends on object properties and contact geometry [24].

Granular Jamming Grippers Brown et al. [25] demonstrate a creative approach to create a universal gripper, employing a fundamentally different principle, using a flexible membrane filled with granular material that transitions between fluid-like and solid-like states through vacuum-induced jamming [26]. In the unpressurized state, the granular material flows freely, allowing the membrane to conform around objects of arbitrary shape. When vacuum is applied, the granular particles jam together, causing the gripper to rigidify while maintaining the conformed shape, thereby securing the object.

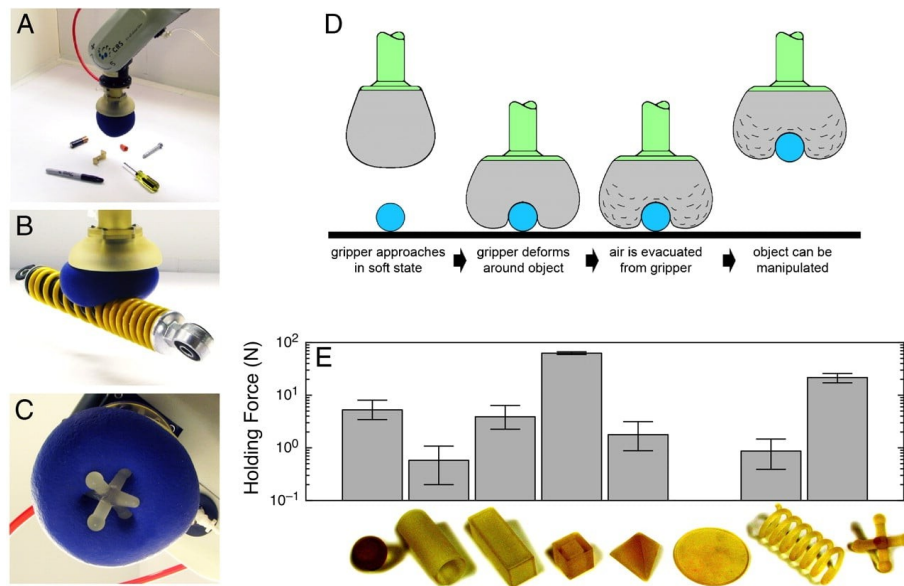


Fig 2.5 – Granular Jamming Gripper [25]

This approach enables universal grasping capabilities, handling objects with geometries that are extremely challenging for other gripper types, including concave shapes, irregular surfaces, and objects without distinct graspable features [27]. The gripper requires no prior knowledge of object geometry and minimal sensing, as the passive conforming behavior handles geometric variability [27]. The compliant nature provides gentle handling suitable for delicate objects.

However, jamming grippers face several limitations. The actuation speed is slow, as both the conforming phase and jamming phase require time for material settling and vacuum establishment [28]. The holding force is generally modest compared to mechanical or vacuum grippers and decreases with increasing object surface irregularity. The granular material experiences wear and degradation over repeated cycles, potentially affecting performance consistency. Precise force control and grasp quality assessment are difficult due to the distributed contact and complex internal mechanics [28].

2.1.3 Adhesive-based Grippers

Adhesive-based grippers utilize surface adhesion forces rather than mechanical constraint, suction, or magnetic attraction to secure objects. This approach offers unique capabilities for handling materials where conventional methods are inadequate, particularly thin, flexible, porous, or fragile objects.

Electroadhesive Grippers generate adhesion through electrostatic forces by applying high voltage across thin electrodes separated by a dielectric material [29,30]. When voltage is applied, an electric field is created that induces polarization in the object surface, resulting in an attractive force between the gripper and object. This mechanism, known as electroadhesion, works on both conductive and insulating materials through different physical processes [30].

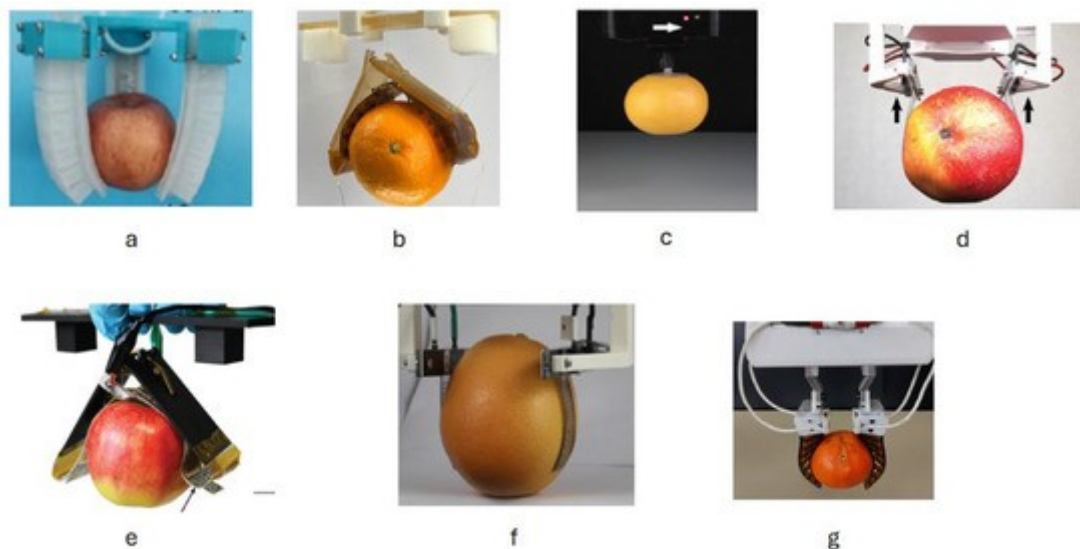


Fig 2.6 – Electroadhesive Gripper [29]

A key advantage of electroadhesive grippers is the rapid switching capability between gripping and releasing states through voltage control, enabling cycle times suitable for industrial automation. The adhesion can be activated and deactivated within milliseconds, much faster than mechanical actuation. Electroadhesive pads can be made thin and flexible, allowing them to conform to curved surfaces and irregular geometries. The technology operates at very low power consumption, since current flow is minimal once the electric field is established [30].

Research has demonstrated electroadhesive grippers integrated with soft, compliant substrates to enhance surface conformability, improving adhesion on non-planar objects. Applications have been explored, ranging from manufacturing automation to wall-climbing robots and satellite servicing.

However, electroadhesive grippers face several limitations. The adhesion force is relatively modest compared to mechanical or vacuum grippers, limiting applications to

lightweight objects or requiring large gripper areas. Performance is highly sensitive to the gap between electrodes and object surface—even sub-millimeter gaps significantly reduce adhesion. Surface contamination (dust, oil, moisture) substantially degrades performance, and accumulated contamination requires periodic cleaning to maintain adhesion capacity. Humidity can affect dielectric properties and cause current leakage, reducing efficiency. The high voltage requirement necessitates appropriate electrical insulation and safety measures, adding system complexity [29,30].

Gecko-Inspired Adhesive Gripper mimics the adhesion mechanism of gecko lizards, which can climb vertical surfaces and even hang inverted using dry adhesion based on van der Waals forces [31,32]. Gecko toe pads are covered with millions of microscopic hair-like structures called setae, which branch into even finer spatula at their tips. The extremely high density of these nano-scale contacts generates sufficient van der Waals forces to support the gecko's weight. Synthetic gecko-inspired adhesives have been developed using microfabricated polymer structures with similar hierarchical geometries, though typically at larger scales than biological setae due to manufacturing constraints. These artificial structures typically feature wedge-shaped or mushroom-shaped tips that maximize contact area when properly loaded [31,32].



Fig 2.7 – Gecko-Inspired Adhesive Gripper [31]

This directional adhesion property enables controllable attachment and detachment through mechanical loading patterns rather than requiring external actuation or power [33]. By designing gripper mechanisms that apply shear load in the "on" direction during grasping and normal or reverse shear load during release, reliable grasp cycles can be achieved. Adhesion automatically engages when forces are applied appropriately, without requiring active sensing or control [33].

Stanford researchers developed gripper systems combining gecko-inspired adhesives with compliant mechanisms, demonstrating successful grasping of objects ranging from delicate eggs to large pipes and basketballs. The ability to grip firmly without squeezing is particularly valuable for handling fragile or deformable objects. Gecko-inspired adhesives can be integrated with soft pneumatic grippers to combine the adaptability of soft actuators with the holding capacity of adhesive contact [31]. This hybrid approach enables grasping of irregular objects with enhanced reliability. Recent work has also been explored combining gecko and electroadhesive mechanisms to achieve enhanced performance [31].

Despite significant progress, gecko-inspired grippers face several challenges. The manufacturing process for micro-structured adhesives is complex and costly, limiting scalability for large-area applications [33]. The adhesive structures are susceptible to contamination, particularly by small particles that lodge between the microfibers and reduce effective contact area. Adhesion performance degrades on rough surfaces where the feature size exceeds the microstructure dimensions, as the fibers cannot fully conform to large-scale irregularities. Material wear and mechanical damage to the delicate structures can occur over many attachment-detachment cycles, though recent designs have demonstrated durability exceeding 30,000 cycles [31,32].

Hot Adhesive Gripper Cohen Ehud and Oren Oz [34] present a gripper that acquires a grasp using thermoplastic adhesives. The gripper is equipped with a hot adhesive extruder, a linking unit, and a ventilation unit. The gripper acquires a grasp on the target object by extruding a fair amount of hot adhesive on the target object. Then the linking unit is attached to the object where the adhesive is set. The heat energy is conducted from the hot adhesive to the linking unit and then to the surrounding air with the aid of the ventilation unit.

High air pressure is applied through the linking unit, to release the target object by causing a failure of the adhesive link.

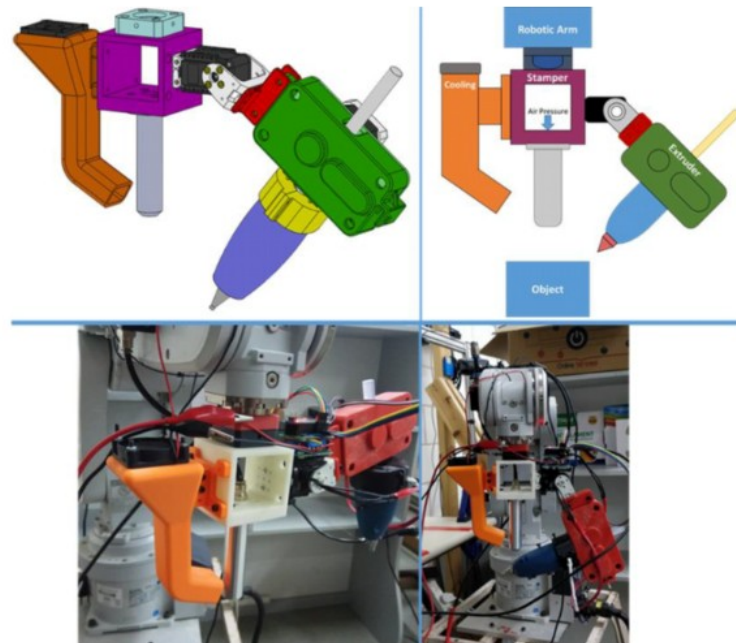


Fig 2.8 – Hot Adhesive Gripper [34]

This grasping principle has not yet proven effective and requires further research and development. The current design suffers from some key drawbacks: the heat is not removed from the linking unit sufficiently fast. The bond between the linking unit and the target object was usually too strong to break by high air pressure alone. Those drawbacks can probably be overcome with further engineering work.

Rigid Stickers Gripper Noam Hasson (2021) [35] introduced an innovative, non-conventional approach to robotic object manipulation by utilizing disposable adhesive agents as a structural translation unit between the gripper and the payload. Designed to bypass the traditional challenges of active force-closure and complex shape adaptation, the **Rigid Sticker Gripper** uses prefabricated, rigid plastic stickers extruded with a uniform profile to minimize production costs. The gripper body (,) mounts to a 6-DOF robotic manipulator and features a dual-solenoid mechanical snapping mechanism that locks onto side sockets in the sticker's plastic skeleton.

To optimize adhesion on varied textures, the base of each rigid sticker features a compliant sponge interlayer under a thin pressure-sensitive adhesive (PSA) film. Built-in micro-switches provide real-time binary feedback on sticker presence, while an internal spring-loaded guide track is paired with an analog force sensor to dynamically measure the compression preload applied during surface acquisition. When the target item is transported to its destination, the solenoids pull the locking handles back, completely releasing the sticker alongside the object without needing to actively peel or break the established bond—a crucial design constraint identified from the failures of earlier hot-thermoplastic grippers.

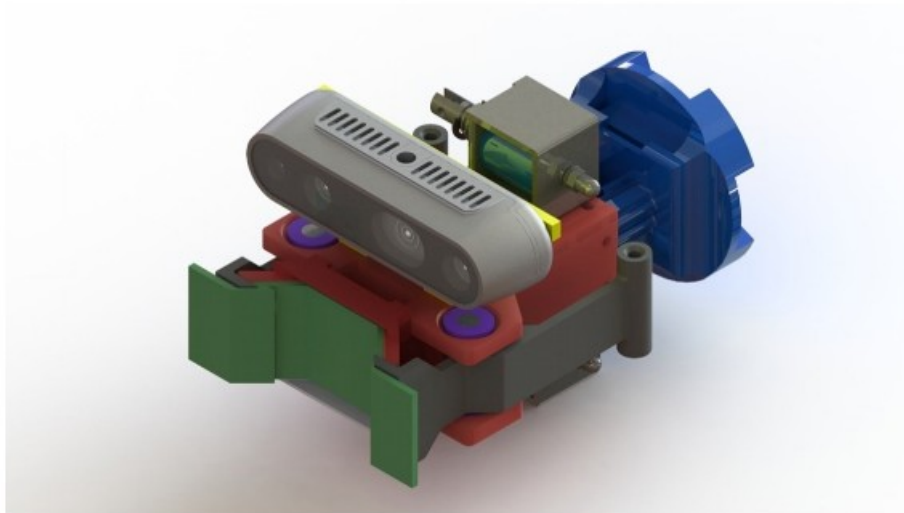


Fig 2.9 – Rigid Stickers Gripper [35]

Despite its successful deployment in collaborative warehouse and bin-picking simulation tasks under the Robot Operating System (ROS) and Gazebo environments, qualitative performance evaluations revealed three primary limitations that restrict the system's operational envelope:

- **Payload Threshold Constraints:** Empirical benchmarking across the Yale-CMU-Berkeley (YCB) object set demonstrated a maximum payload limit of approximately before suffering structural detachment.
- **Geometric Compatibility Bottlenecks:** Because the sticker backing is rigid, it cannot conform to non-planar shapes. While the system achieved a success rate on flat geometries (such as flat tin cans and cardboard boxes), it completely

failed on curved surfaces (such as cylindrical coffee containers) due to the severe lack of available contact area. It also failed on non-rigid plastic bags filled with large internal pieces (e.g., loose Lego blocks), where structural shifts under load immediately destabilized the interface.

- **Dispenser Density and Sustainability Limits:** Each sticker consumes of raw plastic per pick. This creates high material waste and forces a high ecological impact. Furthermore, because the stickers are bulky and must be stacked vertically inside a spring-loaded feeder dispenser, the storage footprint scales linearly with batch size, requiring frequent human intervention to refill the dispenser.

These bottlenecks directly motivate the development of the thin-film flexible adhesive systems explored in this thesis, which reduce the disposable agent's mass to less than while maximizing the effective surface contact area via unconstrained thin-film mechanics.

2.2 Trajectory Planning and Optimization in Grasping

2.2.1 Review of Trajectory Planning Methods

Trajectory planning for robotic manipulation has been addressed through various algorithmic approaches, each with distinct characteristics in terms of completeness guarantees, computational efficiency, and optimality.

Grid-based Methods Classical approaches such as A* and Dijkstra's algorithm operate on discretized configuration spaces [36]. These methods are complete resolution. and can guarantee the shortest paths on the grid [37]. However, they suffer from the curse of dimensionality, becoming computationally prohibitive for high-dimensional configuration spaces typical of robotic arms [38]. Work by Likhachev et al. on Anytime Dynamic A* (AD*) addresses the computational burden by providing suboptimal solutions quickly and improving them over time [39].

Potential Field Methods Artificial potential field approaches, introduced by Khatib [40], represent obstacles as repulsive potentials and goals as attractive potentials, allowing the robot to navigate by following the negative gradient of the total potential [41]. These methods are computationally efficient and provide smooth trajectories but

are prone to local minima and do not provide completeness guarantees [42]. Virtual potential field methods have been applied to grasping by Wang et al. [43], who used them for approach trajectory generation while avoiding obstacles.

Optimization-based Methods Trajectory optimization formulates motion planning as a numerical optimization problem, where a cost functional is minimized subject to kinematic and dynamic constraints [44]. CHOMP (Covariant Hamiltonian Optimization for Motion Planning), developed by Ratliff et al. [45], uses covariant gradient descent to optimize trajectories while maintaining smoothness. TrajOpt, introduced by Schulman et al. [46], formulates trajectory optimization as a sequential convex optimization problem, enabling efficient handling of constraints and demonstrating superior performance in cluttered environments.

These optimization methods can naturally incorporate various constraints and objectives, but require good initial trajectory guesses and may converge to local optima [47]. They have been successfully applied to manipulation tasks where smooth, dynamically feasible trajectories are critical [48].

Sampling-based Methods Rapidly exploring Random Trees (RRT) and Probabilistic Roadmaps (PRM) represent the dominant paradigm for high-dimensional motion planning [49]. LaValle introduced RRT [50], demonstrating probabilistic completeness and efficient exploration of configuration spaces. PRMs build a roadmap of collision-free configurations in a preprocessing phase [51], enabling fast query processing for multiple planning instances.

The key advantage of sampling-based methods is their ability to handle high-dimensional spaces without discretization [50]. However, basic RRT and PRM do not optimize path quality, often producing circuitous routes [51]. This limitation motivated the development of optimal variants, particularly RRT*, which provides asymptotic optimality guarantees [51].

2.2.2 Review of RRT*

RRT*, introduced by Karaman and Frazzoli [51], extends RRT with a rewiring mechanism that progressively refines the search tree to minimize path cost. The algorithm is proven to converge asymptotically to optimal solutions, addressing a

fundamental limitation of basic RRT. Since its introduction, RRT* has been widely adopted and extended for various robotic applications.

Convergence and Performance The asymptotic optimality of RRT* comes at the cost of slower initial convergence compared to RRT [52]. Karaman and Frazzoli analyzed the convergence rate, showing that the probability of finding an ϵ -suboptimal solution approaches one as the number of samples increases [51]. However, practical implementations often require tuning parameters such as the rewiring radius and sampling strategy to achieve good performance within reasonable computation time [53].

Variants and Extensions Numerous variants of RRT* have been proposed to address specific limitations or application requirements. Informed RRT*, developed by Gammell et al. [54], improves convergence by focusing sampling on regions that can potentially improve the current solution, using heuristic guidance from the current best path. This approach significantly reduces the number of samples required to reach near-optimal solutions. Batch Informed Trees (BIT*), also by Gammell et al. [55], combines the asymptotic optimality of RRT* with the multi-query efficiency of PRM. BIT* incrementally constructs a roadmap while exploiting the structure of the current solution to guide edge evaluation, achieving faster convergence than RRT* in many scenarios. Kinodynamic RRT*, introduced by Webb and van den Berg [55], extends RRT* to systems with differential constraints, such as robots with momentum or vehicles with nonholonomic constraints. The algorithm randomly samples control inputs rather than configurations, generating dynamically feasible trajectories. This extension is particularly relevant for trajectory planning with acceleration constraints [56].

Application to Manipulation RRT* has been successfully applied to various manipulation tasks. Berenson et al. [57] used RRT* for manipulator motion planning in cluttered environments, demonstrating improved trajectory quality compared to basic RRT. Their work highlighted the importance of appropriate distance metrics in configuration space, showing that task-space metrics often produce better paths than joint-space metrics.

Devaurs et al. [56] applied RRT* to robot-assisted surgery, where optimal paths are critical due to space constraints and safety requirements. They developed a multi-level RRT* variant that plans simultaneously in different configuration space representations, improving efficiency for redundant manipulators.

For grasping specifically, Shkolnik et al. [58] demonstrated RRT* for planning approach trajectories that minimize the probability of knocking over nearby objects, using a stochastic model of disturbances. Pan et al. developed a GPU-accelerated RRT* variant for real-time manipulation planning, achieving planning times suitable for reactive grasping in dynamic environments.

2.2.3 Force Constraints in Trajectory Planning

Incorporating force constraints into trajectory planning is essential for tasks where excessive forces could cause damage, failure, or unsafe conditions. Most work in this area has focused on contact forces between the robot and its environment rather than internal forces within the gripper system.

Contact Force Constraints Pham et al. [59] addressed trajectory planning under contact force constraints for robotic assembly tasks, where excessive forces can damage parts or cause assembly failure. They formulated the problem as a trajectory optimization with inequality constraints on contact forces, solved using sequential quadratic programming. The approach successfully generated trajectories that maintained contact while respecting force limits.

Suárez et al. [60] developed force-based motion planning for grasp synthesis, where finger contact forces must remain within friction cones to maintain stable grasps. Their approach integrated grasp quality metrics into RRT*, biasing sampling toward configurations that improve force closure properties.

Dynamic Constraints Many trajectory planning approaches implicitly handle force constraints through acceleration limits, since force and acceleration are related by the robot's inertia. Zhang et al. [61] developed a time-optimal trajectory planning method for industrial robots subject to torque limits, formulating the problem as a constrained optimization over path parameterization. Their approach ensures that actuator torques remain within allowable bounds throughout the trajectory.

Hauser and Ng-Thow-Hing [62] extended RRT to handle kinodynamic constraints, ensuring that generated trajectories respect velocity, acceleration, and force limits. Their approach samples control inputs directly, propagating system dynamics to generate new states. While effective, this method requires accurate dynamic models and can be computationally expensive for systems with complex dynamics.

Tool Constraints Some work has considered force constraints related to tools or end-effectors, though this area is less developed than contact force planning. Boisclair et al. [63] examined trajectory planning for welding robots where excessive tool forces could compromise weld quality. They used penalty methods to discourage trajectories that would violate force limits based on tool dynamics.

However, to the best of our knowledge, no prior work has specifically addressed trajectory planning for adhesive grippers where the constraint is the failure load of the adhesive bond. Existing approaches focus on contact forces or actuator limits rather than material failure criteria. This represents a significant gap, as adhesive grippers require fundamentally different planning strategies that account for the complex stress distributions in thin adhesive films and the history-dependent nature of adhesive failure.

3. Background

This chapter provides the fundamental concepts and theoretical foundations necessary for understanding the trajectory planning algorithm and grasp validation system developed in this thesis. The chapter begins with an overview of the theoretical models of adhesion and stress analysis that form the basis for force calculations in the trajectory planning algorithm presented in this thesis. Following this, the RRT* algorithm is introduced as the foundation for path planning, and finally, the principles of sensors and classification methods used in grasp validation are discussed.

3.1 Adhesion and Stress Analysis Models

3.1.1 Fundamentals of Adhesion

Adhesion is the phenomenon where two surfaces are held together by interfacial forces. These forces can arise from various mechanisms, including mechanical interlocking, chemical bonding, electrostatic attraction, and van der Waals forces. In the context of adhesive joints, the adhesion is primarily attributed to molecular interactions at the interface and mechanical interlocking of the adhesive into surface irregularities.

The strength of an adhesive bond depends on several factors, including the surface energy of the adherends, surface roughness, adhesive properties (such as elastic modulus and viscoelasticity), joint geometry, and loading conditions.

For thin adhesive films, used in the gripper presented in this thesis, the mechanical behavior under load is governed by the interplay between adhesive stresses and the deformation of both the adhesive layer and the adherends.

When an adhesive joint is subjected to external loads, stress concentrations develop at the edges of the bonded region. These stress concentrations are the primary sites where failure initiates, typically through either cohesive failure (within the adhesive layer) or adhesive failure (at the interface).

Understanding these stress distributions is essential for predicting the load-bearing capacity of the adhesive grippers.

3.1.2 Pressure-Sensitive Adhesive

Pressure-sensitive adhesives (PSAs), also referred to as self-adhesive or self-stick agents, represent a category of non-reactive materials that establish a bond instantaneously upon the application of mechanical pressure. Unlike many structural adhesives, PSAs require no heat, water, or solvents for activation. They are ubiquitously utilized in products such as labels, note pads, automotive trims, and various high-performance tapes. The integrity of a PSA bond is primarily governed by the magnitude of the compressive force applied during the joining process, though interfacial factors - including surface smoothness, surface free energy, and the removal of contaminants - remain equally vital for ensuring a robust connection [64]. These adhesives are typically engineered to provide stable adhesion at ambient temperatures, though their mechanical response is highly temperature dependent. Specifically, PSAs generally experience a loss of tack in cold environments and a marked reduction in shear holding capacity under elevated temperatures. The underlying physical mechanism of PSA adhesion is rooted in intermolecular Van der Waals interactions, predominantly London dispersion forces. These interactions become effective only at extremely close proximities, typically within a range of 4 angstroms. Beyond this threshold, the adhesion force diminishes according to an inverse sixth-power relationship with distance—a high sensitivity to range that is conceptually analogous to the behavior of the nuclear strong force.¹ To ensure a robust bond is formed across the thin-film interface, an application pressure between 1 and 10 Pa is conventionally required.¹ Once established, these films are capable of supporting complex mechanical loads, which can be resolved into tangential ($P_T = P \cos \theta$) and normal ($P_N = P \sin \theta$) components relative to the bond surface for detailed stress analysis.[65]

3.1.3 Thin Film Peeling Mechanics

In this chapter, we will present the chosen model used to predict the force required to separate a PSA film from a surface.

The basis for the theory is the work of Rivlin (1944) [66] and Spies (1953) [67], and was extended by Kendall (1975) [67]

Kendall derived an equilibrium theory of adhesive bond failure. The theory shows the dependency of the fracture of adhesive bonds on the interfacial surface energy, the geometry and the elastic properties of the system. A simple relationship between the

peeling angle and the critical peeling force is established, based on energy considerations and under the assumption that no viscous processes occur as fracture proceeds. A condition for fracture of an adhesive joint may arrive at by considering the total energy in the system.

If the total energy in a strained joint of dimension a is U_T , then peeling if the joint is possible when

$$(3.1) \quad \frac{dU_T}{da} \geq 0$$

These terms were considered when calculating U_T , and the proposed model and were tested experimentally.

A fracture can proceed at a low rate when the work done in breaking the joint is exactly compensated for by the gain in surface energy of the system:

$$(3.2) \quad \left. \frac{dU_T}{da} \right|_{P=P_{cr}} = 0$$

The total amount of energy stored in a system of thin adherent film is

$$(3.3) \quad U_T = U_s + U_p + U_E$$

Where U_s is the surface energy

$$(3.4) \quad U_s = -aby$$

For a the length of the adherent surface, b the width of the film, and y , the surface energy of the system per unit area.

U_p is the potential energy of the load, which is evaluated as

$$(3.5) \quad U_p = P(a - a \sin \theta)$$

Where θ is the angle between the surface normal and the film.

U_E , the elastic energy of the system is the sum of two terms

$$(3.6) \quad U_E = U_B - \frac{P^2 a}{2btE}$$

Where U_B is the energy stored in the sharp bend in the elastic film, and $P^2a/2btE$ is the energy required to stretch the film while it is removed from the substance.

By using equations (3.4),(3.5),(3.6) on equation (3.3), the expression of the total energy will be

$$(3.7) \quad U_T = -aby + P(a - a \sin \theta) + U_B - \frac{P^2a}{2btE}$$

And for $U_B = \text{const}$ for the entire process of peeling

$$(3.8) \quad U_T = -aby + P(a - a \sin \theta) - \frac{P^2a}{2btE}$$

$P^2a/2btE$ is usually insignificant compared to the other components of the total energy and therefore can be neglected.

By using equations (3.2),(3.8) an expression for the maximum force applied before peeling can be found.

$$(3.9) \quad P_{cr} = \frac{by}{1 - \sin \theta}$$

Where P_{cr} is the critical peeling force, for any $P > P_{cr}$ peeling is expected to occur.

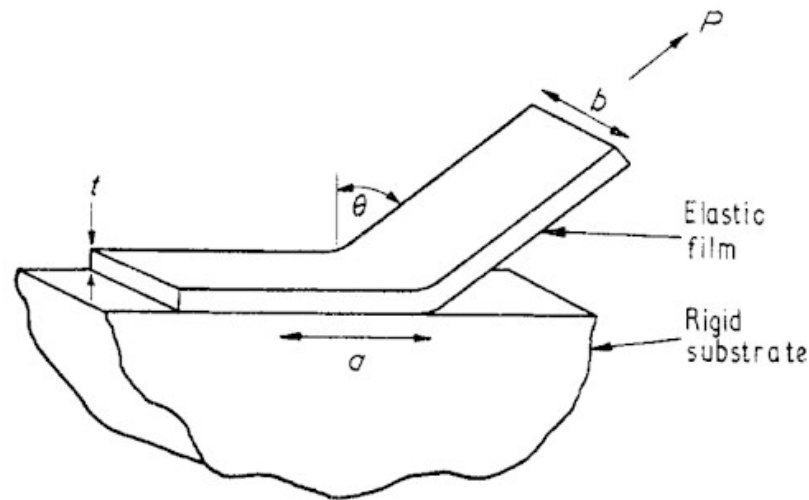


Fig 3.1 – Thin Film Peeling Mechanics [66]

In [68] this model was validated experimentally; the tests were performed over flat surfaces. [69] presents a numerical model and experimental results for the relation

between the peeling force, the peeling angle, and the peeling rate. The elastic term that was neglected at is reconsidered in [70].

3.1.4 Stress Analysis Models for Adhesive Joints

In this chapter, we will present the two models chosen to model stress dispersion through the adhesive joint. The two models will be compared in the scenario presented in the thesis, and the more accurate model will be chosen

3.1.4.1 Kaelble Model

We next present the model developed by Kaelble (1960) [71] to characterize the localized stress distributions within the bond zone of a thin adhesive film. While the Kendall model provides a global energy balance for the critical peeling force, Kaelble proposed a one-dimensional analytical treatment that identifies two potential modes of failure interface propagation: boundary cleavage and boundary shear. Cleavage stresses, normal to the bond plane, are highly localized at the bond boundary where the flexible member undergoes a sudden transition from maximum to zero curvature [71]. The shear stresses, parallel to the bond plane, are distributed throughout the bond.

The model assumes the adhesive layer behaves as a continuous foundation of independent tension and shear springs, commonly referred to as a Winkler foundation [72]. To derive stress functions, we apply the conditions of equilibrium to a differential section of the flexible tape backing of length dx . The internal force and moment relations are established as shown below:

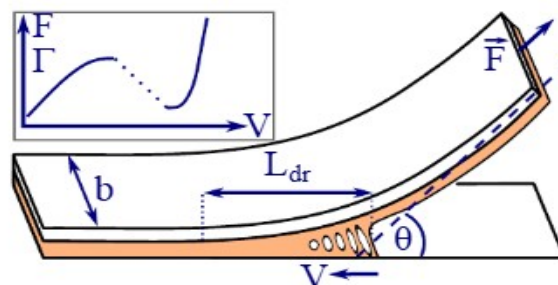


Fig 3.2 – Peel System Relevant to Kaelble Model [92]

- Normal Force Equilibrium:

$$(3.10) \quad ds + f = 0$$

Where f is the incremental cleavage force

● Shear Force Equilibrium:

$$(3.11) \quad dp + q = 0$$

Where q is the incremental shear force

● Moment Equilibrium:

$$(3.12) \quad dM_x + sdx - qh = 0$$

Where:

$$(3.13) \quad dM_{x^c} + dM_{x^s} = dM_x$$

The moment dM_{x^c} is the incremental bonding moment due to bond cleavage and the moment dM_{x^s} is the incremental bending moment due to bond shear

$$(3.14) \quad dM_{x^c} = -sdx$$

$$(3.15) \quad dM_{x^s} = qh$$

Using the equilibria presented, the formulas for bond cleavage stress and bond shear stress could be found.

To deduce the cleavage bond stress function, formulas (3.10), and (3.14) are used.

Bending deformation is governed by the following relation

$$(3.16) \quad EI \left(\frac{d\omega}{ds} \right)_c = M_c$$

Where E, I are Young's modulus and moment of inertia and $\frac{d\omega}{ds}$ is the exact curvature of the flexible member at $x = 0$.

For small bending defections, the following approximation can be used

$$(3.17) \quad \left(\frac{d\omega}{ds} \right)_c = \left(\frac{d^2y}{dx^2} \right)$$

Where $\frac{d^2y}{dx^2}$ is the curvature because of the bending moment M_c .

We will get

$$(3.18) \quad EI \left(\frac{d^2 y}{dx^2} \right) = M_c$$

By differentiating the equation twice, we obtain

$$(3.19) \quad EI \left(\frac{d^4 y}{dx^4} \right)$$

And by integrating equation (3.14) twice we will get

$$(3.20) \quad - \frac{ds}{dx} = \frac{d^2 M_c}{dx^2}$$

By combining equations (3.19), (3.20) the following expression is found

$$(3.21) \quad EI \left(\frac{d^4 y}{dx^4} \right) = - \frac{ds}{dx}$$

From equation (3.10) the following can be deduced:

$$(3.22) \quad ds = - \int \frac{E_a b y dx}{a}$$

Where E_a , a are the Young's modulus and the thickness of the adhesive interlayer, and b is the width of the bond.

Combining equations (3.21), (3.22) the following is obtained

$$(3.23) \quad EI \left(\frac{d^4 y}{dx^4} \right) + \left(\frac{E_a b}{a} \right) y$$

To solve the equation, we will use the standard method for solving linear differential equations of this type

$$(3.24) \quad (D^4 + \beta^4) y$$

$$\text{Where } D = \frac{d}{dx}, \beta^4 = \frac{E_a \cdot b}{aEI \cdot a}$$

As can be seen, for $x \rightarrow 0$, we will get $y = 0$, obtaining

$$(3.25) \quad y = (A \cos(\beta x) + B \sin(\beta x))$$

Where

$$(3.26) \quad A = \frac{\beta M_c + P \sin \omega}{2\beta^3 EI}, B = \frac{M_c}{2\beta^2 EI}$$

Solving equation (3.25) in terms of adhesive interlayer stress, $\sigma = \frac{yE_a}{a}$, the formula for cleavage stress can be obtained

$$(3.27) \quad \sigma = \frac{E_a e^{\beta x}}{2\beta^3 E I a} \beta M_c (\cos(\beta x) + \sin(\beta x)) + P \sin \omega \cos(\beta x)$$

After arranging the equation, and for

$$(3.28) \quad K = \frac{\beta M_c}{\beta M_c + P \sin \omega}$$

We will get the final equation

$$(3.29) \quad \sigma(x) = \sigma_0 (\cos(\beta x) + K \sin(\beta x)) e^{\beta x} @ \sigma_0 = \frac{2\beta P \sin \omega}{b(1-K)}$$

To deduce the cleavage bond stress function, formula (3.11) is used.

The flexible member tensile stress is defined as

$$(3.30) \quad \phi = \frac{P}{2hb}$$

By rearrangement and differentiation, we get

$$(3.31) \quad dp = 2hbd\phi$$

The shear modulus of the adhesive interlayer can be expressed by the following relation

$$(3.32) \quad G = \frac{(-q/bdx)}{(u/a)} = \frac{-qa}{ubdx}$$

$$q = -\frac{Gubdx}{a}$$

For $-q, u$ being the applied shear force and the shear displacement on the interlayer.

Using equation (3.11) with equations (3.31),(3.32), we obtain

$$(3.33) \quad 2hbd\phi - \frac{Gubdx}{a} = 0$$

$$d\phi = \frac{Gudx}{2ha}$$

The total flexible member tensional deformation at x equals the adhesive shear deformation u . Therefore

$$(3.34) \quad \frac{du}{dx} = \frac{\phi}{E}$$

By differentiating equation (3.33), we will get

$$(3.35) \quad \frac{d^2\phi}{dx^2} = \left(\frac{G}{2ah} \right) \left(\frac{du}{dx} \right)$$

Using (3.34), we obtain

$$(3.36) \quad \begin{aligned} \frac{d^2\phi}{dx^2} &= \left(\frac{G}{2Eah} \right) \phi \\ \frac{d^2\phi}{dx^2} - \left(\frac{G}{2Eah} \right) \phi &= 0 \end{aligned}$$

We will solve the equation, while defining

$$(3.37) \quad \alpha = \left(\frac{G}{2Eah} \right)^{0.5}$$

Therefore

$$(3.38) \quad (D^2 - \alpha^2)\phi = 0$$

The known solution for the equation is

$$(3.39) \quad \phi = C_1 e^{-\alpha x} + C_2 e^{\alpha x}$$

For $x \rightarrow 0$, it is seen that $\tau = 0$, we obtain $C_1 = 0$, and

$$(3.40) \quad \begin{aligned} C_1 &= 0 \\ C_2 &= \frac{P \cos \omega}{2bh} \end{aligned}$$

By differentiating equation (3.39), and using $C_1 = 0$ we get

$$(3.41) \quad \frac{d\phi}{dx} = \left(\frac{P \cos \omega}{2bh} \right) \alpha e^{\alpha x}$$

The expression for the adhesion of shear stress is known as

$$(3.42) \quad \tau = \frac{Gu}{a}$$

Which, using equation (3.33), shows the following

$$(3.43) \quad \frac{d\phi}{dx} = \frac{\tau}{2h}$$

By combining equations (3.41),(3.43) we can obtain the formula for shear stress

$$(3.44) \quad \tau(x) = \tau_0 e^{\alpha x} @ \tau_0 = \left(\frac{P \cos \omega}{b} \right) \alpha$$

To bridge the gap between Kaelble's mechanical stress profile and the physical properties of the materials, we utilize the Owens-Wendt-Rabel-Kaelble (OWRK) method (1970) [73]. This approach decomposes the total surface energy γ_l into dispersive and polar, γ_l^p, γ_l^d , components using Young's equation and contact angle measurements.

$$(3.45) \quad \frac{\gamma_l (1 + \cos \theta)}{2\sqrt{\gamma_l^d}} = \sqrt{\gamma_l^p} \sqrt{\frac{\gamma_l^p}{\gamma_l^d}} + \sqrt{\gamma_l^d}$$

By plotting this linear relationship for at least two known probe liquids, researchers can determine the surface energy components of the adherend. This thermodynamic characterization is essential for finding the work of adhesion, W_{sl} , which defines the critical boundary stress, σ_0 , or the energy release rate required to initiate the mechanical peeling described in Kaelble's differential equations.

3.1.4.2 Goland-Reissner Model

For adhesive joints where the load is not perfectly aligned with the bond plane, such as in a single-lap joint (SLJ), the Goland-Reissner (GR) model (1944) provides a necessary advancement over linear theories [74]. This model was the first to account for the geometric nonlinearity caused by the rotation of the adherends under eccentric tensile loading.

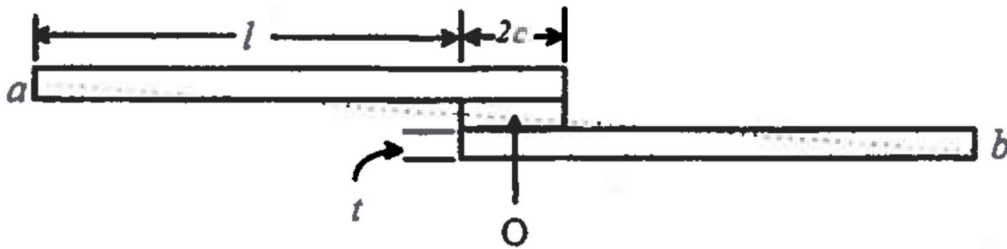


Fig 3.3 – Basic Goland Reissner Model System [74]

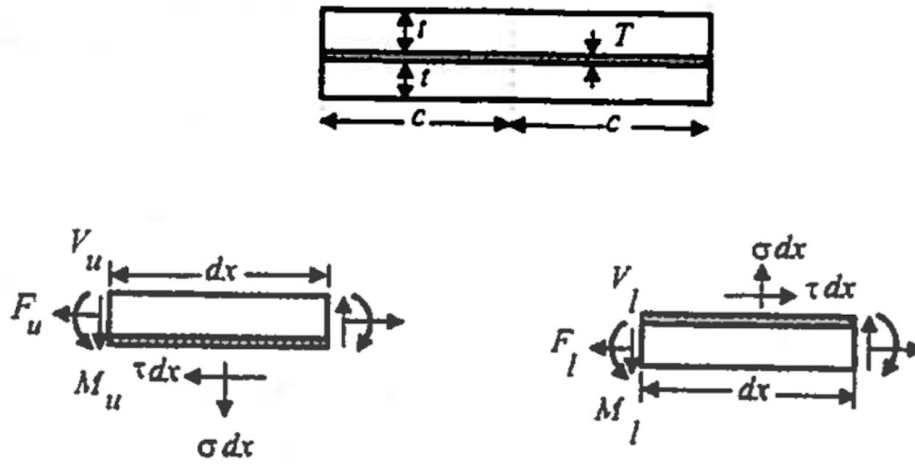


Fig 3.4 – Small Element View of Goland Reissner Model System [74]

Goland-Reissner analysis (1944) [75] for single lap joints was generalized by Bigwood and Crocombe (1989) [76] for any arbitrary end-loaded single overlap configuration. This results in a model that can be applied for many configurations and not only single overlap joints, as shown in **Fig 3.3**.

For the analysis, we will consider substrate loads and adhesive stresses acting on small elements of the upper and lower substrates, as seen in **Fig 3.4**.

Force and moment equilibrium of the upper and lower substrates are as seen below

$$(3.46) \quad \frac{dT_i}{dx} = \tau, \frac{dV_i}{dx} = \sigma, \frac{dM_i}{dx} - V_i + \tau \frac{t_i}{2} = 0$$

Where, for the upper and lower substrate respectively, and all other terms are as shown in **Fig 3.4**.

The expressions for the axial and transverse displacement, u_i, v_i , of each substrate can be produced in terms of the substrate surface strain and moments.

$$(3.47) \quad \frac{du_i}{dx} = \frac{(1 - \nu_i^2)}{E_i t_i} \left(P_i + \frac{M_i t_i^2}{2I_i} \right), \frac{d^2 v_i}{dx^2} = \frac{(1 - \nu_i^2) M_i}{E_i I_i}$$

Where t_i, ν_i, I_i, E_i are the thickness, Poisson's ratio, Moment of inertia and Young's modulus for substrate i.

An expression for the adhesive shear and transverse direct stresses in terms of the adhesive strains and hence substrate displacements can be produced

$$(3.48) \quad \tau = G\gamma = G \frac{(u_1 - u_2)}{\eta}, \sigma = G\varepsilon = G \frac{(v_1 - v_2)}{\eta}$$

Where τ is the adhesive thickness and G is the shear modulus.

By differentiating equation (3.48) the needed number of times, and substituting with equation (3.47). We will get an expression that can be differentiated again, and after substituting with equation (3.46), we will get the governing differential equations

$$(3.49) \quad \frac{d^7\tau}{dx^7} - K_1 \frac{d^5\tau}{dx^5} + K_3 \frac{d^3\tau}{dx^3} - K_5 \frac{d\tau}{dx} = 0$$

$$(3.50) \quad \frac{d^6\sigma}{dx^6} - K_1 \frac{d^4\tau}{dx^4} + K_3 \frac{d^2\tau}{dx^2} - K_5 \sigma = 0$$

Where K_1, K_3, K_5 are constants.

The governing differential equations are solvable only upon the application of suitable boundary conditions. While the complex adhesive sandwich described in Fig. x cannot be solved analytically and necessitates numerical computation, the axially loaded single lap joint offers a more accessible case. In this configuration, the twelve unknown loads at the interface ends are reduced to six, specifically at the distal ends of the upper and lower substrates. The seminal analytical solution for this streamlined setup was pioneered by Goland and Reissner (1944) [75], and its methodology is detailed below.

The 1944 Goland and Reissner approach to this joint geometry is divided into two distinct components: the determination of the load-related boundary conditions and the formulation of mathematical expressions for the adhesive shear, τ , and peel, σ , stresses. As the generalized sandwich analysis presented in the preceding section serves as a direct extension of their stress-state derivations, we will avoid redundant repetition of that derivation here. Instead, our focus will shift to the first phase of the analysis - solving for the boundary load states, a derivation fundamentally characterized by the introduction of what is now recognized as the bending moment factor.

A simplified single lap joint is shown in Fig 3.3. For a sufficiently long joint, we can assume the following expressions for the end loads

$$(3.51) \quad T_{11} = T_{22} = P$$

$$(3.52) \quad M_{11} = M_{22} = -\frac{P}{0.5t}$$

The transverse shear forces, V_{11} and V_{22} , are established to ensure the preservation of moment equilibrium within the joint. In physical application, the joint undergoes rotation upon loading, which shifts the line of action of the force closer to the central axis of the substrate. This geometric realignment effectively attenuates the internal moment, a phenomenon the Goland-Reissner model addresses by introducing a corrective factor k , known as the bending moment factor, to scale the nominal bending moment

$$(3.53) \quad M_{11} = M_{22} = -\frac{kPt}{2}$$

In a single-overlap joint subjected to axial loading, the bending moment within the substrate exerts a substantial influence on the adhesive stress state; consequently, the application of precise moment values is paramount for accurate analysis.

To mathematically capture the effects of joint rotation, the Goland-Reissner framework partitions the assembly into two distinct domains: the unbonded substrate extension (the free region) and the bonded overlap region, as illustrated in Fig 3.4 Transverse deflection is subsequently evaluated independently for each of these sections.

Apart from the vertical deflections of the neutral axes within these regions (w_a, w_b) , all physical parameters are defined, enabling the formulation of governing equations for the bending moments and the associated curvature for both segments.

$$(3.54) \quad w_a' = -\frac{M_a}{(EI)_a} = -\frac{P(\alpha x_a - w_a)}{(EI)_a}$$

$$(3.55) \quad w_b' = -\frac{M_b}{(EI)_a} = -\frac{P((\alpha L_a + x_b) - w_a - 0.5(t + \eta))}{(EI)_b}$$

While the transverse shear forces (V_{11}, V_{22}) are established to satisfy the requirements of moment equilibrium, physical joints inherently undergo rotation during the loading phase.

This geometric shift effectively linearizes the load path by moving the force's line of action closer to the substrate's neutral axis, which leads to a measurable reduction in the internal moment.

To mathematically capture this nonlinear behavior, the Goland-Reissner model introduces a corrective term k , known as the bending moment factor, to scale the nominal bending moment, shown in equation, In the traditional Goland-Reissner methodology, the influence of adhesive thickness η is typically omitted from the analysis. The governing system consists of second-order differential equations for the deflections w_a, w_b , representing the neutral axis displacements of the free and overlap regions, respectively.

These equations are analytically solvable by applying boundary conditions that stipulate zero deflection at the distal terminals of regions a and b, alongside the maintenance of continuous deflections and slopes at the transition interface. Under the physically sound assumption of significant free substrate length, these expressions can be simplified to derive the effective sandwich moment

$$M_{11} = -\frac{kPt}{0.5}$$

Where

$$(3.57) \quad k = \frac{\cosh(u_2 c)}{\cosh(u_2 c) + 2\sqrt{2} \sinh(u_2 c)}$$

$$(3.58) \quad u_2 = \sqrt{\frac{3(1 - \nu^2)P}{2\eta E_s}}$$

The parameter k serves as a scaling factor for the undeformed bending moment, transitioning from a value of 1.0 under low-load conditions to a minimum threshold of approximately 2.6 as the applied load increases.

In tandem, the transverse shear forces V_{11}, V_{22} are determined to ensure the preservation of moment equilibrium within the joint assembly.

By utilizing these boundary conditions in conjunction with the governing elasticity equations for the adhesive layer, we derive the following closed-form expressions for the shear and peel stresses in an axially loaded single-overlap joint.

The equation found for the shear stress is

$$\frac{\tau(x)}{\tau_{av}} = \frac{1}{4} \left[\frac{\beta c}{t} (1+3k) \frac{\cosh\left(\frac{\beta c}{t} \frac{x}{c}\right)}{\sinh\left(\frac{\beta c}{t}\right)} + 3(1-k) \right] \quad (3.59)$$

For $\tau_{av} = \frac{P}{2bc}$, we finally get

$$\tau(x) = \frac{P}{8bc} \left[\frac{\beta c}{t} (1+3k) \frac{\cosh\left(\frac{\beta c}{t} \frac{x}{c}\right)}{\sinh\left(\frac{\beta c}{t}\right)} + 3(1-k) \right] \quad (3.60)$$

Where

$$\beta^2 = \frac{8tG}{E_s \eta} \quad (3.61)$$

The equation found for the peel stress is

$$\begin{aligned} \frac{\sigma(x)}{\sigma_{av}} = \frac{t}{R_1 c^2} & \left[\left(R_2 \lambda^2 \frac{k}{2} - \lambda k w \cos \lambda \cosh \lambda \right) \cosh\left(\frac{\lambda x}{c}\right) \cos\left(\frac{\lambda x}{c}\right) \right. \\ & \left. + \left(R_3 \lambda^2 \frac{k}{2} - \lambda k w \sin \lambda \sinh \lambda \right) \sinh\left(\frac{\lambda x}{c}\right) \sin\left(\frac{\lambda x}{c}\right) \right] \end{aligned} \quad (3.62)$$

For $\sigma_{av} = \frac{P}{b}$, we finally get

Taken together, these two models give the failure criterion this thesis relies on throughout: the Kaelble formulation provides the tensile peel-stress distribution ahead of the bond edge, and the Goland-Reissner formulation provides the corresponding shear stress distribution across the overlap. The critical peeling force derived here, P_{cr} , is what Chapter 5 evaluates numerically for the measured adhesive properties, and it is

what ultimately bounds the tensile capacity used in the planner's hard feasibility gate in Chapter 6. The length of the derivation reflects that this single quantity is downstream of the adhesive's full stress distribution, not an assumption -- every term in the final expression traces back to a physically measured property from Chapter 5.

$$(3.63) \quad \sigma(x) = \frac{Pt}{R_1 b c^2} \left[\left(R_2 \lambda^2 \frac{k}{2} - \lambda k w \cos \lambda \cosh \lambda \right) \cosh \left(\frac{\lambda x}{c} \right) \cos \left(\frac{\lambda x}{c} \right) + \left(R_3 \lambda^2 \frac{k}{2} - \lambda k w \sin \lambda \sinh \lambda \right) \sinh \left(\frac{\lambda x}{c} \right) \sin \left(\frac{\lambda x}{c} \right) \right]$$

Where

$$(3.64) \quad \lambda = \frac{c}{t} \left(\frac{6Et}{E_s \eta} \right)^{0.25}$$

$$(3.65) \quad R_1 = \frac{\sinh 2\lambda + \sin 2\lambda}{2}$$

$$(3.66) \quad R_2 = \sinh \lambda \cos \lambda - \cosh \lambda \sin \lambda$$

$$(3.67) \quad R_3 = \sinh \lambda \cos \lambda + \cosh \lambda \sin \lambda$$

3.2 RRT and RRT* Algorithms

3.2.1 Rapidly – exploring Random Tree (LaValle, 1998)

The Rapidly exploring Random Tree (RRT) is a randomized data structure designed to efficiently explore high-dimensional configuration spaces by probing the free space with a sampling scheme. It is particularly effective for robotic manipulators with many degrees of freedom where explicit mapping of obstacles is computationally prohibitive. [48]

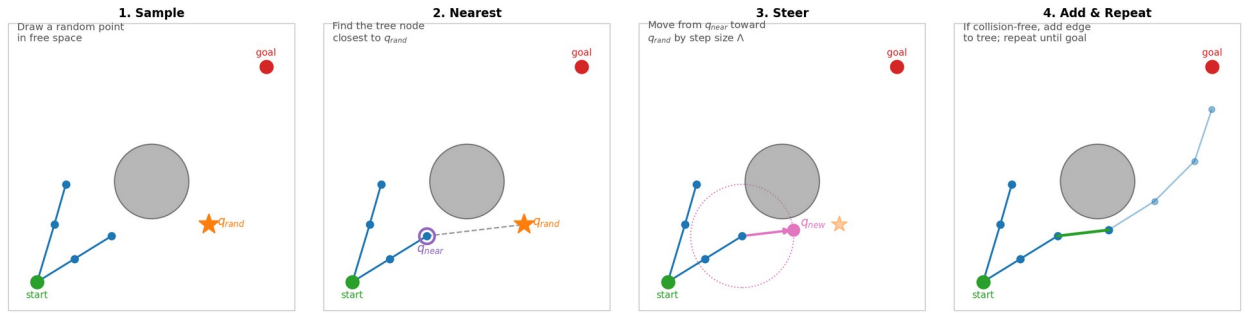


Fig 3.8 – RRT growth: one iteration (sample, find nearest node, steer toward the sample, add the new node if collision-free)

This single cycle, sample, nearest, steer, add repeats until a node falls within the goal region or an iteration budget is reached; the tree in the rightmost panel is the starting point for the next iteration.

3.2.1.1 Problem Definition

The motion planning problem for RRT is defined as follows

Input:

- A configuration space $C \stackrel{def}{=} \mathbb{R}^d$ for some $d \geq 2$.
- A free space $F \subseteq C$ representing all collision-free configurations.
- Initial and target configurations $s, t \in F$.

Task: Find a continuous path $\sigma: \rightarrow F$ such that $\sigma(0) = s, \sigma(1) = t$

Objective: Identify any feasible solution that connects the start to the goal region without intersecting the obstacle space C_{obs}

3.2.1.2 Algorithm Overview

The algorithm builds a tree rooted at q_{init} by performing n iterations. In each iteration, a random configuration q_{rand} is sampled uniformly from the bounded configuration space X . The algorithm identifies the existing node q_{near} in the current tree T that is closest to q_{rand} using a distance metric ρ , typically Euclidean: $d = \|q_{rand} - q_{near}\|_2$. A new configuration q_{new} is then generated by moving a fixed step size δ from q_{near} toward q_{rand} . If the line segment connecting q_{near} and q_{new} is collision-free, verified by a black-box collision detection module $D(x)$, the new node is added to the tree and an edge is established. This iterative expansion ensures the tree grows rapidly toward large unexplored regions of the state space.

3.2.1.3 Pseudo Code

In this subchapter, we present the pseudocode for the RRT algorithm described in the subchapter above.

Algorithm 1 Rapidly-exploring Random Tree (RRT)

Require: Initial configuration q_{init} , number of iterations n , step size Δ , target region F_{target}

Ensure: A collision-free path τ from q_{init} to the target region, or FAILURE

```

1:  $T.init(q_{init})$  ▷ Initialize tree with root node
2: for  $i = 1$  to  $n$  do
3:    $q_{rand} \leftarrow \text{SAMPLE-FREE}$ 
4:    $q_{near} \leftarrow \text{NEAREST}(T, q_{rand})$ 
5:    $q_{new} \leftarrow \text{STEER}(q_{near}, q_{rand}, \Delta)$ 
6:   if  $\text{COLLISION-FREE}(q_{near}, q_{new})$  then
7:      $T.addVertex(q_{new})$ 
8:      $T.addEdge(q_{near}, q_{new})$ 
9:     if  $q_{new} \in F_{target}$  then
10:      return  $\text{PATH}(T, q_{new})$ 
11:     end if
12:   end if
13: end for
14: return FAILURE

```

Fig 3.5 – Pseudocode: RRT Algorithm

3.2.1.4 Key Components of RRT

The RRT implementation relies on four primary sub-functions:

- **SAMPLE-FREE:** Returns a sample q_{rand} from the free space F uniformly at random.

- **NEAREST:** Identifies the vertex $q_{near} \in V$ that is closest to q_{rand} based on a distance metric, typically the L_2 norm.
- **STEER:** Generates a new configuration q_{new} by moving from q_{near} toward q_{rand} by an incremental step size δ .
- **COLLISION-FREE:** A "black box" module that verifies whether the line segment connecting q_{near} and q_{new} intersects any obstacles.

3.2.1.5 Optimality and Completeness

RRT is proven to be probabilistically complete, meaning that if a feasible path exists, the probability that the algorithm will find it converges to one as the number of samples approaches infinity. However, RRT is not optimal. Because new points are attached to the nearest node without additional optimization, the resulting paths are often tortuous and far from the shortest possible distance.

3.2.1.6 Complexity Analysis of RRT

The computational complexity of the RRT is $O(n^2)$ relative to the number of iterations. The nearest neighbor search, if using a brute-force approach, is $O(n)$, but can be reduced to $O\left(n^{1-\frac{1}{d}}\right)$ using spatial partitioning data structures such as kd-trees.

3.2.1.7 Theoretical Properties of RRT

The primary theoretical advantage of RRT is that its distribution of vertices approaches the sampling distribution, leading to consistent exploratory behavior. Furthermore, an RRT always remains connected despite a minimal number of edges, making it an efficient module for complex planning systems.

3.2.1.8 RRT Example

Consider a 2D environment with a single obstacle. Initially, the tree contains only the root q_{start} . In the first iteration, a q_{rand} is sampled; if it is in an open region, the tree extends toward it. After 500 iterations, the uniform sampling strategy naturally draws the tree toward the largest unexplored Voronoi regions, effectively bypassing the obstacle until a branch enters the goal region.

A classic illustration of RRT exploring free space versus an environment with a single obstacle after 500 iterations, as demonstrated in the work of LaValle (1998). This example highlights how the uniform sampling strategy and Voronoi bias enable the tree to naturally bypass obstructions to reach the goal.

3.2.2 Rapidly – exploring Random Tree Star (Karaman and Frazzoli, 2011)

To address the sub-optimality of RRT, the Rapidly - exploring Random Tree Star (RRT*) algorithm was introduced as an asymptotically optimal variant. It maintains the exploratory power of RRT while introducing local graph-rewiring techniques to refine path quality. [77]

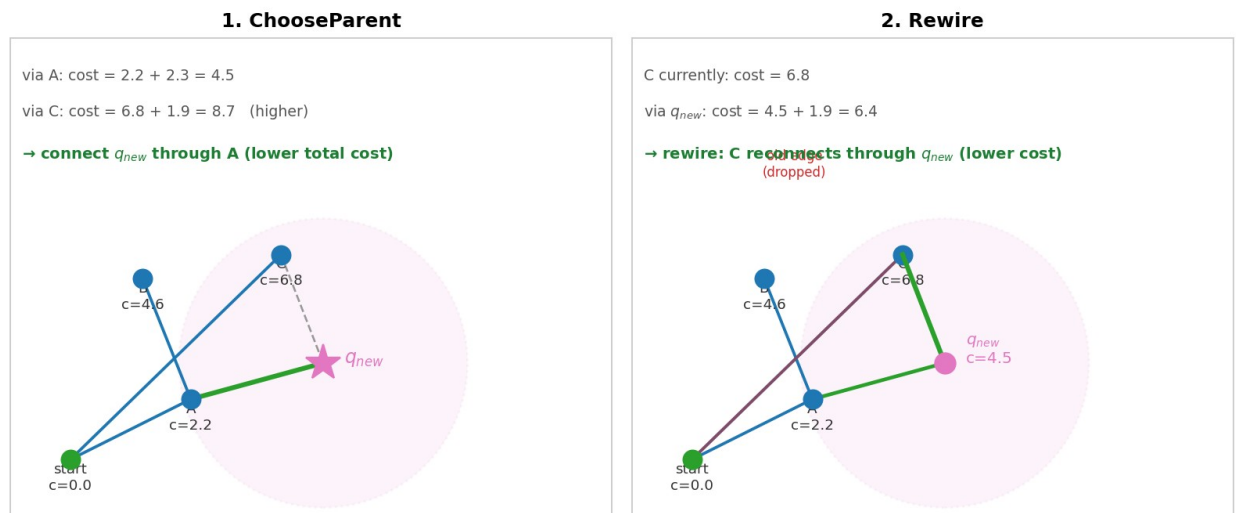


Fig 3.9 – RRT*: the two steps added beyond RRT. *ChooseParent* (left) connects a new node through whichever nearby node gives the lowest accumulated cost, not simply the nearest one; *Rewire* (right) then checks whether that new node offers a cheaper path to any of its other neighbors, and reconnects them if so.

3.2.2.1 Problem Definition

The problem definition for RRT* is identical to RRT but includes an explicit quality measure

Cost Function: $c(\sigma)$, typically defined as the total path length (Hausdorff measure).

Goal: Find an optimal path σ^* such that

$$(3.68) \quad c(\sigma^*) = \inf \left\{ c(\sigma) \mid \sigma \in \sum_{F,s,t} \text{ \& } \sigma = \text{robust} \right\}$$

3.2.2.2 Algorithm Overview

Every RRT* iteration begins with a standard RRT-style expansion step to create a candidate node q_{new} . However, after q_{new} is successfully generated and verified as collision-free, the algorithm performs a neighborhood search within a shrinking radius r . It utilizes a ChooseParent procedure to evaluate all nodes in the neighborhood X_{near} returning the subset of vertices within a min-radius ball. The algorithm selects the node q_{min} that yields the minimum cumulative cost-to-come from the root to q_{new} . After the node is inserted, a Rewire procedure iterates through the same neighborhood to check if the path to any existing node can be improved by routing it through the newly added q_{new} . If a cheaper path is found, the tree is restructured by updating the neighboring node's parent pointer.

3.2.2.3 Pseudo Code

In this subchapter, we present the pseudocode for the RRT algorithm described in the subchapter above.

Algorithm 2 Rapidly-exploring Random Tree Star (RRT*)

Require: Initial configuration q_{init} , number of iterations n , step size Δ , connection radius function $r(n)$

Ensure: A tree T containing an asymptotically optimal path from q_{init}

```
1:  $T.init(q_{init})$ 
2: for  $i = 1$  to  $n$  do
3:    $q_{rand} \leftarrow \text{SAMPLE-FREE}$ 
4:    $q_{near} \leftarrow \text{NEAREST}(T, q_{rand})$ 
5:    $q_{new} \leftarrow \text{STEER}(q_{near}, q_{rand}, \Delta)$ 
6:   if  $\text{COLLISION-FREE}(q_{near}, q_{new})$  then
7:      $X_{near} \leftarrow \text{NEAR}(T, q_{new}, r(n))$   $\triangleright$  Vertices within shrinking radius  $r(n)$ 
8:      $q_{min} \leftarrow \text{CHOOSEPARENT}(X_{near}, q_{near}, q_{new})$ 
9:      $T.addVertex(q_{new})$ 
10:     $T.addEdge(q_{min}, q_{new})$ 
11:     $T \leftarrow \text{REWIRE}(T, X_{near}, q_{min}, q_{new})$ 
12:   end if
13: end for
14: return  $T$ 
```

Fig 3.6 – Pseudocode: RRT* Algorithm

3.2.2.4 Key Components of RRT

RRT* enhances the standard RRT with two critical optimization procedures:

- **ChooseParent:** Instead of connecting q_{new} to the closest node q_{near} , the algorithm searches a neighborhood X_{near} within a radius r around q_{new} . It selects the parent q_{min} that results in the minimum cumulative cost from the root:

$$(3.69) \quad \text{Cost}(q_{new}) = \min_{q \in X_{near}} \{ \text{Cost}(q) + c(q, q_{new}) \}$$

- **Rewire:** After adding q_{new} , the algorithm checks if any existing nodes in X_{near} can reach the root with a lower total cost if they are reconnected through q_{new} . If $\text{Cost}(q_{new}) + c(q_{near}, q_{neighbor}) < \text{Cost}(q_{neighbor})$, the old edge is deleted, and a new one is established.

3.2.2.5 Optimality and Completeness

RRT* inherits the probabilistic completeness of RRT and adds asymptotic

optimality. As the number of iterations $n \rightarrow \infty$, $\text{Cost}(q_{new}) \rightarrow \text{Cost}^*$ with probability 1.

$$(3.70) \quad r(n) = \gamma \left(\frac{\log n}{n} \right)^{1/d}$$

.Where d is the dimension and $\mathcal{V}$ is a constant related to the free space volume

3.2.2.6 Complexity Analysis of RRT*

The time complexity of RRT* is $O(n \log n)$ per iteration due to the nearest-neighbor and range searches. While RRT* requires more computation than RRT per sample, it remains more efficient than grid-based solvers for high-dimensional problems because it avoids explicit state-space discretization.

3.2.2.7 Theoretical Properties of RRT*

- Voronoi Bias: Like RRT, it explores large open regions first.
- Anytime Capability: It finds an initial feasible path quickly and then continuously improves it during the remaining execution time.
- Pruning: Advanced variants (e.g., Informed RRT*) further improve efficiency by restricting sampling to a hyperspheroid once an initial solution is found.

3.2.2.8 RRT* Example

In a multi-obstacle environment, RRT* quickly identifies a "zigzag" path to the goal. As sampling continues, the ChooseParent and Rewire steps eliminate unnecessary turns. For example, if a new sample is placed between two waypoints, RRT* will "short-cut" the path through this new node, gradually straightening the trajectory toward the shortest global distance.

The effectiveness of the RRT* local rewiring and optimization is demonstrated in the work of Solovey et al. (2019) [78] and the comparative analysis in the Jackal robot environment. These examples show that in a 100-by-100-meter workspace with obstacles, RRT* produces significantly shorter paths compared to standard RRT by modifying its tree structure on-the-fly.

3.3 Sensor and Classification Principles

The ability of a gripper to validate its own grasp depends on the physical principles of its sensors and the robustness of its classification logic.

3.4 Photoresistors

A photoresistor (LDR) is a semiconductor device whose resistance decreases as incident light intensity increases. In a robotic finger, a photoresistor can be paired with an LED; when an object is grasped, it blocks or reflects the light, creating a signature change in the electrical signal. This is a low-cost, high-speed alternative to camera-based vision, though it requires calibration to account for ambient lighting conditions and the differing reflectivity of materials. [79]

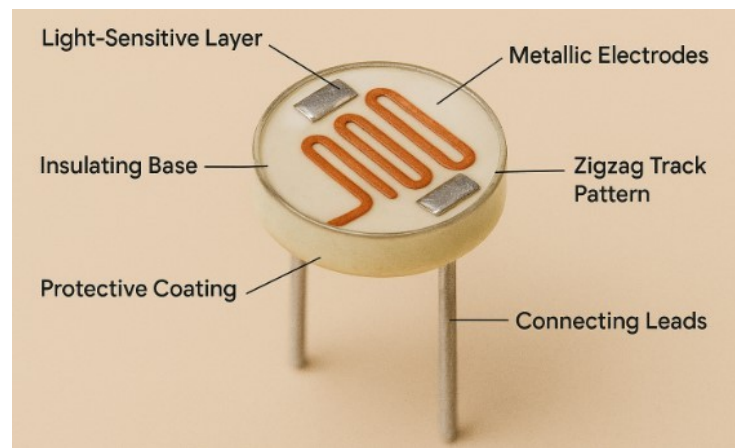


Fig 3.7 – Photoresistor (LDR) Construction and Key Components

3.5 Neural Networks

Artificial Neural Networks (NNs) are computational models inspired by biological neural networks, organized as interconnected groups of units called neurons. These units are typically aggregated into layers: an input layer to receive features, hidden layers to extract internal representations, and an output layer for final predictions.

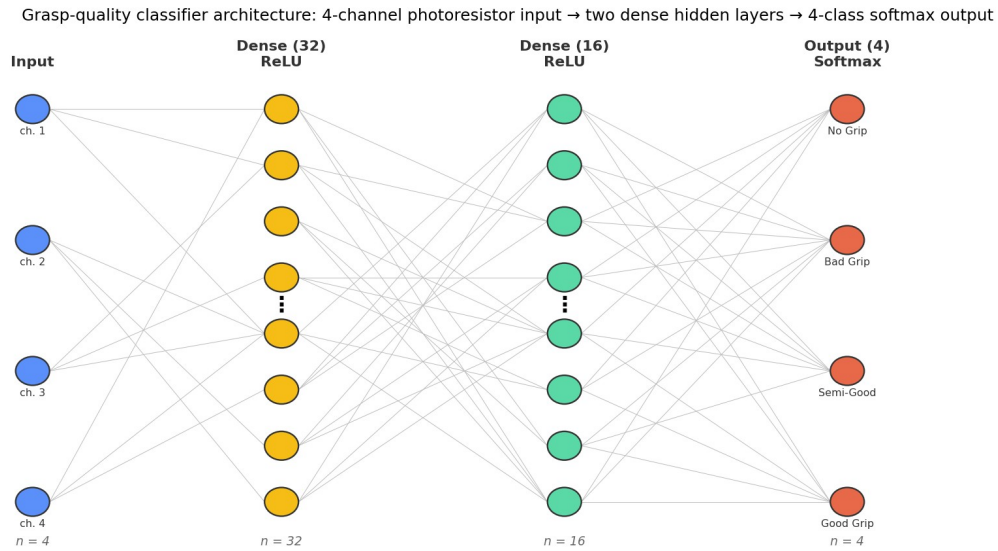


Fig 3.10 – Grasp-quality classifier architecture: 4-channel photoresistor input, two dense hidden layers (32, 16 units, ReLU), 4-class softmax output

3.5.1 Artificial Neuron and Non-Linearity [80]

Each artificial neuron receives inputs x_i , which are multiplied by weights w_i and summed with a bias b to form an affine transformation: $z = \sum w_i x_i + b$. To enable the network to learn complex, non-linear relationships, an **activation function** $\phi(z)$ is applied to the result. Common functions include:

- **Sigmoid:** An S-shaped function defined as $\phi(z) = \frac{1}{1 + e^{-z}}$, producing outputs between 0 and 1.
- **Rectified Linear Unit (ReLU):** A thresholding function $\phi(z) = \max(0, z)$ that accelerates the convergence of gradient descent.
- **Hyperbolic Tangent (tanh):** An antisymmetric function ranging from -1 to 1.

3.5.2 The Universal Approximation Theorem

A pivotal result in neural network theory is the Universal Approximation Theorem (UAT). It states that a feedforward network with a single hidden layer and a finite number of neurons can approximate any continuous function on a compact subset of $\mathbb{R}^n$ to an arbitrary degree of

accuracy, provided it uses a non-polynomial activation function. This theorem provides the mathematical justification for using ANNs to model the intricate, often unpredictable interactions encountered in unstructured robotic tasks. [81]

3.5.3 Training Mechanics and Complexity

Networks "learn" by adjusting their parameters to minimize the difference (empirical risk) between predicted and actual values. The standard training process involves two phases:

1. **Forward Pass:** Propagates input data through the layers to generate a prediction.

The computational complexity is $O(\omega)$, where ω is the total number of weights.

2. **Backward Pass (Backpropagation):** Computes the gradient of a loss function with respect to each weight using the chain rule of calculus. Parameters are then updated via Gradient Descent by moving in the opposite direction of the gradient to reach a global minimum.

The total algorithmic complexity for training over e epochs and m samples is $O(\omega \cdot m \cdot e)$, reflecting a linear growth relative to the dataset size and iteration count.

4. Problem Definition

The primary objective of this research is to develop a robust trajectory planning and validation framework for a robotic manipulator performing pick-and-place tasks using adhesive interfaces. Unlike conventional mechanical clamping, the integrity of an adhesive grasp is a dynamic function of the robot's motion. This chapter outlines the environment, the specific hardware specifications, and the mathematical constraints that define a "successful" manipulation.

4.1 Adhesion Based Pick and Pack Task

The manipulation task is modeled as a transformation of a target object from an initial state q_{init} to a target goal region G within a workspace W . In the context of a warehouse environment, the system must navigate cluttered bins or shelves to retrieve items of varying geometries, ranging from planar boxes to convex bottles and textured produce.

The task cycle is defined as follows:

1. **Sticker Formation:** The system generates a disposable adhesive agent.
2. **Acquisition:** The gripper approaches the object surface and establishes a bond under a controlled compressive preload P .
3. **Transport:** The robot moves the object along a trajectory $\sigma(t)$ while maintaining the inertial loads below the adhesive failure threshold.
4. **Release:** The gripper deactivates its locking mechanism, releasing the agent and the object at the destination.

4.2 Hardware Specifications – Robotic Arm

The proposed system is designed to be platform-agnostic and is compatible with any industrial or collaborative 6-degree-of-freedom (6-DOF) robotic arm. This DOF count is the minimum requirement to ensure full manipulability and the ability to approach surfaces from arbitrary orientations within the workspace.

For any chosen manipulator, the configuration space C is defined by the joint angle vector $q \in \mathbb{R}^6$. The free space $F \subseteq C$ is defined as the subset of configurations where neither the robot nor the grasped object intersects with environmental obstacles. While specific reach and

payload limits are arm-dependent, the trajectory planning algorithm accounts for these parameters as hard kinematic boundaries.

4.3 Custom Thin-Film Adhesive End-Effector

In this thesis, a custom adhesive gripper, developed by Pickcommerce AI Robotics is used. The gripper, shown in FIG, is built from a rectangular shell, within an active and passive rollers are located, the adhesive film travels on the outside of the shell from side to side. The bottom of the gripper is flat and is where the lifted object is attached to the gripper. The object is lifted by forcing contact with the gripper by pushing it onto the object using the robotic arm. After the object is moved to the desired location it is released from the gripper using a piston activated pusher, polling the adhesive grip away from the object and breaking the adhesive bond.

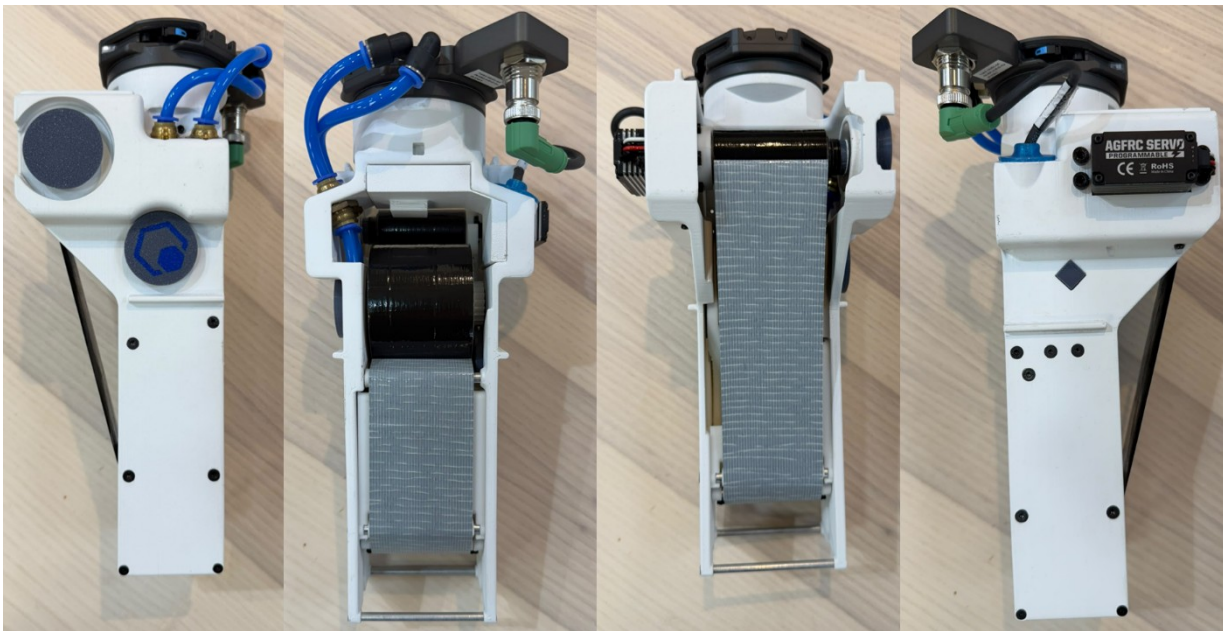


Fig 4.1 –Pickcommerce’s Custom Thin-Film Adhesive Gripper

4.4 Integrated Grasping Validation System

Autonomous operation requires real-time feedback regarding the status of the adhesive interface. The system incorporates an Integrated Grasping Validation System utilizing a distributed photoresistor (LDR) array. This sub-system performs two essential functions:

1. **Grasp Quality Assessment:** Upon contact, the array detects light occlusion and reflection signatures produced by the object. A feed-forward Artificial Neural

Network (ANN) processes these signals to classify the grasp as "Stable" or "Unstable" based on contact area and centering.

2. **Slippage Detection:** During motion, the system samples sensor data at high frequency. A Long Short-Term Memory (LSTM) network is employed to identify temporal "signatures" of signal drift or micro-oscillations that indicate incipient slip. This enables the system to implement reactive safety maneuvers before a catastrophic loss of grip occurs.

4.5 Operating Constraints and Failure Limits

1. The planning problem is defined by hard constraints that must be satisfied at every time step t along the path σ :
2. **Kinematic Boundaries:** Joint positions, velocities, and accelerations must remain within the specific bounds of the deployed robotic arm.
3. **Adhesive Limit Surface:** The instantaneous force P on the bond must not exceed the material's "limit surface". This force is resolved into tangential $P_T = P \cos \theta$ and normal components relative to the bond surface.
4. **Interfacial Stress Thresholds:** Failure initiates if peak stresses exceed the critical energetic threshold. As derived from the **Goland-Reissner model**, the shear stress $\tau(x)$ incorporates the $(1-3k)$ reduction term for load alignment, and both stresses are governed by the stiffness-based concentration factor β :

$$(4.1) \quad \beta = \sqrt{\frac{G_{ad}}{E_p h_p^3 h_{ad}}}$$

Where h_p^3 accounts for the flexural rigidity of the adherend during motion.

4.6 Problem Objective

The goal is to develop a force-aware path planning algorithm that identifies an asymptotically optimal path σ^* minimizing travel cost while strictly avoiding configurations that would result in adhesive failure or unrecoverable slippage, as verified by the integrated validation system.

5. Adhesive Force Based Acceleration Calculation

To ensure the reliability of the pick-and-place task, the trajectory planner must account for the physical limits of the adhesive interface. Unlike mechanical grippers, where the grip force is actively controlled by an actuator, the capacity of an adhesive gripper is a passive property dictated by the material's limit surface and the dynamic loads applied during robot motion. This chapter derives the mathematical expressions for maximum allowable linear and angular accelerations based on the mechanical response of the thin-film bond.

5.1 Theoretical Limit Surfaces

The concept of a Limit Surface (L_{cm}) is fundamental to characterizing the grasping envelope of an adhesive end-effector. We refer to the gripper's capabilities as a convex set of attainable forces and moments that can be exerted on a target object without causing bond detachment. For the purpose of this development, the gripper is assumed to be rigidly connected to the object until the maximum adhesive capacity, defined by the boundary of the limit surface, is exceeded. [89], [90]

In a 2D plane, the net force and moment vector $F_{cm} = \begin{bmatrix} F_x \\ F_y \\ M \end{bmatrix}$ must lie within L_{cm} . The boundary of this set is determined by the adhesive's failure criteria in both shear and peel modes. If the resultant force vector from the robot's acceleration and the object's weight shifts outside this surface, the bond initiates a peeling process, leading to immediate or gradual loss of the grasp. [89]

5.2 Linear and Angular Acceleration Derivations

In this section, we formulate the relationship between the manipulator's kinematics and the resulting stresses at the adhesive interface. The total force P acting on the bond is a vector sum of the gravitational force mg and the inertial force ma imparted by the robotic arm.

5.2.1 Development of the General Acceleration Derivation

We will first consider the most general scenario, where the gripper is tilted in a specific angle, and the gripper is moving in a nonspecific direction, meaning in an angle compared to the Cartesian Coordinate System.

This scenario is illustrated in **Fig 5.1**.

Free-body diagram: gripper tilted at angle θ , accelerated in an arbitrary direction $\vec{a}$

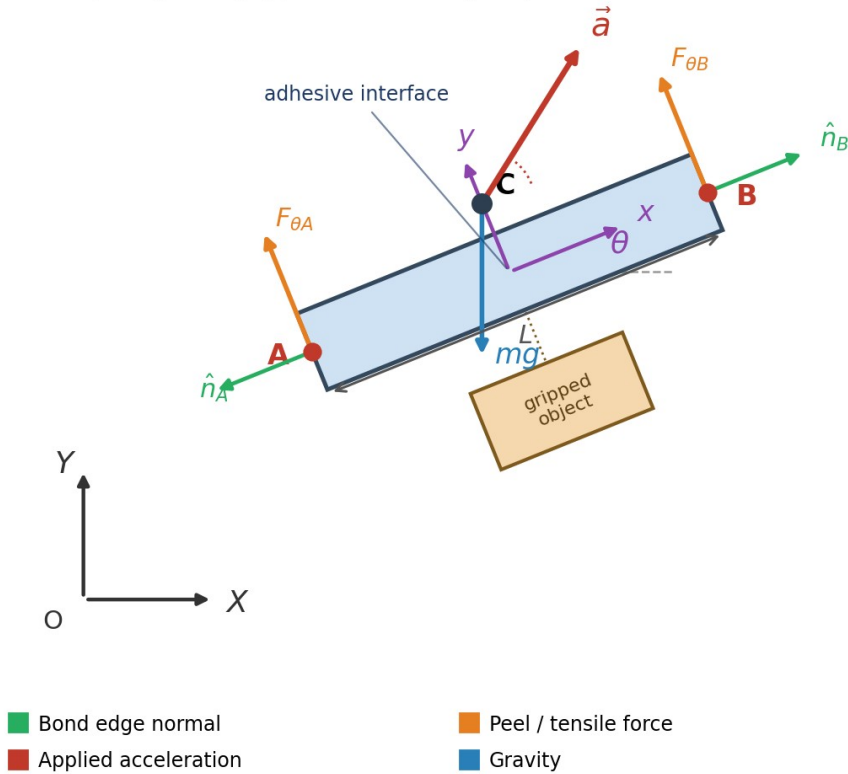


Fig 5.1 – Tilted Gripper Accelerated in a Nonspecific Direction

When the gripper is tilted at a specific angle, the components of gravity and acceleration are redistributed across the shear and peel axes. This orientation significantly alters the "limit surface" as the effective weight of the object now contributes to both tangential and normal stresses.

For the adhesive gripper being used in this thesis, it can be seen that the tilt angle of the center of the gripper and the edges of the gripper are the same, and when assuming good adhesion

between the object and the gripper, we can assume that the face of the object is parallel to the face of the gripper, and therefore the tilt angle is the same.

We will define the axis of the gripper as follows:

$$(5.1) \quad \hat{n}_{\perp} = \hat{r} = R_{(\varphi)} \cdot \hat{i}$$

where $R(\varphi)$ rotates by the tilt angle φ

$$(5.2) \quad \hat{n}_A = \hat{n}_B = \hat{n} = R_{(\varphi)} \cdot \hat{j} = R_{(\varphi+90^\circ)} \cdot \hat{i}$$

Written this way, both the contact-normal direction and the tangential direction are simply the global basis vectors $\hat{i}$ and $\hat{j}$ rotated by the gripper tilt angle φ , the same rotation $R(\varphi)$ applied to each. Section 5.2.1 previously expanded every such vector into its $\hat{i}$, $\hat{j}$ components by hand at each step; expressing them instead as a rotation of a fixed basis makes the dependence on φ explicit and avoids re-deriving the same trigonometric expansion for every subsequent vector in this section.

Where φ is the tilt angle of the gripper compared to the horizon, $\hat{n}$ is the unit vector perpendicular to the contact area and $\hat{r}$ is the unit vector parallel to the contact area.

In the general scenario, the direction of movement is not specific, and therefore it could be in an angle compared to the gripper, we will define the acceleration vector as follows:

$$(5.3) \quad \hat{a} = \cos \delta \cdot \hat{n} + \sin \delta \cdot \hat{r}$$

Where δ is the angle between the acceleration direction and the gripper's own local axes.

By inserting eq. (5.2),(5.1) into eq. (5.3), we can develop the correlation between the direction of movement and the Cartesian Coordinate System:

$$(5.4) \quad \hat{a} = \cos \delta \cdot R_{(\varphi)} \cdot \hat{j} + \sin \delta \cdot R_{(\varphi)} \cdot \hat{i}$$

$$(5.5) \quad \hat{a} = R_{(\varphi)} (\cos \delta \cdot \hat{j} + \sin \delta \cdot \hat{i})$$

To find the expression for a , we will use the force equilibrium equation and the moment equilibrium equation.

We know that according to Newton's Second Law $\Sigma \vec{F} = m\hat{a}$, and for the scenario presented in this chapter:

$$(5.6) \quad \vec{F}_{\theta} + \vec{F}_r + \vec{F}_{mg} = m \cdot a \cdot \hat{a}$$

We can define each vector of the system separately and develop an expression of linear acceleration as a factor of the forces in the system.

First, we will consider the gravitational forces, the forces of gravity in this scenario are dictated by the mass of the lifted object, therefore:

$$(5.7) \quad \bar{F}_{mg} = m_{obj} \cdot \bar{g}, \bar{g} = -g \cdot \hat{j}$$

Now, let us consider the expression of acceleration; for acceleration as well, the relevant mass is the mass of the object. We also know that $\sum \bar{F}_y = m_{obj} \cdot a_y$, and by using eq. (5.7), we get:

To find the expressions for the forces due to tensile stress, $\bar{F}_\theta$, and due to shear stress, $\bar{F}_r$, we must consider the following. Tensile forces are only perpendicular to the contact area, therefore they are purely in the direction $\hat{n}$, while shear forces work only parallel to the contact area, therefore they are purely in direction $\hat{r}$.

In the adhesive gripper, there are two edges to the contact area as can be seen in Fig 5.1, which means there are two sets of tensile and shear forces, one from each end. It is important to know that the expression of the tensile and shear forces is an expression of stress distribution, which will be developed in a later chapter.

The expression for the tensile forces is:

$$(5.8) \quad \bar{F}_\theta = F_\theta \cdot \hat{n}$$

After inserting eq. we get:

$$(5.9) \quad \bar{F}_\theta = F_\theta \cdot R(\varphi) \cdot \hat{j}$$

The expression for the shear forces is:

$$(5.10) \quad \bar{F}_r = F_r \cdot \hat{r}$$

Depending on the direction of movement.

After inserting eq. (5.2) we get:

$$(5.11) \quad \bar{F}_r = F_r \cdot R(\varphi) \cdot \hat{i}$$

By inserting eq. (5.6), (5.8), (5.11) into eq. (5.5) and projecting onto the gripper's own local axes with $R(-\varphi)$, we get a set of two equations for the acceleration:

$$(5.13) \quad F_r \cdot \hat{i} + F_\theta \cdot \hat{j} - m_{obj} \cdot g \cdot (\sin \omega \cdot \hat{i} + \cos \omega \cdot \hat{j}) = m \cdot a (\sin \delta \cdot \hat{i} + \cos \delta \cdot \hat{j})$$

After rearranging the expressions, we get:

$$(5.14) \quad F_r = m \cdot a \cdot \sin \delta + m_{obj} \cdot g \cdot \sin \omega$$

$$(5.15) \quad F_\theta = m \cdot a \cdot \cos \delta + m_{obj} \cdot g \cdot \cos \omega$$

For finding the expression for the angular acceleration, we will use the moment equilibrium equation.

We know that according to Newton's Second Law for rotation $\sum \bar{M} = I \cdot \alpha$. In this chapter we are discussing the calculation in 2D, therefore the rotation will only be around the $\hat{k}$ axis. For a calculation done around C, the geometrical center of the lifted object, we will get:

$$(5.16) \quad \sum \bar{M}_C = \bar{M}_{mg} + \bar{M}_a + \bar{M}_\theta + \bar{M}_r = I \cdot \alpha$$

From the equation above, we can develop an expression of angular acceleration as a factor of the moments acting on the object.

First, we will consider the moment due to gravitational forces:

$$(5.17) \quad \bar{M}_{mg} = \bar{r}_g \times \bar{F}_{mg}$$

Where the arm of the moment, $\bar{r}_g$, is regarding the gripper, and therefore is in the direction of the gripper's own tangential axis:

$$(5.18) \quad \bar{r}_g = \left(\frac{L}{2} \right) \cdot \hat{r}$$

Using eq. (5.7), (5.17) we get:

$$(5.19) \quad \begin{aligned} \bar{M}_{mg} &= \left(\frac{L}{2} \right) \cdot \hat{r} \times (-m_{obj} \cdot g \cdot \hat{j}) \\ \bar{M}_{mg} &= - \left(\frac{L}{2} \right) \cdot m_{obj} \cdot g \cdot \cos \varphi \cdot \hat{k} \end{aligned}$$

Now, let us consider the expression for the moment due to the linear acceleration of the gripped object:

$$(5.20) \quad \bar{M}_a = \bar{r}_g \times (m \cdot a \cdot \hat{a})$$

Using eq. (5.5), (5.17) we can write the equation as:

$$\bar{M}_a = \left(\frac{L}{2} \right) \cdot \hat{r} \times (m \cdot a \cdot R(\varphi) (\sin \delta \cdot \hat{i} + \cos \delta \cdot \hat{j}))$$

And finally we get:

$$(5.21) \quad \bar{M}_a = \left(\frac{L}{2} \right) \cdot m \cdot a \cdot \cos \delta \cdot \hat{k}$$

For the moment created by the tensile and the shear forces, we must consider that both forces are both a result of stress distribution, meaning they change with the distance from the edge of the bond. Therefore, the moment from those forces will be produced by integration:

$$(5.22) \quad \bar{M}_\theta = \int_0^L \bar{r}(x) \times (\sigma_\theta(x) \cdot \hat{n}) \cdot w \cdot dx$$

$$(5.23) \quad \bar{M}_\theta = \int_0^L \bar{r}(x) \times (\tau_{r(x)} \cdot \hat{r}) \cdot w \cdot dx$$

where w is the bond width and x runs along the bond length L

Where $\vec{r}(x) = x \cdot \hat{r}$ is the position-dependent moment arm in the gripper's own coordinate system.

By using the definition of $\vec{r}(x)$ and inserting equations (5.11), (5.12) into eq. (5.22), (5.23) accordingly we will get:

$$(5.24) \quad \bar{M}_\theta = \int_0^L x \cdot \hat{r} \times (\sigma_\theta(x) \cdot R(\varphi) \cdot \hat{j}) \cdot w \cdot dx$$

$$(5.25) \quad \bar{M}_r = \int_0^L x \cdot \hat{r} \times (\tau_{rx}(x) \cdot R(\varphi) \cdot \hat{i}) \cdot w \cdot dx$$

Since $\hat{r} = R(\varphi) \cdot \hat{i}$, both cross products reduce to rotation-invariant scalars:

After vectorial multiplication and organizing the equations, we will get:

$$(5.26) \quad \bar{M}_\theta = \int_0^L x \cdot \sigma_\theta(x) \cdot w \cdot dx \cdot \hat{k}$$

$$(5.27) \quad \bar{M}_r = 0$$

(since $\hat{r} \times \hat{r} = 0$: shear acts along the same line as its own arm)

Substituting (5.17), (5.21), (5.26), (5.27) into (5.16) gives α explicitly

We have found an expression for α as a function of the properties of the object and the tensile and shear forces, and therefore it can be numerically solved for a specific object and specific adhesive.

By using either Kaelble or GR model to find the maximum forces before failure, we can find the acceleration limit a_limit and the angular acceleration limit α_limit .

5.2.2 Examples for specific scenarios

In this chapter, a version of the general calculation will be shown for specific chosen scenarios, and simplified equations will be presented.

5.2.2.1.1 Pure Horizontal Movement with a Horizontal Gripper

For this specific scenario we can determine:

$$\varphi = 0, \quad \delta = 90^\circ$$

As seen in **Fig 5.2**

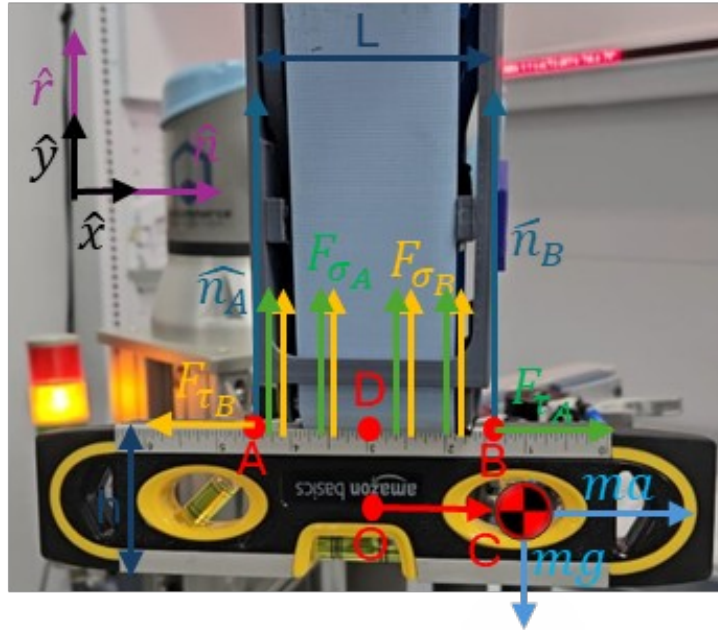


Fig 5.2 – Horizontal Gripper Accelerated Horizontally

Therefore, we can simplify equations (5.13), (5.14) to be:

$$F_r = m \cdot a, F_\theta = m_{obj} \cdot g$$

Substituting $\varphi = 0, \delta = 90^\circ$ into eq. (5.6): $m \cdot a \cdot \hat{i}$

$$\bar{F}_\theta = (\bar{F}_{\theta A} + \bar{F}_{\theta B}) \cdot \hat{j} \quad (\text{tensile forces at the two edges, summed})$$

$$\sum F_x = m \cdot a_x, \sum F_y = m \cdot a_y$$

This matches intuition directly: with a horizontal gripper accelerated purely horizontally, the shear force alone supports the inertial load and the tensile force alone supports the object's weight.

$$\bar{F}_r = (\bar{F}_{rA} - \bar{F}_{rB}) = m_{obj} \cdot a \quad (\text{shear forces at the two edges, opposing})$$

$$\sum F_x : F_{rA} - F_{rB} = m_{obj} \cdot a$$

$$a = (F_{rA} - F_{rB}) / m_{obj}$$

$$\sum F_y : F_{\theta A} - F_{\theta B} - m_{obj} \cdot g = 0$$

$$F_{\theta A} + F_{\theta B} = m_{obj} \cdot g$$

For the expression for the angular acceleration, we will use the moment equilibrium eq as shown before; with $\varphi = 0$, $\delta = 90^\circ$, the calculation is done around C, the geometrical center of the lifted object, and we will get:

$$\bar{M}_g = - \left(\frac{L}{2} \right) \cdot m_{obj} \cdot g \cdot \hat{k}, \bar{M}_a = 0$$

$$\sum \tau_O = \tau_\theta + \tau_r + \tau_{mg} + \tau_a = I \cdot \alpha_z$$

$$\alpha_z = \frac{(\tau_\theta + \tau_r + \tau_{mg} + \tau_a)}{I_{obj}}$$

(5.28)

By using eq. (5.21), (5.26), (5.27) and (5.28):

(since $\cos \delta = \cos 90^\circ = 0$, the inertial moment $\bar{M}_a$ vanishes in this scenario)

We can get the following simplified equation:

$$I \cdot \alpha = - \left(\frac{L}{2} \right) \cdot m_{obj} \cdot g \cdot \hat{k} + \bar{M}_\theta$$

(5.29)

$$\alpha = \frac{\left[\bar{M}_\theta - \left(\frac{L}{2} \right) \cdot m_{obj} \cdot g \right]}{I}$$

(5.30)

5.2.2.2 Pure Vertical Movement with a Horizontal Gripper

For this specific scenario we can determine:

$$\varphi = 0, \quad \delta = 0^\circ$$

As seen in **Fig 5.3**

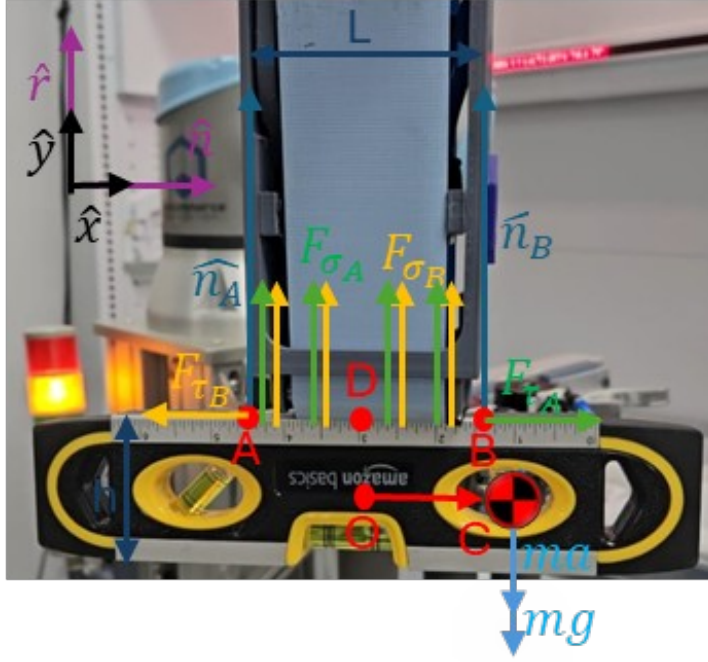


Fig 5.3 – Horizontal Gripper Accelerated Vertically

Therefore, we can simplify equations (5.13), (5.14) to be:

$$F_r = 0, F_\theta = m \cdot a + m_{obj} \cdot g$$

Substituting $\varphi = 0, \delta = 0^\circ$ into eq. (5.6): $m \cdot a \cdot \hat{j}$

$$\bar{F}_\theta = (F_{\theta A} + F_{\theta B}) \cdot \hat{j} \text{ (as in eq. (5.19) above)}$$

$$\sum F_x = m \cdot a_x, \sum F_y = m \cdot a_y$$

This again matches intuition: with the gripper accelerated purely vertically, the shear force vanishes entirely and the tensile force alone supports both the inertial load and the object's weight together.

$$\sum F_x = F_{rA} - F_{rB} = 0$$

$F_{rA} = F_{rB}$ (shear forces balance exactly: no horizontal acceleration to resist)

$$\sum F_y = F_{\theta A} + F_{\theta B} - m_{obj} \cdot g = m \cdot a$$

$$a = \frac{(F_{\theta A} + F_{\theta B} - m_{obj} \cdot g)}{m_{obj}}$$

For the expression for the angular acceleration, we will use the moment equilibrium eq as shown before; with $\varphi = 0$, $\delta = 0^\circ$, the calculation is done around C, the geometrical center of the lifted object, and we will get:

$$\bar{M}_{mg} = -\left(\frac{L}{2}\right) \cdot m_{obj} \cdot g \cdot \hat{k}, \bar{M}_a = \left(\frac{L}{2}\right) \cdot m \cdot a \cdot \hat{k} \quad (5.31)$$

$$\sum \tau_O = \tau_\theta + \tau_r + \tau_{mg} + \tau_a = I \cdot \alpha_z$$

$$\alpha_z = \frac{(\tau_\theta + \tau_r + \tau_{mg} + \tau_a)}{I_{obj}}$$

By using eq. (5.21), (5.26), (5.27) and (5.31):

We can get the following simplified equation:

$$I \cdot \alpha = \left(\frac{L}{2}\right) \cdot (m \cdot a - m_{obj} \cdot g) \cdot \hat{k} + \bar{M}_\theta \quad (5.31)$$

$$\alpha = \left[\bar{M}_\theta + \left(\frac{L}{2}\right) \cdot (m \cdot a - m_{obj} \cdot g) \right] / I \quad (5.32)$$

5.2.2.3 Movement in a non-specific direction with a Horizontal Gripper

For this specific scenario we can determine:

$$\varphi = 0, \quad \delta \text{ arbitrary}$$

As seen in **Fig 5.4**

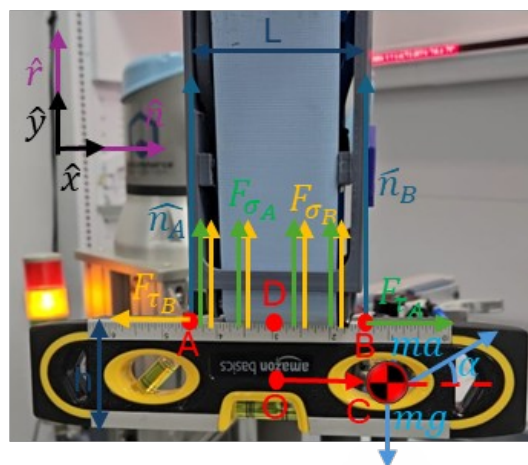


Fig 5.4 – Horizontal Gripper Accelerated in a Nonspecific Direction

Therefore, we can simplify equations (5.13), (5.14) to be:

$$F_r = m \cdot a \cdot \sin \delta, F_\theta = m \cdot a \cdot \cos \delta + m_{obj} \cdot g$$

Substituting $\varphi = 0$ into eq. (5.6):

$$\bar{F}_\theta = (F_{\theta A} + F_{\theta B}) \cdot \hat{j} \quad (\text{as in eq. (5.19) above})$$

$$\sum F_x = m \cdot a_x, \sum F_y = m \cdot a_y$$

And by using the force equilibrium equation we will get:

$$(5.33) \quad \hat{a} = \sin \delta \cdot \hat{i} + \cos \delta \cdot \hat{j} \quad (\text{since } R(0) = \text{identity})$$

For the expression for the angular acceleration, we will use the moment equilibrium eq as shown before; with $\varphi = 0$, the calculation is done around C, the geometrical center of the lifted object, and we will get:

$$(5.34) \quad \bar{M}_{mg} = -\left(\frac{L}{2}\right) \cdot m_{obj} \cdot g \cdot \hat{k}, \bar{M}_a = \left(\frac{L}{2}\right) \cdot m \cdot a \cdot \cos \delta \cdot \hat{k}$$

By using eq. (5.21), (5.26), (5.27) and (5.34):

We can get the following simplified equation, recovering the general result of eq. (5.16) with $\varphi = 0$ throughout:

$$(5.35) \quad I \cdot \alpha = \left(\frac{L}{2}\right) \cdot (m \cdot a \cdot \cos \delta - m_{obj} \cdot g) \cdot \hat{k} + \bar{M}_\theta$$

$$(5.36) \quad I \cdot \alpha = \left[\bar{M}_\theta + \left(\frac{L}{2}\right) \cdot (m \cdot a \cdot \cos \delta - m_{obj} \cdot g) \right] / I$$

5.2.3 Expressions for Stress Variables

To calculate the failure point of the adhesion, we utilize expressions for the force resulting from tensile, peel, and shear stress. In this chapter, we will show the development of the force expression based on Kaelble model and on GR model.

5.2.3.1 Correlation between Force and Stress

In the final expression we developed, both the linear and angular acceleration are dependent on the force from the stress. In both Kaelble and GR model, the stress is measured as adhesive work, therefore the units of stress are $J/m^2 = F \cdot m/m^2 = F/m$. Because this is a 2D scenario, the force is the concentrated force for the width of the gripper:

$$(5.65) \quad F_{\sigma}(x) = \int_0^b \sigma(x) dz$$

$$(5.66) \quad F_{\tau}(x) = \int_0^b \tau(x) dz$$

Assuming the adhesion on the edges of the gripper is complete and the object is centered regarding the z axis, we can assume that the peel angle on the edges is 0 and therefore, the distribution of stress is symmetric, and as a result we can determine:

$$(5.67) \quad F_{\sigma}(x) = b\sigma(x)$$

$$(5.68) \quad F_{\tau}(x) = b\tau(x)$$

5.2.3.2 Kaelble Model

Let us remember that as seen in eq. (3.29), the expression for the tensile stress as a factor of distance from peeling edge is:

$$\sigma(x) = \sigma_0 (\cos(\beta x) + K \sin(\beta x)) e^{\beta x}$$

Where:

$$\sigma_0 = \frac{2\beta P \sin(\theta)}{b(1-K)}; K = \frac{\beta M_c}{\beta M_c + P \sin(\theta)}; \beta = \left(\frac{E_a b}{4EIa} \right)^{0.25}$$

For failure, the peeling force in the equation is $P = P_{cr}$ as shown in eq. (3.9).

By inserting eq. (3.29) into eq. (5.67) we get:

$$(5.69) \quad F_{\sigma}(x) = b\sigma_0 (\cos(\beta x) + K \sin(\beta x)) e^{\beta x}$$

In the gripper used in this thesis there are two peeling edges, on the opposite side of the bond, therefore the x value for one is in the opposite direction:

$$(5.70) \quad \begin{aligned} F_{\sigma_A}(r) &= b\sigma_{0_A} (\cos(\beta_A r) + K_A \sin(\beta_A r)) e^{\beta_A r} \\ F_{\sigma_B}(r) &= b\sigma_{0_B} (\cos(\beta_B (L - r)) + K_B \sin(\beta_B (L - r))) e^{\beta_B (L - r)} \end{aligned}$$

By inserting eq. (5.70) into eq. (5.9) we can get the expression of the tensile force according to Kaelble:

$$(5.71) \quad \begin{aligned} \bar{F}_\sigma(r) &= b \left[\sigma_{0_A} (\cos(\beta_A r) + K_A \sin(\beta_A r)) e^{\beta_A r} \right. \\ &\quad \left. + \sigma_{0_B} (\cos(\beta_B (L - r)) + K_B \sin(\beta_B (L - r))) e^{\beta_B (L - r)} \right] \hat{n} \end{aligned}$$

Using this equation and equation (5.67), we can develop the expression for torque because of the tensile force. We will assume $\sigma_{0A} = \sigma_{0B} = \sigma_0$; $\beta_A = \beta_B = \beta$; $K_A = K_B = K$ to simplify the equation:

$$\begin{aligned} \tau_{F_\sigma} &= \left[\int_0^L \left(r - \frac{L}{2} \right) b\sigma_0 \left[(\cos(\beta r) + K \sin(\beta r)) e^{\beta r} + (\cos(\beta (L - r)) + K \sin(\beta (L - r))) e^{\beta (L - r)} \right] dr \right] \hat{k} \\ \tau_{F_\sigma} &= \left[b\sigma_0 \int_0^L \left[\left(r - \frac{L}{2} \right) (\cos(\beta r) + K \sin(\beta r)) e^{\beta r} + \left(r - \frac{L}{2} \right) (\cos(\beta (L - r)) + K \sin(\beta (L - r))) e^{\beta (L - r)} \right] dr \right] \hat{k} \end{aligned}$$

We will solve the integral in two parts:

$$\tau_{F_\sigma} = b\sigma_0 (\tau_1 + \tau_2)$$

We will start with the expression τ_1 :

$$\tau_1 = \int_0^L \left(r - \frac{L}{2} \right) (\cos(\beta r) + K \sin(\beta r)) e^{\beta r} dr$$

$$\tau_1 = \int_0^L \left[r \cos(\beta r) + rK \sin(\beta r) - \frac{L}{2} \cos(\beta r) - \frac{L}{2} K \sin(\beta r) \right] e^{\beta r} dr$$

$$\begin{aligned} \int_0^L \left[-\frac{L}{2} \cos(\beta r) e^{\beta r} \right] dr &= -\frac{L}{2} \int_0^L \cos(\beta r) e^{\beta r} dr = \\ &= -\frac{L}{2} \left[\frac{e^{\beta r} \sin(\beta r) + e^{\beta r} \cos(\beta r)}{4\beta} \right]_0^L = -\frac{L}{4\beta} [e^{\beta L} \sin(\beta L) + e^{\beta L} \cos(\beta L) - 1] \end{aligned}$$

$$\begin{aligned} \int_0^L \left[-\frac{LK}{2} \sin(\beta r) e^{\beta r} \right] dr &= -\frac{LK}{2} \int_0^L \sin(\beta r) e^{\beta r} dr = \\ &= -\frac{LK}{2} \left[\frac{-e^{\beta r} \cos(\beta r) + e^{\beta r} \sin(\beta r)}{4\beta} \right]_0^L = \frac{L}{4\beta} [e^{\beta L} \sin(\beta L) - e^{\beta L} \cos(\beta L) - 1] \end{aligned}$$

$$\begin{aligned} \int_0^L [r \cos(\beta r) e^{\beta r}] dr &= \frac{1}{2\beta} \int_0^L \left[x \sin(\beta r) e^{\beta r} + x \cos(\beta r) e^{\beta r} - \frac{\sin(\beta r) e^{\beta r}}{\beta} \right]_0^L = \\ &= \frac{1}{2\beta} \left[L \sin(\beta L) e^{\beta L} + L \cos(\beta L) e^{\beta L} - \frac{\sin(\beta L) e^{\beta L}}{\beta} \right] \end{aligned}$$

$$\begin{aligned} \int_0^L [rK \sin(\beta r) e^{\beta r}] dr &= \frac{K}{2} \int_0^L [x \sin(\beta r) e^{\beta r}] dr = \\ \frac{K}{\beta} \left[r \sin(\beta r) e^{\beta r} - \frac{r(\sin(\beta r) + \cos(\beta r)) e^{\beta r}}{2} + \frac{\cos(\beta r) e^{\beta r}}{2\beta} \right]_0^L &= \\ = \frac{K}{2\beta^2} [\beta L \sin(\beta L) e^{\beta L} - \beta L \cos(\beta L) e^{\beta L} + \cos(\beta L) e^{\beta L} - 1] \end{aligned}$$

After combining the equations and rearranging we will get:

$$\begin{aligned} \tau_1 &= \frac{1}{2\beta} \left\{ 0.5L [\sin(\beta L)(1+K) + \cos(\beta L)(1-K)] e^{\beta L} + 1 + K \right\} \\ &\quad + K \cos(\beta L) e^{\beta L} - \frac{K + \sin(\beta L) e^{\beta L}}{\beta} \end{aligned} \quad (5.72)$$

Now, let us do the same process for I_2 :

$$\tau_2 = \int_0^L \left[r - \frac{L}{2} \right] \left(\cos(\beta(L-r)) + K \sin(\beta(L-r)) \right) e^{\beta(L-r)} dr$$

$$\tau_2 = \int_0^L \left[r \cos(\beta(L-r)) + rK \sin(\beta(L-r)) - \frac{L}{2} \cos(\beta(L-r)) - \frac{L}{2} K \sin(\beta(L-r)) \right] e^{\beta r} dr$$

$$\begin{aligned} \int_0^L r \cos(\beta(L-r)) e^{\beta r} dr &= \frac{1}{2\beta^2} \left[(\beta r - 1) \sin(\beta(r-L)) e^{\beta r} + \beta r \cos(\beta(r-L)) e^{\beta r} \right]_0^L \\ &= \frac{Le^{\beta L}}{2\beta} - \frac{\sin(\beta L)}{2\beta^2} \end{aligned}$$

$$\begin{aligned} \int_0^L rK \sin(\beta(L-r)) e^{\beta r} dr &= -\frac{K}{2\beta^2} \left[\beta r \sin(\beta(r-L)) e^{\beta r} + (1-\beta r) \cos(\beta(r-L)) e^{\beta r} \right]_0^L \\ &= \frac{K}{2\beta^2} \left[\cos(\beta L) + (\beta L - 1) e^{\beta L} \right] \end{aligned}$$

$$\begin{aligned} \int_0^L \left[-\frac{L}{2} \cos(\beta(L-r)) e^{\beta r} \right] dr &= -\frac{L}{4\beta^2} \left[\beta \cos(\beta(L-r)) e^{\beta r} - \beta \sin(\beta(r-L)) e^{\beta r} \right]_0^L \\ &= -\frac{L}{4\beta} \left[\sin(\beta L) - \cos(\beta L) + e^{\beta L} \right] \end{aligned}$$

$$\begin{aligned} \int_0^L \left[-\frac{L}{2} K \sin(\beta(L-r)) e^{\beta r} \right] dr &= -\frac{KL}{4\beta^2} \left[\beta \sin(\beta(L-r)) e^{\beta r} + \beta \cos(\beta(r-L)) e^{\beta r} \right]_0^L \\ &= \frac{KL}{4\beta} \left[\sin(\beta L) + \cos(\beta L) - e^{\beta L} \right] \end{aligned}$$

After combining the equations and rearranging we will get:

$$\begin{aligned} \tau_2 &= \frac{1}{2\beta^2} \left\{ \left[\frac{\beta L}{2} (K+1) - 1 \right] e^{\beta L} - \beta \left[1 + L \left(\frac{1-K}{2} \right) \right] \sin(\beta L) \right. \\ &\quad \left. + \left[K + \beta L \left(\frac{1+K}{2} \right) \right] \cos(\beta L) \right\} \end{aligned} \quad (5.73)$$

Now, let us remember that eq. (3.44) defines the expression of shear stress as a factor of peeling edge.

$$\tau(x) = \tau_0 e^{\alpha x}$$

Where:

$$\tau_0 = \left(\frac{P \cos \omega}{b} \right) \alpha, \alpha = \left(\frac{G}{2Eah} \right)^{0.5}$$

For failure, the peeling force in the equation is $p = p_{cr}$ as shown in eq. (3.9).

By inserting eq. (3.29) into eq. (5.67) we get:

$$(5.74) \quad F_r(x) = b\tau_0 e^{\alpha x}$$

In the gripper used in this thesis there are two peeling edges, on the opposite side of the bond, therefore the x value for one is in the opposite direction:

$$(5.75) \quad \begin{aligned} F_{\tau_A}(r) &= b\tau_{0_A} e^{\alpha_A r} \\ F_{\tau_B}(r) &= b\tau_{0_B} e^{\alpha_B(L-r)} \end{aligned}$$

By inserting eq. (5.74) into eq. (5.10) we can get the expression of the tensile force according to Kaelble:

$$(5.76) \quad \bar{F}_r(r) = b(\tau_{0_A} e^{\alpha_A r} \pm \tau_{0_B} e^{\alpha_B(L-r)}) \hat{r}$$

Using this equation and equation (5.26), we can develop the expression for the torque because of the tensile force. We will assume $\tau_{0_A} = \tau_{0_B} = \tau_0$; $\alpha_A = \alpha_B = \alpha$ to simplify the equation:

$$(5.77) \quad \tau_{F_r} = \left[- \left(\frac{h}{2} \right) \int_0^L b\tau_0 (e^{\alpha r} \pm e^{\alpha(L-r)}) dr \right] \hat{k} = \left[- \frac{hb\tau_0}{2} \int_0^L (e^{\alpha r} \pm e^{\alpha(L-r)}) dr \right] \hat{k}$$

The integral inside the expression is simpler in this case and can be solved instantly:

$$\int_0^L (e^{\alpha r} \pm e^{\alpha(L-r)}) dr = \frac{1}{\alpha} \left[(e^{\alpha r} \mp e^{\alpha(L-r)}) \right]_0^L$$

$$\int_0^L (e^{\alpha r} \pm e^{\alpha(L-r)}) dr = \frac{2(e^{\alpha L} - 1)}{\alpha}$$

or

$$\int_0^L (e^{\alpha r} \pm e^{\alpha(L-r)}) dr = 0$$

Therefore:

$$\tau_{F_x} = \left[\frac{-hb\tau_0(e^{\alpha L} - 1)}{\alpha} \right] \hat{k}$$

or

$$\tau_{F_x} = 0$$

(5.78)

5.2.3.3 GR Model

Let us remember that as seen in eq. (3.63), the expression for tensile stress as a factor of distance from peeling edge is:

$$\sigma(x) = \sigma_0 \left[\left(R_2 \lambda^2 \frac{k}{2} - \lambda k w \cos \lambda \cosh \lambda \right) \cosh \left(\frac{\lambda x}{c} \right) \cos \left(\frac{\lambda x}{c} \right) \right. \\ \left. + \left(R_3 \lambda^2 \frac{k}{2} - \lambda k w \sin \lambda \sinh \lambda \right) \sinh \left(\frac{\lambda x}{c} \right) \sin \left(\frac{\lambda x}{c} \right) \right]$$

Where:

$$\sigma_0 = \frac{Pt}{R_1 b c^2}; R_1 = \frac{\sinh 2\lambda + \sin 2\lambda}{2}; R_2 = \sinh \lambda \cos \lambda - \cosh \lambda \sin \lambda;$$

$$R_3 = \sinh \lambda \cos \lambda + \cosh \lambda \sin \lambda; \lambda = \frac{c}{t} \left(\frac{6Et}{E_s \eta} \right)^{0.25}$$

It is important to notice none of the shortened expressions is dependent on x, and therefore can be left shortened for the entire calculation, and can be written in an even shortened expression:

$$(5.79) \quad \sigma(x) = \sigma_0 \left[R_4 \cosh\left(\frac{\lambda x}{c}\right) \cos\left(\frac{\lambda x}{c}\right) + R_5 \sinh\left(\frac{\lambda x}{c}\right) \sin\left(\frac{\lambda x}{c}\right) \right]$$

Where:

$$R_4 = \left(R_2 \lambda^2 \frac{k}{2} - \lambda k w \cos \lambda \cosh \lambda \right); R_5 = \left(R_3 \lambda^2 \frac{k}{2} - \lambda k w \sin \lambda \sinh \lambda \right)$$

By inserting eq. (3.29) into eq. (5.79) we get:

$$(5.80) \quad F_{\sigma}(x) = b \sigma_0 \left[R_4 \cosh\left(\frac{\lambda x}{c}\right) \cos\left(\frac{\lambda x}{c}\right) + R_5 \sinh\left(\frac{\lambda x}{c}\right) \sin\left(\frac{\lambda x}{c}\right) \right]$$

In the gripper used in this thesis there are two peeling edges, on the opposite side of the bond, therefore the x value for one is in the opposite direction:

$$(5.81) \quad \begin{aligned} F_{\sigma_a}(r) &= b \sigma_{0_a} \left[R_{4A} \cosh\left(\frac{\lambda_A r}{c}\right) \cos\left(\frac{\lambda_A r}{c}\right) + R_{5A} \sinh\left(\frac{\lambda_A r}{c}\right) \sin\left(\frac{\lambda_A r}{c}\right) \right] \\ F_{\sigma_n}(r) &= b \sigma_{0_n} \left[R_{4B} \cosh\left(\frac{\lambda_B r}{c}\right) \cos\left(\frac{\lambda_B (L-r)}{c}\right) + \right. \\ &\quad \left. + R_{5B} \sinh\left(\frac{\lambda_B (L-r)}{c}\right) \sin\left(\frac{\lambda_B (L-r)}{c}\right) \right] \end{aligned}$$

By inserting eq. (5.81) into eq. (5.9) we can get the expression of the tensile force according to GR:

(5.82)

$$\begin{aligned} \bar{F}_{\sigma}(r) &= b \left[\sigma_{0_A} \left[R_{4A} \cosh\left(\frac{\lambda_A r}{c}\right) \cos\left(\frac{\lambda_A r}{c}\right) + R_{5A} \sinh\left(\frac{\lambda_A r}{c}\right) \sin\left(\frac{\lambda_A r}{c}\right) \right] \right. \\ &\quad \left. + \sigma_{0_B} \left[R_{4B} \cosh\left(\frac{\lambda_B r}{c}\right) \cos\left(\frac{\lambda_B (L-r)}{c}\right) + R_{5B} \sinh\left(\frac{\lambda_B (L-r)}{c}\right) \sin\left(\frac{\lambda_B (L-r)}{c}\right) \right] \right] \hat{n} \end{aligned}$$

Using this equation and equation (5.26), we can develop the expression for the torque

because of the tensile force. We will assume $\sigma_{0A} = \sigma_{0B} = \sigma_0$; $\beta_A = \beta_B = \beta$; $R_{4A} =$

$R_{4B} = R_4$; $R_{5A} = R_{5B} = R_5$ to simplify the equation:

$$\tau_{F_\sigma} = \left[\int_0^L \left(r - \frac{L}{2} \right) b \sigma_0 \left[R_4 \cosh \left(\frac{\lambda r}{c} \right) \cos \left(\frac{\lambda r}{c} \right) + R_5 \sinh \left(\frac{\lambda r}{c} \right) \sin \left(\frac{\lambda r}{c} \right) + \right. \right. \\ \left. \left. + R_4 \cosh \left(\frac{\lambda (L-r)}{c} \right) \cos \left(\frac{\lambda (L-r)}{c} \right) + R_5 \sinh \left(\frac{\lambda (L-r)}{c} \right) \sin \left(\frac{\lambda (L-r)}{c} \right) \right] dr \right] \hat{k}$$

$$\tau_{F_\sigma} = \left[b \sigma_0 \int_0^L \left(r - \frac{L}{2} \right) \left[R_4 \cosh \left(\frac{\lambda r}{c} \right) \cos \left(\frac{\lambda r}{c} \right) + \right. \right. \\ \left. \left. + R_5 \sinh \left(\frac{\lambda r}{c} \right) \sin \left(\frac{\lambda r}{c} \right) \right] dr + \int_0^L \left(r - \frac{L}{2} \right) \left[R_4 \cosh \left(\frac{\lambda (L-r)}{c} \right) \cos \left(\frac{\lambda (L-r)}{c} \right) + \right. \right. \\ \left. \left. + R_5 \sinh \left(\frac{\lambda (L-r)}{c} \right) \sin \left(\frac{\lambda (L-r)}{c} \right) \right] dr \right] \hat{k}$$

We will solve the integral in two parts:

$$\tau_{F_\sigma} = b \sigma_0 (\tau_1 + \tau_2)$$

We will start with the expression I_2 :

$$\tau_1 = \int_0^L \left(r - \frac{L}{2} \right) \left[R_4 \cosh \left(\frac{\lambda r}{c} \right) \cos \left(\frac{\lambda r}{c} \right) + R_5 \sinh \left(\frac{\lambda r}{c} \right) \sin \left(\frac{\lambda r}{c} \right) \right] dr$$

$$\tau_1 = \int_0^L \left[r R_4 \cosh \left(\frac{\lambda r}{c} \right) \cos \left(\frac{\lambda r}{c} \right) + \right. \\ \left. + r R_5 \sinh \left(\frac{\lambda r}{c} \right) \sin \left(\frac{\lambda r}{c} \right) - \frac{L R_4}{2} \cosh \left(\frac{\lambda r}{c} \right) \cos \left(\frac{\lambda r}{c} \right) - \frac{L R_5}{2} \sinh \left(\frac{\lambda r}{c} \right) \sin \left(\frac{\lambda r}{c} \right) \right] dr$$

$$\int_0^L r R_4 \cosh \left(\frac{\lambda r}{c} \right) \cos \left(\frac{\lambda r}{c} \right) = R_4 \int_0^L r \cosh \left(\frac{\lambda r}{c} \right) \cos \left(\frac{\lambda r}{c} \right) \\ = \frac{R_4 c}{4 \lambda^2} \left[e^{-\frac{\lambda r}{c}} \left[\left((\lambda r - c) e^{\frac{2 \lambda r}{c}} + \lambda r + c \right) \sin \left(\frac{\lambda r}{c} \right) + \left((\lambda r) e^{\frac{2 \lambda r}{c}} - \lambda r \right) \cos \left(\frac{\lambda r}{c} \right) \right] \right]_0^L \\ = \frac{R_4 c e^{-\frac{\lambda L}{c}}}{4 \lambda^2} \left[\left((\lambda L - c) e^{\frac{2 \lambda L}{c}} + \lambda L + c \right) \sin \left(\frac{\lambda L}{c} \right) + \left(\lambda L e^{\frac{2 \lambda L}{c}} - \lambda L \right) \cos \left(\frac{\lambda L}{c} \right) \right]$$

$$\begin{aligned}
& \int_0^L r R_5 \sinh\left(\frac{\lambda r}{c}\right) \sin\left(\frac{\lambda r}{c}\right) = R_5 \int_0^L r \sinh\left(\frac{\lambda r}{c}\right) \sin\left(\frac{\lambda r}{c}\right) \\
& = -\frac{R_5 c}{4\lambda^2} \left[e^{-\frac{\lambda r}{c}} \left[\left((\lambda r) e^{\frac{2\lambda r}{c}} + \lambda r \right) \sin\left(\frac{\lambda r}{c}\right) + \left((c - \lambda r) e^{\frac{2\lambda r}{c}} + \lambda r + c \right) \cos\left(\frac{\lambda r}{c}\right) \right] \right]_0^L \\
& = \frac{R_5 c e^{-\frac{\lambda L}{c}}}{4\lambda^2} \left[\left(\lambda L e^{\frac{2\lambda L}{c}} + \lambda L \right) \sin\left(\frac{\lambda L}{c}\right) + \left((c - \lambda L) e^{\frac{2\lambda L}{c}} + \lambda L + c \right) \cos\left(\frac{\lambda L}{c}\right) - 2c e^{\frac{\lambda L}{c}} \right]
\end{aligned}$$

$$\begin{aligned}
& \int_0^L \frac{L R_4}{2} \cosh\left(\frac{\lambda r}{c}\right) \cos\left(\frac{\lambda r}{c}\right) = -\frac{L R_4}{2} \int_0^L \cosh\left(\frac{\lambda r}{c}\right) \cos\left(\frac{\lambda r}{c}\right) \\
& = -\frac{L R_4 c}{8\lambda} \left[e^{-\frac{\lambda r}{c}} \left[\left(e^{\frac{2\lambda r}{c}} + 1 \right) \sin\left(\frac{\lambda r}{c}\right) + \left(e^{\frac{2\lambda r}{c}} - 1 \right) \cos\left(\frac{\lambda r}{c}\right) \right] \right]_0^L \\
& = -\frac{L R_4 c e^{-\frac{\lambda L}{c}}}{8\lambda} \left[\left(e^{\frac{2\lambda L}{c}} + 1 \right) \sin\left(\frac{\lambda L}{c}\right) + \left(e^{\frac{2\lambda L}{c}} - 1 \right) \cos\left(\frac{\lambda L}{c}\right) \right] \\
& \int_0^L \frac{L R_5}{2} \sinh\left(\frac{\lambda r}{c}\right) \sin\left(\frac{\lambda r}{c}\right) = -\frac{L R_5}{2} \int_0^L \sinh\left(\frac{\lambda r}{c}\right) \sin\left(\frac{\lambda r}{c}\right) \\
& = -\frac{L R_5 c}{8\lambda} \left[e^{-\frac{\lambda r}{c}} \left[\left(e^{\frac{2\lambda r}{c}} + 1 \right) \sin\left(\frac{\lambda r}{c}\right) + \left(1 - e^{\frac{2\lambda r}{c}} \right) \cos\left(\frac{\lambda r}{c}\right) \right] \right]_0^L \\
& = -\frac{L R_5 c e^{-\frac{\lambda L}{c}}}{8\lambda} \left[\left(e^{\frac{2\lambda L}{c}} + 1 \right) \sin\left(\frac{\lambda L}{c}\right) + \left(1 - e^{\frac{2\lambda L}{c}} \right) \cos\left(\frac{\lambda L}{c}\right) \right]
\end{aligned}$$

After combining the equations and rearranging we will get:

$$\begin{aligned}
\tau_1 = & \frac{c e^{-\frac{\lambda L}{c}}}{4\lambda^2} \left\{ \frac{\lambda L}{2} \left(e^{\frac{2\lambda L}{c}} - 1 \right) \left[R_4 \cos\left(\frac{\lambda L}{c}\right) + R_5 \sin\left(\frac{\lambda L}{c}\right) \right] \right. \\
& + \left(\frac{\lambda L}{2} + c \right) \left[R_4 \sin\left(\frac{\lambda L}{c}\right) + R_5 \cos\left(\frac{\lambda L}{c}\right) \right] - 2c R_5 e^{\frac{\lambda L}{c}} \\
& \left. + \left[R_4 \left(\frac{\lambda L}{2} - c \right) \sin\left(\frac{\lambda L}{c}\right) + R_5 \left(c - \frac{3\lambda L}{2} \right) \cos\left(\frac{\lambda L}{c}\right) \right] e^{\frac{2\lambda L}{c}} \right\}
\end{aligned} \tag{5.83}$$

Now, let us do the same process for τ_2 :

$$\tau_2 = \int_0^L \left(r - \frac{L}{2} \right) \left[R_4 \cosh \left(\frac{\lambda(L-r)}{c} \right) \cos \left(\frac{\lambda(L-r)}{c} \right) + R_5 \sinh \left(\frac{\lambda(L-r)}{c} \right) \sin \left(\frac{\lambda(L-r)}{c} \right) \right] dr$$

$$\tau_2 = \int_0^L \left[r R_4 \cosh \left(\frac{\lambda(L-r)}{c} \right) \cos \left(\frac{\lambda(L-r)}{c} \right) + \right. \\ \left. + r R_5 \sinh \left(\frac{\lambda(L-r)}{c} \right) \sin \left(\frac{\lambda(L-r)}{c} \right) + \right. \\ \left. - \frac{LR_4}{2} \cosh \left(\frac{\lambda(L-r)}{c} \right) \cos \left(\frac{\lambda(L-r)}{c} \right) - \frac{LR_5}{2} \sinh \left(\frac{\lambda(L-r)}{c} \right) \sin \left(\frac{\lambda(L-r)}{c} \right) \right] dr$$

$$\int_0^L r R_4 \cosh \left(\frac{\lambda(L-r)}{c} \right) \cos \left(\frac{\lambda(L-r)}{c} \right) = R_4 \int_0^L r \cosh \left(\frac{\lambda(L-r)}{c} \right) \cos \left(\frac{\lambda(L-r)}{c} \right)$$

$$= \frac{R_4 c}{4\lambda^2} \left[e^{-\frac{\lambda(r-L)}{c}} \left[\left((\lambda r - c) e^{\frac{2\lambda r}{c}} + \lambda r e^{\frac{2\lambda L}{c}} + c e^{\frac{2\lambda L}{c}} \right) \sin \left(\frac{\lambda(r-L)}{c} \right) + \left(\lambda r e^{\frac{2\lambda r}{c}} - \lambda r e^{\frac{2\lambda L}{c}} \right) \cos \left(\frac{\lambda(r-L)}{c} \right) \right] \right]_0^L$$

$$= \frac{R_4 c^2 e^{-\frac{\lambda L}{c}}}{4\lambda^2} \left(e^{\frac{2\lambda L}{c}} - 1 \right) \sin \left(\frac{\lambda L}{c} \right)$$

$$\int_0^L r R_5 \sinh \left(\frac{\lambda(L-r)}{c} \right) \sin \left(\frac{\lambda(L-r)}{c} \right) = R_5 \int_0^L r \sinh \left(\frac{\lambda(L-r)}{c} \right) \sin \left(\frac{\lambda(L-r)}{c} \right)$$

$$= \frac{R_5 c}{4\lambda^2} \left[e^{-\frac{\lambda(r-L)}{c}} \left[\left(\lambda r e^{\frac{2\lambda r}{c}} - \lambda r e^{\frac{2\lambda L}{c}} \right) \sin \left(\frac{\lambda(r-L)}{c} \right) + \left((c - \lambda r) e^{\frac{2\lambda r}{c}} + \lambda r e^{\frac{2\lambda L}{c}} + c e^{\frac{2\lambda L}{c}} \right) \cos \left(\frac{\lambda(r-L)}{c} \right) \right] \right]_0^L$$

$$= -\frac{R_5 c^2 e^{-\frac{\lambda L}{c}}}{4\lambda^2} \left(\left(e^{\frac{2\lambda L}{c}} + 1 \right) \cos \left(\frac{\lambda L}{c} \right) - 2e^{\frac{\lambda L}{c}} \right)$$

$$\int_0^L -\frac{LR_4}{2} \cosh \left(\frac{\lambda(L-r)}{c} \right) \cos \left(\frac{\lambda(L-r)}{c} \right) = -\frac{LR_4}{2} \int_0^L \cosh \left(\frac{\lambda(L-r)}{c} \right) \cos \left(\frac{\lambda(L-r)}{c} \right)$$

$$= -\frac{LR_4 c}{8\lambda} \left[e^{-\frac{\lambda(r-L)}{c}} \left[\left(e^{\frac{2\lambda(r-L)}{c}} + 1 \right) \sin \left(\frac{\lambda(r-L)}{c} \right) + \left(e^{\frac{2\lambda(r-L)}{c}} - 1 \right) \cos \left(\frac{\lambda(r-L)}{c} \right) \right] \right]_0^L$$

$$= -\frac{LR_4 c e^{-\frac{\lambda L}{c}}}{8\lambda} \left[\left(e^{\frac{2\lambda L}{c}} + 1 \right) \sin \left(\frac{\lambda L}{c} \right) + \left(e^{\frac{2\lambda L}{c}} - 1 \right) \cos \left(\frac{\lambda L}{c} \right) \right]$$

$$\begin{aligned}
& \int_0^L -\frac{LR_5}{2} \sinh\left(\frac{\lambda(L-r)}{c}\right) \sin\left(\frac{\lambda(L-r)}{c}\right) = -\frac{LR_5}{2} \int_0^L \sinh\left(\frac{\lambda(L-r)}{c}\right) \sin\left(\frac{\lambda(L-r)}{c}\right) \\
& = -\frac{LR_5 c}{8\lambda} \left[e^{-\frac{\lambda(r-L)}{c}} \left[\left(e^{\frac{2\lambda(r-L)}{c}} + 1 \right) \sin\left(\frac{\lambda(r-L)}{c}\right) + \left(1 - e^{\frac{2\lambda(r-L)}{c}} \right) \cos\left(\frac{\lambda(r-L)}{c}\right) \right] \right]_0^L \\
& = -\frac{LR_5 c e^{-\frac{\lambda L}{c}}}{8\lambda} \left[\left(e^{\frac{2\lambda L}{c}} + 1 \right) \sin\left(\frac{\lambda L}{c}\right) + \left(1 - e^{\frac{2\lambda L}{c}} \right) \cos\left(\frac{\lambda L}{c}\right) \right]
\end{aligned}$$

After combining the equations and rearranging we will get:

$$\begin{aligned}
\tau_2 = & \frac{ce^{-\frac{\lambda L}{c}}}{4\lambda^2} \left\{ \left[cR_4 \left(e^{\frac{2\lambda L}{c}} - 1 \right) - \frac{\lambda L}{2} (R_4 + R_5) \left(e^{\frac{2\lambda L}{c}} + 1 \right) \right] \sin\left(\frac{\lambda L}{c}\right) \right. \\
& \left. + \left[\frac{\lambda L}{2} (R_5 - R_4) \left(e^{\frac{2\lambda L}{c}} - 1 \right) - cR_5 \left(e^{\frac{2\lambda L}{c}} + 1 \right) \right] \cos\left(\frac{\lambda L}{c}\right) + 2cR_5 e^{\frac{\lambda L}{c}} \right\}
\end{aligned}
\tag{5.84}$$

Now, let us remember that eq. (3.60) defines the expression of shear stress as a factor of peeling edge.

$$\tau(x) = \tau_0 \left[\frac{\beta c}{t} (1 + 3k) \frac{\cosh\left(\frac{\beta c x}{t c}\right)}{\sinh\left(\frac{\beta c}{t}\right)} + 3(1 - k) \right]$$

Where:

$$\begin{aligned}
\tau_0 &= \frac{P}{8bc}; \beta = \left(\frac{8tG}{E_s \eta} \right)^{0.5}; k = \frac{\cosh(u_2 c)}{\cosh(u_2 c) + 2\sqrt{2} \sinh(u_2 c)} \\
u_2 &= \sqrt{\frac{3(1 - \nu^2)P}{2\eta E_s}}
\end{aligned}$$

It is important to notice none of the shortened expressions is dependent on x, and therefore can be left shortened for the entire calculation, and can be written in an even shortened expression:

$$(5.85) \quad \tau(x) = \tau_0 \left[R_6 \cosh \left(\frac{\beta c x}{t c} \right) + R_7 \right]$$

Where:

$$R_6 = \frac{\beta c (1 + 3k)}{t \sinh \left(\frac{\beta c}{t} \right)}; R_7 = 3(1 - k)$$

For failure, the peeling force in the equation is $p = p_{cr}$ as shown in eq. (3.9).

By inserting eq. (3.29) into eq. (5.85) we get:

$$(5.86) \quad F_\tau(x) = b\tau_0 \left[R_6 \cosh \left(\frac{\beta x}{t} \right) + R_7 \right]$$

In the gripper used in this thesis there are two peeling edges, on the opposite side of the bond, therefore the x value for one is in the opposite direction:

$$(5.87) \quad \begin{aligned} F_{\tau_A}(r) &= b\tau_{0A} \left[R_{6A} \cosh \left(\frac{\beta_A r}{t} \right) + R_{7A} \right] \\ F_{\tau_B}(r) &= b\tau_{0B} \left[R_{6B} \cosh \left(\frac{\beta_B (L - r)}{t} \right) + R_{7B} \right] \end{aligned}$$

By inserting eq. (5.75) into eq. (5.11) we can get the expression of the tensile force according to GR:

$$(5.88) \quad \bar{F}_\tau(r) = b \left[\tau_{0A} \left(R_{6A} \cosh \left(\frac{\beta_A r}{t} \right) + R_{7A} \right) \pm \tau_{0B} \left(R_{6B} \cosh \left(\frac{\beta_B (L - r)}{t} \right) + R_{7B} \right) \right] \hat{r}$$

Using this equation and equation (5.11), we can develop the expression for the torque

because of the tensile force. We will assume $\sigma_{0A} = \sigma_{0B} = \sigma_0$; $\beta_A = \beta_B = \beta$; $R_{6A} =$

$R_{6B} = R_6$; $R_{7A} = R_{7B} = R_7$ to simplify the equation:

(5.89)

$$\begin{aligned}\tau_{F_x} &= \left[- \left(\frac{h}{2} \right) \int_0^L b \tau_0 \left[\left(R_6 \cosh \left(\frac{\beta r}{t} \right) + R_7 \right) \pm \left(R_6 \cosh \left(\frac{\beta (L - r)}{t} \right) + R_7 \right) \right] dr \right] \hat{k} \\ &= \left[- \frac{hb\tau_0}{2} \left[\int_0^L R_6 \cosh \left(\frac{\beta r}{t} \right) + R_7 \right] dr \pm \int_0^L R_6 \cosh \left(\frac{\beta (L - r)}{t} \right) + R_7 \right] dr \right] \hat{k}\end{aligned}$$

We will solve each integral separately and find the two optional results:

$$\begin{aligned}\int_0^L \left(R_6 \cosh \left(\frac{\beta r}{t} \right) + R_7 \right) dr &= \left[\frac{R_6 t}{\beta} \sinh \left(\frac{\beta r}{t} \right) + R_7 r \right]_0^L \\ &= \frac{R_6 t}{\beta} \sinh \left(\frac{\beta L}{t} \right) + R_7 L\end{aligned}$$

$$\begin{aligned}\int_0^L \left(R_6 \cosh \left(\frac{\beta (L - r)}{t} \right) + R_7 \right) dr &= \left[R_7 r - \frac{R_6 t}{\beta} \sinh \left(\frac{\beta (L - r)}{t} \right) \right]_0^L \\ &= \frac{R_6 t}{\beta} \sinh \left(\frac{\beta L}{t} \right) + R_7 L\end{aligned}$$

Therefore, the expression for the moment as result of shear stress is:

$$\tau_{F_x} = \frac{R_6 t}{\beta} \sinh \left(\frac{\beta L}{t} \right) + R_7 L \pm \left(\frac{R_6 t}{\beta} \sinh \left(\frac{\beta L}{t} \right) + R_7 L \right)$$

$$\tau_{F_x} = \frac{2R_6 t}{\beta} \sinh \left(\frac{\beta L}{t} \right) + 2R_7 L$$

or

$$\tau_{F_x} = 0$$

(5.90)

The result of this section is the pair of quantities the rest of this thesis calls a_{limit} and α_{limit} : the maximum linear and angular acceleration the adhesive bond can sustain for a given gripper orientation and applied load, accounting for the combined tensile, shear, and gravitational contributions derived above. These are not used directly as a single scalar limit -- section 5.1 and the Reciprocal-Norm criterion of Chapter 6 combine them with the peeling force of section 3.1 to produce the full adhesive limit surface that the Force-Aware RRT* planner enforces as a hard constraint in Chapter 6. The length of this derivation is what makes that later constraint a physically grounded acceleration limit rather than an empirically tuned threshold.

5.3 Material Characterization Experiments

Accurate failure prediction requires precise empirical inputs for the adhesive and backing materials. The following experiments were conducted to characterize the system parameter.

5.3.1 Direct Thickness Measurement

To validate the manufacturer's technical specifications and ensure consistency within the specific batch of adhesive agent, direct thickness measurements were conducted.

5.3.1.1 Experimental Setup and Methodology

The total thickness of the 3M™ Value Duct Tape 1900 was measured using a high-precision digital micrometer (Insize IP65, 0–25 mm range, 0.001 mm resolution). To capture batch-specific variability, ten random locations were selected along a single roll of the silver tape. Each measurement was taken by carefully placing a flat tape segment between the micrometer anvils and rotating the ratchet stop to apply a uniform, repeatable contact force. This setup minimizes user-induced measurement error and accounts for the potential compressibility of the soft PSA layer.

5.3.1.2 Experimental Results

The test was done 10 times, the Figures below present the test equipment and an example of a single run.



Fig 5.5 – Micrometer Used for the Experiment

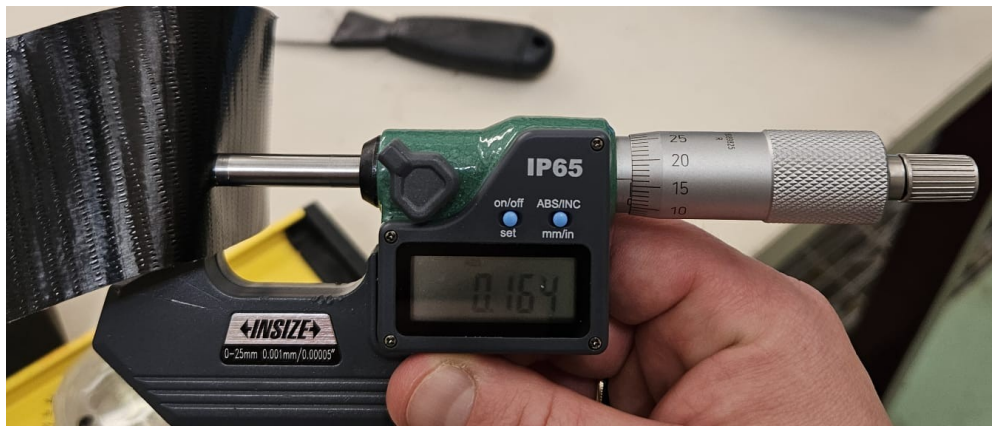


Fig 5.6 – Run 3 of the Direct Thickness Measurement Experiment

The table below shows the results for all 10 runs

Table 5.1 – Direct Thickness Measurement Results

measurement	-	1	2	3	4	5	6	7	8	9	10
a	$[mm]$	0.162	0.164	0.164	0.162	0.163	0.161	0.160	0.163	0.159	0.163

5.3.1.3 Analysis

The mean thickness derived from the experimental data is **0.1621 mm**. While the digital micrometer provides an accurate measurement of the composite structure, the Kaelble and Goland-Reissner stress models require separate inputs for the backing thickness ($2h$) and the adhesive layer thickness (a).

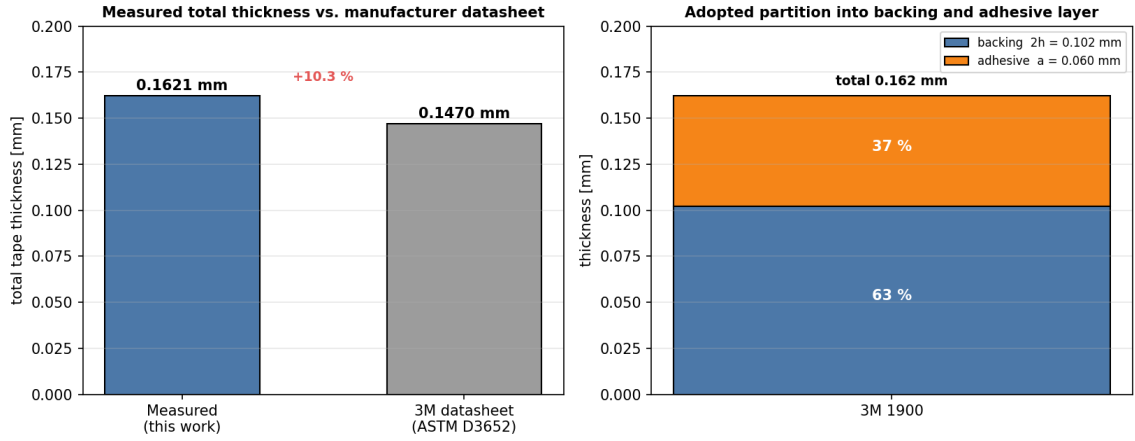


Fig 5.7 – Direct Thickness Measurement — measured total vs. datasheet, and adopted backing/adhesive partition

Based on technical standards for utility-grade duct tapes and the 3M™ technical datasheet[91], the total measured thickness is partitioned according to the typical ratios of polyethylene-laminated scrim to synthetic rubber adhesive. For a total thickness of ~ 0.162 mm, a backing thickness ($2h$) of **0.102 mm** (approximately 4.0 mils) and an adhesive thickness (a) of **0.060 mm** (approximately 2.4 mils) are adopted. This 63/37% split accounts for the reinforced cloth scrim within the polyethylene layer, which provides the primary flexural rigidity (EI) for the backing component in Kaelble's derivation.

5.3.1.4 Conclusions

The direct thickness measurements successfully established an empirical baseline of **0.162 mm** for the specific 3M™ 1900 tape batch. By partitioning this value into $2h = 0.102$ mm and $a = 0.060$ mm, we ensure the analytical stress models are physically grounded. These batch-specific parameters are critical for calculating the stress concentration factors α and β , which directly define the robot's dynamic Limit Surface during high-speed manipulation.

5.3.2 Simple Tensile Test

To provide accurate inputs for the Kaelble and Goland-Reissner stress models, it is necessary to determine the mechanical properties of the adhesive-backing system. We conduct a static load elongation test, a form of simple tensile testing, to empirically derive the Young's Modulus (E) of the 3M Value Duct Tape 1900.

5.3.2.1 Experimental Setup and Methodology

The experimental apparatus utilizes a vertical guide tower (**Fig 5.**). The top anchor is a static grey plastic arm fixed to the tower's apex. A "double-lap" configuration is employed to ensure symmetric loading and to mitigate secondary bending effects.

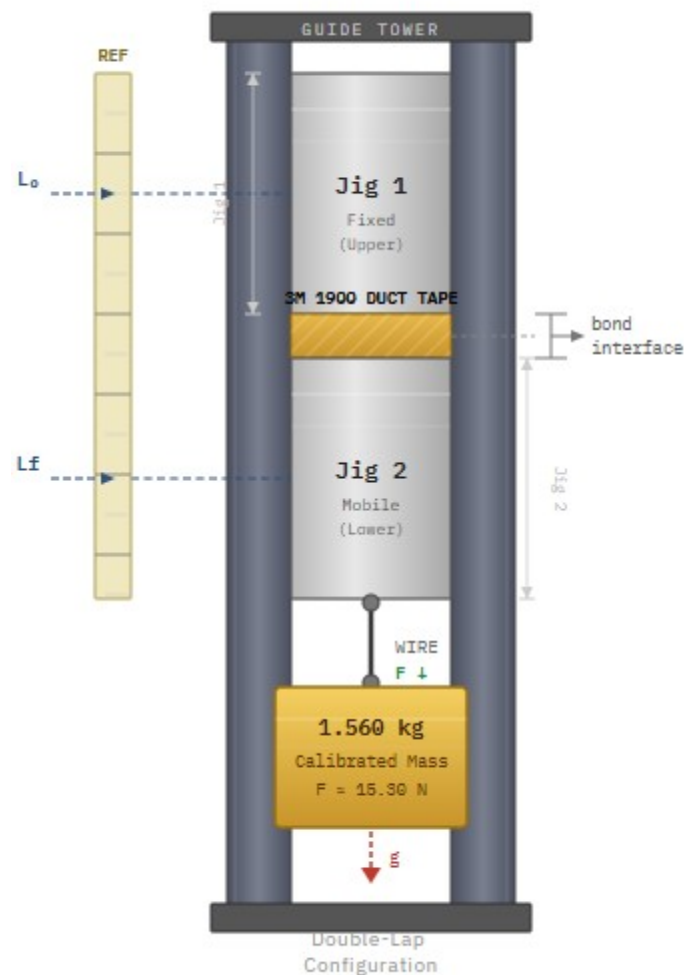


Fig 5.8 – Simple Tensile Test Setup

The setup consists of the following components:

1. **Adherends:** Two 3D-printed PLA bricks (Fig 5.) serve as the substrates. The PSA tape is bonded around the bricks ensuring the tensile load is shared equally.
2. **Guide Tower:** A vertical, multi-colored guide tower provides a visual reference for measuring extension and ensuring the load path remains perfectly vertical (Fig 5.).
3. **Loading System:** Tension is applied by suspending a calibrated mass from the bottom PLA brick using high-strength fishing wire. A digital scale confirms the hanging mass is (Fig 5.), generating a constant force.
4. **Measurement:** The initial length and final length under tension of the silver active segment are recorded using a steel ruler (Fig 5.8).



Fig 5.9 – Simple Tensile Test Equipment: Weight (Top Left), Ruler (Top Right), PLA Bricks (Bottom Left), Adhesive (Bottom Right)

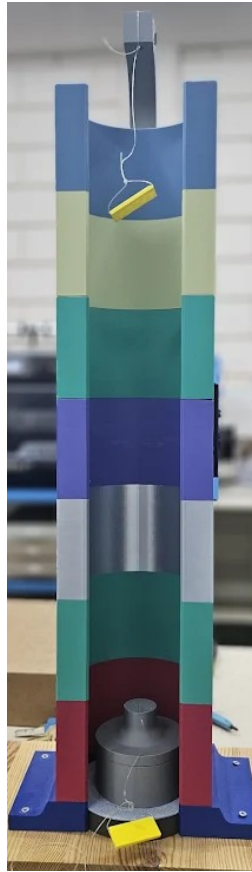


Fig 5.10 – Simple Tensile Test Guide Tower Setup

5.3.2.2 Governing Equations

The mechanical response is analyzed using engineering stress-strain (σ / ϵ) relationships. The total cross-sectional area (A) for the two strips is:

$$(5.91) \quad F = m_{\text{weight}} \cdot g [N]$$

$$(5.92) \quad A = b \cdot t [m^2]$$

$$(5.93) \quad \sigma = \frac{F}{A} [Mpa]$$

$$(5.94) \quad \epsilon = \frac{L - L_0}{L_0}$$

$$(5.95) \quad E = \frac{\sigma}{\epsilon} [Mpa]$$

5.3.2.3 Experimental Results

Four experimental runs were conducted to account for the viscoelastic variability of the duct tape.

In this chapter we will show the calculations and the results of the experiment.

The figure below shows a single run done in the experiment

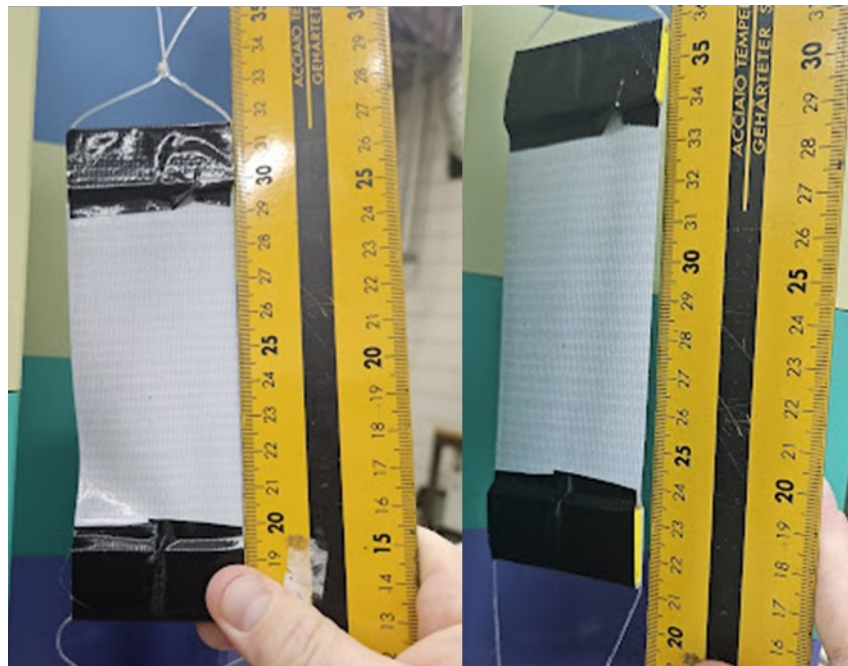


Fig 5.11 – Run 3 of the Simple Tensile Test Experiment

First, the stress applied on the PSA strip was calculated, using the parameters shown in **Table 5.2**

Table 5.2 – Experiment Setup Parameters

m_{weight}	b	t
$[Kg]$	$[m]$	$[m]$
1.56	0.05	$1.47 \cdot 10^{-4}$

Using the parameter above and eq. (5.91), (5.92), (5.93), the stress can be calculated:

$$F = 1.56 [Kg] \cdot 9.81 \left[\frac{N}{Kg} \right] = 15.3036 [N]$$

$$A = 0.05 \cdot 1.47 \cdot 10^{-4} = 7.35 \cdot 10^{-6}$$

$$\sigma = \frac{F}{A} = \frac{15.3036}{7.35 \cdot 10^{-6}} = 2082122.45 [pa] = 2.0821 [Mpa]$$

To calculate the strain, the strip was measured before tension was applied and under tension, as shown in **Fig 5.11** for run number 4. The results are shown in **Table 5.3** below

Table 5.3 – Strip Length Measurement

<i>Run</i>	L_0	L_{finale}
<i>Units</i>	[cm]	[cm]
1	8.5	8.6
2	10.4	10.6
3	20.8	21.1
4	23.0	23.3

Using the measurements from the experiment and eq. (5.94), (5.95), the Young's modulus of the PSA can be calculated, an example is shown for run 2:

$$\varepsilon = \frac{10.7 - 10.5}{10.5} = \frac{0.2}{10.5} = 0.019$$

$$E = \frac{\sigma}{\varepsilon} = \frac{2.0821}{0.019} = 109.584 [Mpa]$$

The results are shown in **Table 5.4** below

Table 5.4 – Calculated Results

<i>Run</i>	ε	E
<i>Units</i>	[]	[Mpa]
1	0.0118	176.98
2	0.0192	108.27
3	0.0144	144.36
4	0.0130	159.63

5.3.2.4 Analysis

The tensile testing of 3M™ Value Duct Tape 1900 revealed a significant variation in the measured effective modulus, ranging from **108.27 MPa** to **176.98 MPa**. The variability observed across the four runs is characteristic of the complex, viscoelastic behavior of composite tapes, where the polymer backing and adhesive layer respond to load with both elastic and time-dependent components.

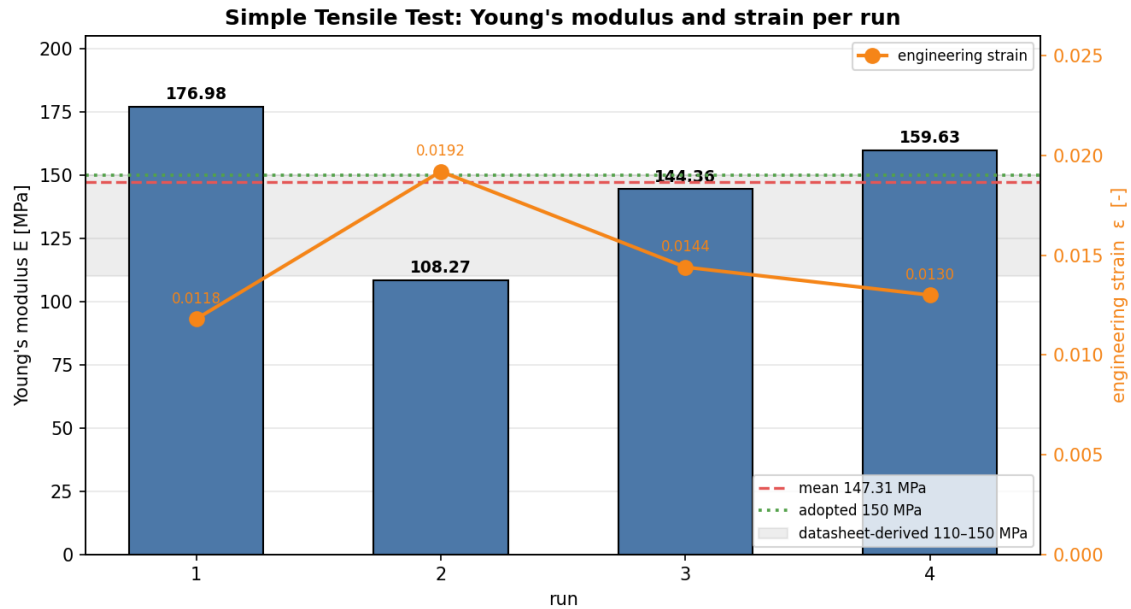


Fig 5.12 – Simple Tensile Test: Young's modulus and engineering strain per run

A critical observation from the dataset is the relationship between strain and apparent stiffness. Run 2 exhibited the highest engineering strain ($\epsilon = 0.0192$), resulting in the lowest calculated modulus (108.27 MPa). This outcome is likely indicative of "parasitic extension" occurring at the adhesive grips or within the unbonded regions of the tape. In contrast, Run 1 showed the lowest strain ($\epsilon = 0.0118$) and the highest stiffness (176.98 MPa). This spread demonstrates that for specimens with shorter initial lengths, any localized slippage or compliance in the experimental fixture represents a larger proportion of the total extension, thereby artificially inflating the measured strain and deflating the apparent modulus.

To reconcile these findings with industrial expectations, we refer to the 3M™ 1900 technical datasheet, which specifies a tensile strength of 33 N/cm and an elongation at break of 15%. A theoretical secant modulus derived from these limits falls within the **110–150 MPa** range. The experimental results from Runs 3 and 4 (144.36 MPa and 159.63 MPa, respectively) show excellent convergence toward this theoretical expectation. The high value in Run 1 likely reflects the material's instantaneous

stiffening before creep effects dominate, while Run 2 reflects the influence of test-fixture compliance.

5.3.2.5 Conclusions

The tensile experiments on 3M™ Value Duct Tape 1900 confirm that the material exhibits an effective stiffness broadly consistent with manufacturer specifications when analyzed under pick-and-place stress levels. Despite the scatter inherent in simple tensile setups for flexible composites, the dataset provides a reliable baseline for mechatronic modeling.

By averaging the results from the four runs, an effective Young's modulus of approximately **147.31 MPa** is found. For the purposes of the subsequent analytical modeling in this thesis, we adopt a rounded average value of **150 MPa**. This parameter is used to calculate the stress concentration factor β in Kaelble's model, ensuring that the predicted "Limit Surface" for the robot's trajectory planner accurately reflects the stiffening behavior of the composite tape under high-speed manipulation.

In the context of the thesis, the chosen modulus of 150 MPa should be regarded as an effective engineering parameter rather than a fundamental material constant. It provides consistent and physically plausible stiffness value for use in subsequent analytical models and numerical simulations involving the tape, while acknowledging that the true response is viscoelastic and rate-dependent. Future work could refine this parameter by performing standardized tensile tests with controlled strain rate, more accurate displacement measurement (for example, extensometers in the gauge section), and multiple load levels, enabling a clearer separation between elastic, viscoelastic, and fixture-related contributions to the overall deformation.

5.3.3 Simple Shear Test

The shear holding capacity of the PSA interface is a critical determinant of grasp stability during horizontal transport. To empirically derive the Shear Modulus (G), a dead-weight lap-shear experiment was conducted using the 3M™ Value Duct Tape 1900.

5.3.3.1 Experimental Setup and Methodology

The experimental apparatus utilizes a vertical guide tower to ensure pure shear loading without parasitic bending moments. A "double-lap" configuration is employed where two identical adhesive strips share the tensile load symmetrically.

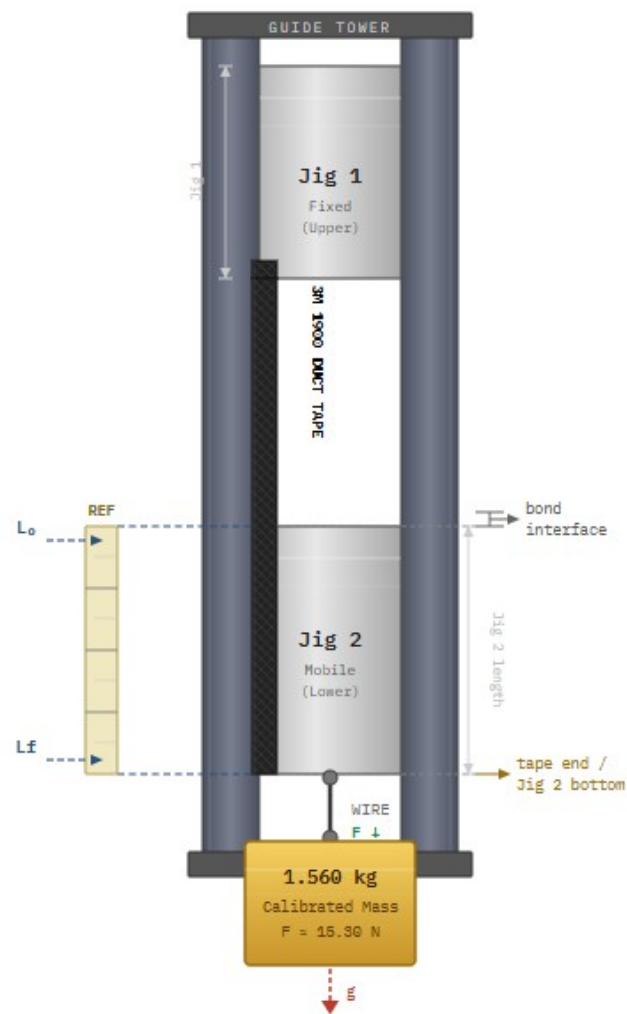


Fig 5.13 – Simple Shear Test Setup

The setup consists of the following components:

1. **Adherends:** Two 3D-printed PLA bricks serve as the substrates. "Jig 1" (Fig 5.) is fixed to the apex of the tower, while "Jig 2" (Fig 5.10) is mobile and suspended below.
2. **Bond Interface:** The PSA tape is bonded to the side faces of the bricks with a precisely measured initial overlap length (L). (Fig 5.15)
3. **Loading System:** Tension is applied by suspending a calibrated mass of 1.56 kg from the mobile Jig 2 using high-strength fishing wire (Fig 5.15).
4. **Measurement:** The displacement (δ) of the mobile brick relative to the guide tower is recorded using a digital caliper. This displacement represents the total shear deformation of the adhesive layer under the constant force $F = mg$, Fig (5.11).



Fig 5.14 – Simple Shear Test Equipment: Weight (Top Left), Caliper & Ruler (Top Right), PLA Bricks (Bottom Left), Adhesive (Bottom Right)

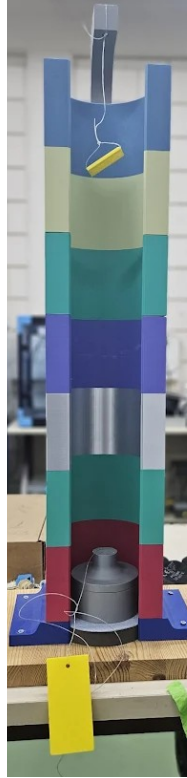


Fig 5.15 – Simple Shear Test Guide Tower Setup

5.3.3.2 Governing Equations

The mechanical response of the shear joint is analyzed using engineering shear stress-strain relationships. The total force F exerted on the adhesive assembly is:

$$(5.96) \quad F = m \cdot g [N]$$

Where m is the suspended mass. The cross-sectional area A_{Shear} of the adhesive interface, accounting for the double-lap configuration (two strips), is calculated as:

$$(5.97) \quad A_{\text{Shear}} = 2 \cdot b \cdot L [mm^2]$$

Where b is the width of the adhesive strip (50 mm) and L is the overlap length between the adhesion strip and Jig 2. The resulting shear stress τ is:

$$(5.98) \quad \tau = \frac{F}{A_{\text{Shear}}} [MPa]$$

To determine the Shear Modulus (G), the shear strain γ is computed as the ratio of measured displacement to the adhesive layer thickness (δ_{sh}) as:

$$(5.99) \quad \gamma = \frac{\delta_{sh}}{a}$$

The displacement measured in the experiment is the total displacement (δ_{tot}) of both the backing (δ_{back}) and the adhesive layer (δ_{sh}), therefore:

$$(5.100) \quad \delta_{sh} = \delta_{tot} - \delta_{back} [mm]$$

δ_{back} is a result of tensile stretching of the tape between the anchor and the start of the bond interface and δ_{sh} is derived from the shear of the adhesive layer.

For finding δ_{back} , Hooke's Law was used:

$$(5.101) \quad \delta_{back} = \frac{F}{k_{back}} @ k_{back} = \frac{E \cdot A_{back}}{L_{free}} \rightarrow \delta_{back} = \frac{F \cdot L_{free}}{E \cdot A_{back}} [mm]$$

Where a is the adhesive thickness derived from the previous characterization (0.060 mm). Finally, the effective Shear Modulus G of the PSA is found via:

$$(5.102) \quad G = \frac{\tau}{\gamma} [MPa]$$

This parameter G serves as a fundamental input for the shear stress concentration factor α in both the Kaelble and Goland-Reissner models.

5.3.3.3 Experimental Results

Four experimental runs were conducted across increasing overlap bounds (Calibers 1 through 4) to capture the compliance scale of the duct tape. The raw values were systematically extracted by evaluating the position shift on the digital micrometer display across corresponding states.

The results from the experiment and the setup parameters are compiled in the tables below.

The figure below shows a single run done in the experiment

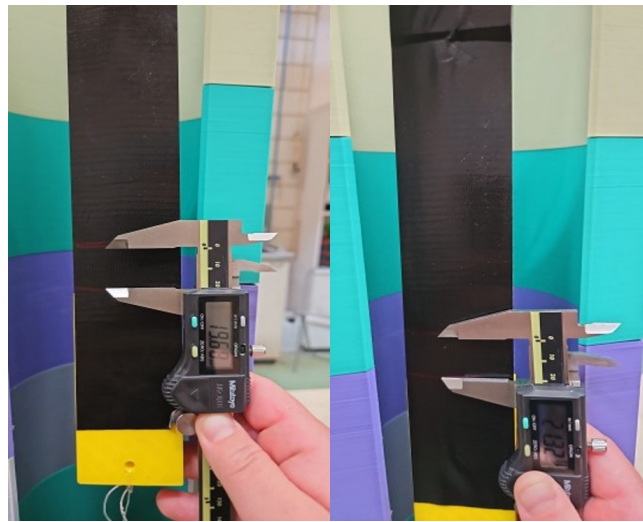


Fig 5.16 – Run 3 of the Simple Shear Test Experiment: No weight (Left) Vs. With weight (Right)

First, the stress applied on the PSA strip was calculated, using the parameters shown in **Table 5.5**.

Table 5.5 – Experiment Setup Parameters

m_{weight}	b	a
$[Kg]$	$[m]$	$[mm]$
1.56	0.05	0.060

Using the parameter above and eq. (5.96), (5.97), (5.98), the shear stress can be calculated:

$$F = 1.56 [Kg] \cdot 9.81 \left[\frac{N}{Kg} \right] = 15.3036 [N]$$

$$A_{Shear} = 2 \cdot 0.05 \cdot L = 0.1 \cdot L$$

$$\tau = \frac{F}{A} = \frac{15.3036}{0.1 \cdot L} = \frac{153.036}{0.1 \cdot L} [pa]$$

It can be seen that the shear stress is a factor of the Overlap length, therefore it will be calculated for each experiment run separately.

The total displacement was extracted by comparing the position of the red horizontal reference markings in the "Initial" and "Final" states for each trial, as shown in Fig 5.15 for run number 4. The results are shown in Table 5.6 below

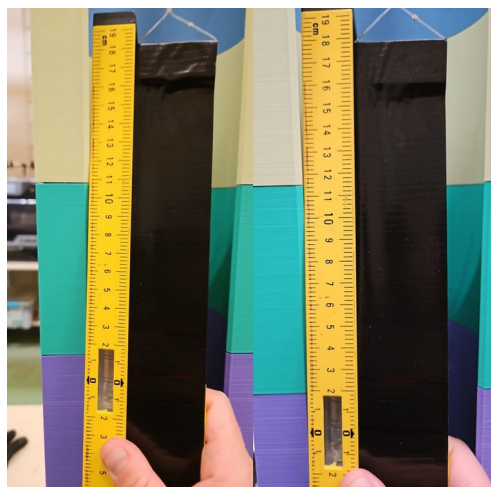


Fig 5.17 – Run 4 of the Simple Shear Test Experiment: Change in total length No weight (Left) Vs. With Weight (Right)

Table 5.6 – Strip Length Measurement

Run	L	$X_{NoWeight}$		δ_{tot}
Units	$[mm]$	$[mm]$	$[mm]$	$[mm]$
1 (Run 1)	25.0	17.87	19.74	1.87
2 (Run 2)	50.0	19.44	20.02	0.58
3 (Run 3)	75.0	19.68	21.32	1.64
4 (Run 4)	120.0	21.86	22.51	0.65

Using the measurements from the experiment and eq. (5.99), (5.100), (5.101), the displacement of the adhesive layer can be found, as shown for run 1 of the experiment:

$$A_{back} = 2 \cdot b_{back} \cdot 2h = 2 \cdot 50 \cdot 0.102 = 10.2 [mm^2]$$

$$\delta_{back} = \frac{F \cdot L_{free}}{E \cdot A_{back}} = \frac{15.3 \cdot 54.4}{150 \cdot 10.2} = 0.544 [mm]$$

$$\delta_{sh} = \delta_{tot} - \delta_{back} = 1.87 - 0.544 = 1.326 [mm]$$

Table 5.7 presents the calculations for all runs:

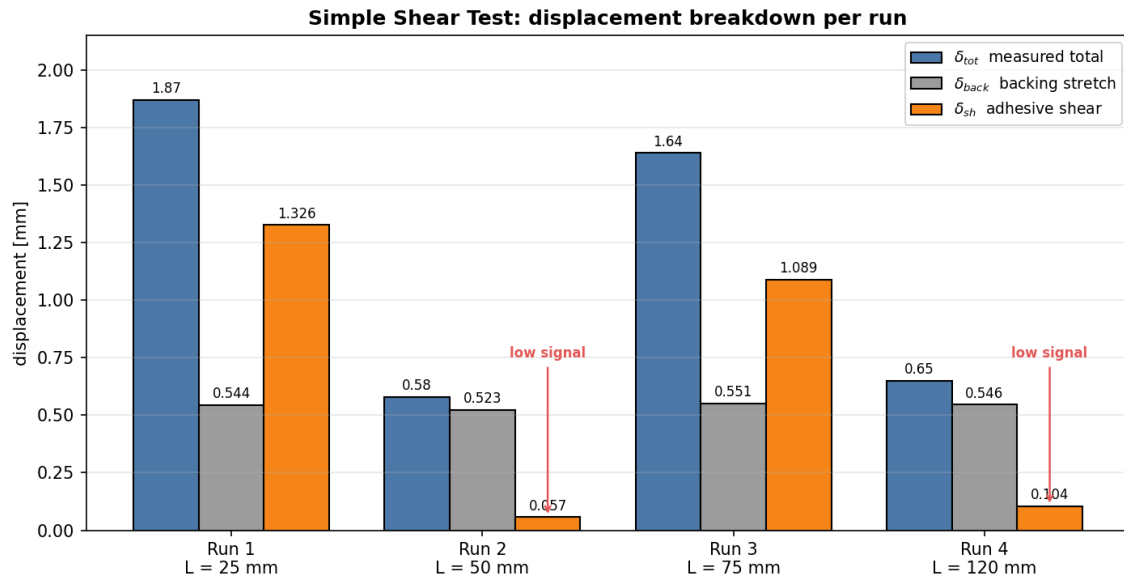


Fig 5.18 – Simple Shear Test: displacement breakdown per run (δ_{tot} , δ_{back} , δ_{sh})

Table 5.7 – Calculated Results

Run	δ_{tot}	δ_{back}	δ_{sh}
Units	[mm]	[mm]	[mm]
1	1.87	0.544	1.326
2	0.58	0.523	0.057
3	1.64	0.551	1.089
4	0.65	0.546	0.104

Using the measurements from the experiment and eq. (5.98), (5.99), (5.100) the Shear Modulus of the PSA can be calculated, an example is shown for run 1:

$$\tau = \frac{F}{A} = \frac{15.3036}{0.1 \cdot 0.05} = 3060.72 [Pa] \approx 3.06 [kPa]$$

$$\gamma = \frac{\delta_{sh}}{a} = \frac{2.6}{0.06} = 43.33$$

$$G = \frac{\tau}{\gamma} = \frac{3.06}{43.33} = 0.07 [MPa]$$

The results are shown in Table 5.8 below.

Table 5.8 – Calculated Results

<i>Run</i>	γ	τ	G
<i>Units</i>	$[\]$	$[kPa]$	$[Mpa]$
1	6.121	22.098	0.277
2	3.061	0.956	3.201
3	2.041	18.148	0.112
4	1.275	1.731	0.737

5.3.3.4 Analysis

The dead-weight lap-shear testing of 3M™ Value Duct Tape 1900 yielded computed Shear Modulus (G) values ranging from 0.112 MPa to 3.201 MPa across the four runs. However, Runs 2 and 4 exhibited very small total displacements ($\delta_{tot} = 0.58[mm] / 0.65[mm]$ respectively), and after subtracting the backing-stretch contribution (δ_{back}), the remaining shear displacement (δ_{sh}) for these two runs was small enough (0.057mm and 0.104mm) to be comparable to the digital caliper's reading uncertainty. Since G is computed from δ_{sh} via a ratio, small errors in these two runs propagate into large errors in the derived modulus, this is the most likely explanation for Run 2's outlying $G = 3.201 MPa$. Runs 2 and 4 are therefore excluded from the headline Shear Modulus estimate; Runs 1 and 3, which had substantially larger and more reliable δ_{sh} signals (1.326mm and 1.089mm), give G

= 0.277 MPa and $G = 0.112$ MPa respectively, averaging to $G \approx 0.195 \text{ MPa}$. This value is consistent with the rheological expectations for high-tack pressure-sensitive adhesives and aligns with the Dahlquist Criterion, which stipulates that for a PSA to exhibit "quick tack" (instantaneous surface wetting), the shear modulus should be on the order of 0.1 MPa at the application frequency.

The dataset shows displacement values that do not scale monotonically with overlap length as a simple shear-lag model might predict (e.g. Run 3, with a mid-range overlap of 75mm, shows the second-largest total displacement). This is consistent with the small-sample size ($n=4$, one run per overlap length) and the sensitivity of the two shorter- δ_{sh} runs discussed above; a larger dataset with repeated trials per overlap length would be needed to confidently characterize any true shear-lag trend and to statistically justify excluding outliers rather than relying on the reading-uncertainty argument used here.

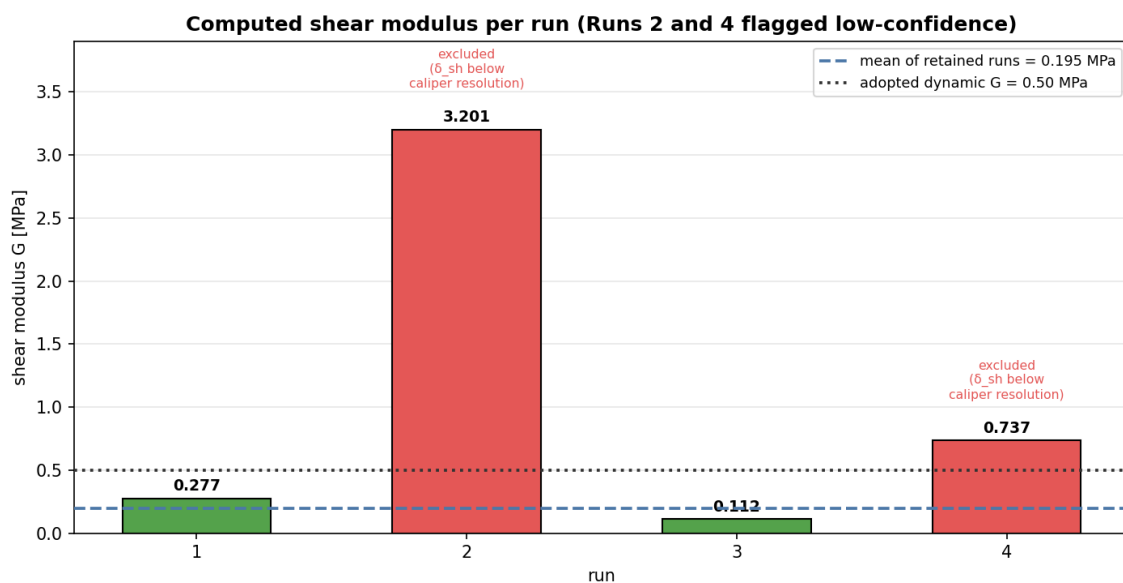


Fig 5.19 – Computed shear modulus per run (Runs 2 and 4 flagged low-confidence)

5.3.3.5 Conclusions

The simple shear test successfully established a baseline for the lateral stability of the adhesive interface. While the static experiments provide a "lower bound" modulus of ~ 0.195 MPa (averaged from Runs 1 and 3; see Analysis above), this value represents the material in its most compliant, creep-dominated state.

For the purpose of dynamic mechatronic modeling, an effective Shear Modulus (G) of **0.50 MPa** is adopted. This adjustment accounts for two critical factors: first,

industrial robotic maneuvers occur at high frequencies (milliseconds), where polymer adhesives exhibit significantly higher stiffness than in static dead-weight trials due to time-temperature superposition principles. Second, a value of 0.50 MPa aligns with typical shear modulus benchmarks for bulk synthetic rubbers and elastomers, which generally fall between **0.3 MPa and 1.0 MPa**. Adopting this rounded value provides a physically robust and conservative input for calculating the shear stress concentration factor α , ensuring the robot's trajectory planning algorithm maintains a reliable safety margin against horizontal slippage.

5.3.4 Surface Energy Test

5.3.4.1 Experimental Setup and Methodology

To empirically determine the interfacial surface energy (γ) of the 3M™ Value Duct Tape 1900 adhesive, a stationary dead-weight peel test was conducted at a 90° peel angle. A strip of tape was adhered to a test substrate, and calibrated dead-weights were incrementally suspended from the free end via a wire/string arrangement until peeling initiated. The mass at which peeling just began (the critical mass, m_w) was recorded for each trial. This methodology follows the stationary energy-conservation approach established for adhesive tape characterization [83], applied here using the peel-mechanics framework already derived in 3.1.3 (Kendall's model).

Eight trials were conducted in total: four on plain, uncoated cardboard substrate (5.3.4.3, Table 5.9) and four on a coated cardboard substrate (Table 5.10), allowing a direct comparison of adhesive performance between the two surface conditions.

5.3.4.2 Governing Equations

From 3.1.3, Kendall's critical peeling force relation (Eq. 3.9) is:

$$(5.103) \quad P_{cr} = \frac{\gamma b}{(1 - \sin\theta)}$$

where θ is the angle between the surface normal and the peeled film. For a 90° peel (film pulled perpendicular to the substrate), $\theta = 0$, simplifying to:

$$(5.104) \quad P_{cr} = \gamma b \Rightarrow \gamma = \frac{P_{cr}}{b} = \frac{m_w g}{b}$$

with $b = 50\text{mm}$, the standard width of the 3M™ 1900 tape used throughout this thesis (5.3.1, Table 5.5).

5.3.4.3 Experiment Results

Table 5.9 presents the critical mass, critical force, and computed surface energy for each uncoated-cardboard trial.

Table 5.9 – Surface Energy Test Results (Uncoated Cardboard)

<i>Test</i>	m_w	P_{cr}	γ
<i>Units</i>	[Kg]	[N]	$\left[\frac{J}{m^2} \right]$
1	1.605	15.745	314.90
2	1.75	17.168	343.35
3	1.10	10.791	215.82
4	1.42	13.930	278.60

Average: $\gamma \approx 288.2 \text{ J/m}^2$

Table 5.10 presents the equivalent results for the coated-cardboard trials.

Table 5.10 – Surface Energy Test Results (Coated Cardboard)

<i>Test</i>	m_w	P_{cr}	γ
<i>Units</i>	[Kg]	[N]	$\left[\frac{J}{m^2} \right]$
1	2.4245	23.784	475.69
2	2.5085	24.608	492.17
3	2.5085	24.608	492.17
4	2.1685	21.273	425.46

Average: $\gamma \approx 471.4 \text{ J/m}^2$

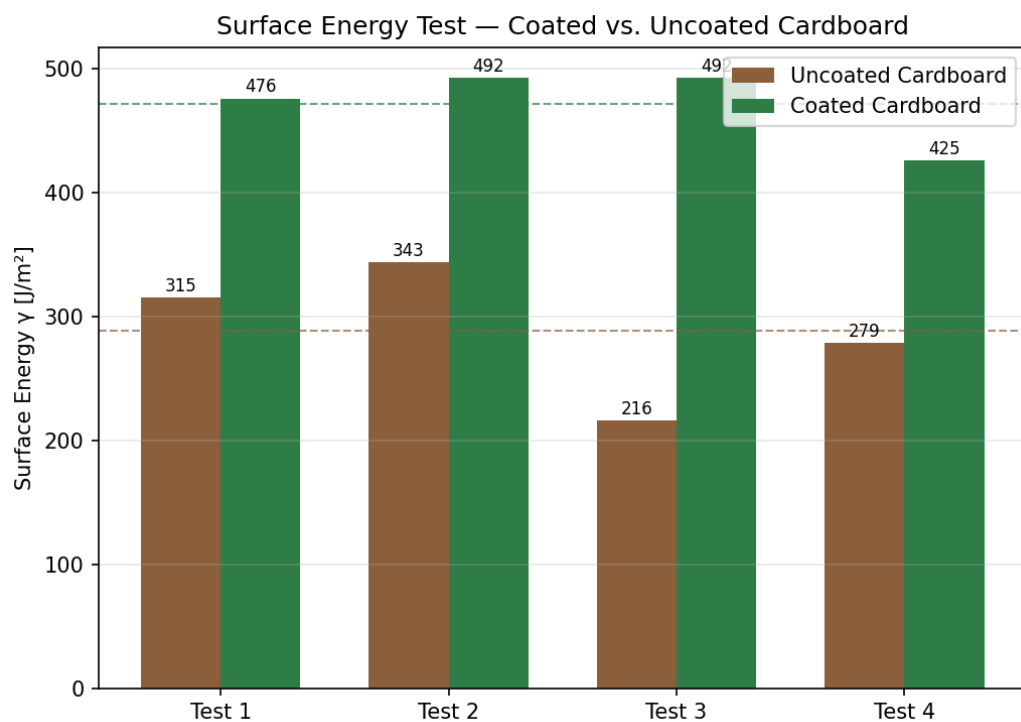


Fig 5.20 – Surface Energy Comparison, Coated vs. Uncoated Cardboard

5.3.4.4 Analysis

The dead-weight peel testing of 3M™ Value Duct Tape 1900 on uncoated cardboard yielded computed surface energy (γ) values ranging from 215.82 to 343.35 J/m² across the four trials, with a mean of $\gamma \approx 288.2$ J/m². This result is orders of magnitude larger than the thermodynamic surface energies typically associated with van der Waals interfacial bonding (on the order of 0.01–1 J/m²), and also substantially exceeds the $\gamma \approx 14$ J/m² reported for a low-tack stationery tape tested under an analogous stationary dead-weight method [83].

This discrepancy is expected and is well documented in the pressure-sensitive-adhesive literature. The measured fracture/peel energy (G) of a PSA can be decomposed into a true interfacial surface energy term (G_0) and a dissipative term (Ψ) associated with viscoelastic and plastic deformation of the adhesive during debonding; Ψ is typically three to four orders of magnitude larger than G_0 [86]. The value calculated in this section should therefore be understood as an apparent or practical peel energy, dominated by viscoelastic dissipation in the adhesive layer rather than the intrinsic thermodynamic work of adhesion.

Independent literature support for the magnitude of this result comes from commercial duct tape peel-adhesion specifications and from PSA de-adhesion energy studies. A patented duct tape adhesive formulation reports a 90° peel adhesion of 853–872 N/m [84] — dimensionally equivalent to γ under the present formulation ($\text{N/m} \equiv \text{J/m}^2$) — which exceeds the uncoated-cardboard value measured here. A separate mandrel-peel study of PSA tapes on cellulose, PVC, aluminum, and Teflon backings reported de-adhesion energies of 60–160 J/m² [85], below the present result but of the same order of magnitude. Taken together, these references place the measured γ values for 3M™ 1900 within the range expected for an aggressive industrial-grade PSA.

The coated-cardboard trials yielded a mean $\gamma \approx 471.4$ J/m², approximately 1.64× higher than the uncoated-cardboard result. This is consistent with the coating increasing the effective contact area and/or wetting quality between the adhesive and the substrate surface, allowing more of the PSA's intrinsic tack to be converted into measurable peel resistance — a mechanism also noted in the tack literature, where substrate surface free energy is identified as a dominant factor governing PSA tack and peel performance [86]. Table 5.10 also shows Tests 2 and 3 sharing an identical critical mass (2.5085 kg);

with only $n=4$ trials per surface condition, this is not unexpected but does mean the coated-cardboard average should be treated with the same caution as any small-sample estimate.

5.3.4.5 Conclusions

The stationary dead-weight peel test successfully established apparent surface energies for the 3M™ 1900 adhesive on two cardboard substrate conditions, computed via Kendall's 90° peel-energy relation (Eq. 5.104): $\gamma \approx 288.2 \text{ J/m}^2$ for uncoated cardboard and $\gamma \approx 471.4 \text{ J/m}^2$ for coated cardboard. Given the dominant role of viscoelastic dissipation in this measurement, these values should be treated as practical, application-relevant peel energies rather than fundamental material constants.

The consistent increase in γ on the coated substrate indicates that surface preparation is a meaningful lever for adhesive performance in this application and should be considered when selecting or specifying substrate treatments for the adhesive gripper's target objects.

6. Force-Constrained Trajectory Planning

Algorithm

The transition from material characterization to autonomous robotic operation requires a motion planning framework that can translate interfacial stress limits into dynamic path constraints. In conventional industrial pick-and-place applications, trajectory planning typically focuses on geometric obstacle avoidance and the kinematic limits of the manipulator, such as joint velocity and torque saturation. However, when utilizing an adhesive-based end-effector, the primary bottleneck for system performance is not the mechanical capacity of the arm, but the energetic stability of the thin-film bond.

Unlike mechanical grippers that utilize active force-closure to secure an object, the adhesive gripper presented in this thesis relies on a passive, surface-energy-dependent interface. As established in the characterization of the 3M™ Value Duct Tape 1900, this bond is highly sensitive to the direction and magnitude of the inertial loads imparted during transport. Abrupt accelerations can lead to localized stress concentrations at the peeling fronts, triggering a catastrophic failure of the grasp even if the robot is operating well within its own kinematic boundaries.

This chapter details the development of a force-aware motion planner based on the Rapidly Exploring Random Tree Star (RRT^*) algorithm. The standard RRT^* is augmented with a mechatronic validation layer that evaluates the feasibility of every potential trajectory segment. By integrating the Kaelble and Goland-Reissner stress models directly into the planning loop, the algorithm ensures that the resultant force vector - composed of gravitational and inertial components - remains strictly within the **Adhesive Limit Surface** (L_{cm}).

The proposed framework utilizes the empirical constants derived in the previous chapter to compute peak boundary stresses in real-time. The high-level logic of the expanded algorithm is illustrated in **Fig 6.1**.

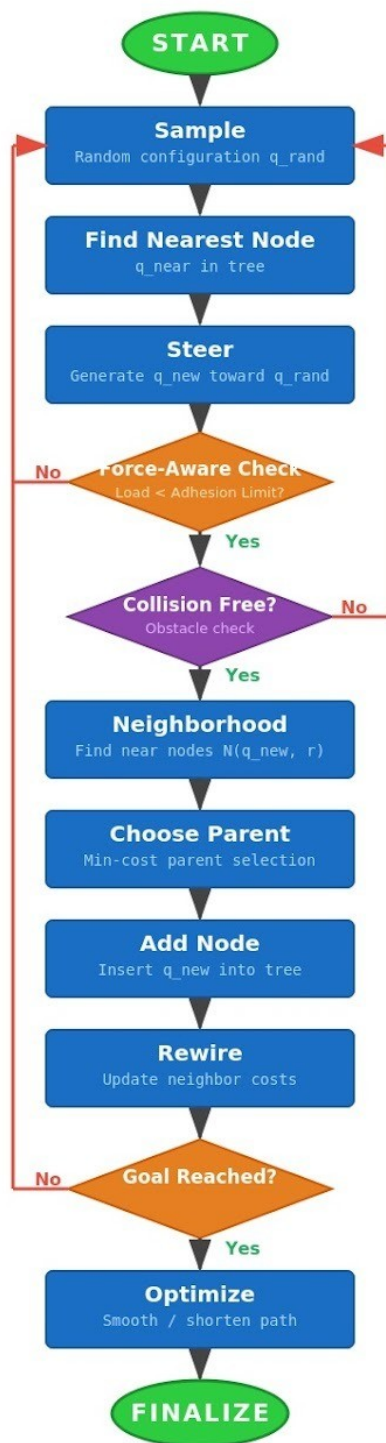


Fig 6.1 – RRT Block Diagram*

6.1 Problem Formulation and State Space Definition

To effectively navigate a cluttered environment while protecting the adhesive bond, the motion planning problem must be rigorously defined within a mathematical framework that accounts for both geometric and dynamic constraints.

6.1.1 Configuration Space (C)

The configuration of the robotic manipulator is represented by a state vector $q \in C$, where C is the configuration space. For the 3-DOF planar system utilized in this implementation, a configuration is defined by the joint angles:

$$(6.1) \quad q = [q_1, q_2, q_3]^T$$

The configuration space is bounded by the physical limits of the actuators. Based on the experimental setup, the joint limits are defined as:

- **Joint 1 (q_1):** $[-\pi, \pi]$ rad
- **Joint 2 (q_2):** $[-\pi/2, \pi/2]$ rad
- **Joint 3 (q_3):** $[-\pi, \pi]$ rad

A configuration is considered valid only if it resides within the free space C_{free} , which is the subset of C that does not result in a collision between the robot, the grasped object, and the obstacle space C_{obs} .

The robot's physical structure, visualized in **Figure 6.2**, dictates the reach and the transformation of joint movements into gripper-frame accelerations.

6.1.2 Dynamic Path Constraints

Unlike standard path planning where any continuous curve in C_{free} is acceptable, adhesive manipulation introduces a set of non-convex dynamic constraints. A path $\sigma_{(s)} \rightarrow C_{free}$ must satisfy the **Orthogonal Constraint Bottleneck** at every point s :

1. **Linear Acceleration Limit:** The inertial and gravitational force vector must not exceed the interfacial capacity $f_{max}(\sigma)$ derived from the Kaelble or Goland-Reissner models.
2. **Angular Acceleration Limit:** The rotational stability must be maintained such that $\alpha < \alpha_{limit}$, ensuring the tipping moment does not initiate edge peeling.

These constraints are evaluated in the gripper's local frame $(\hat{u}_t, \hat{u}_n)$, where the current tilt angle ϕ and the direction of movement δ dictate the available "force budget" for the segment.

6.1.3 The Objective Function (J)

The goal of the RRT* algorithm is to identify an asymptotically optimal path σ^* that minimizes a multi-objective cost function. In this research, we move beyond simple Euclidean distance to include a "Force Risk" penalty. The cost J of a path is defined as:

$$(6.2) \quad J(\tau) = \int_0^L (1 + \gamma \cdot R(a(s))) ds$$

Where:

- L is the total Euclidean path length in joint space.
- γ is the **Cost Weighting Factor**, which defines the priority of force safety over path efficiency.
- $R(a)$ is the **Risk Function**, which increases as the required acceleration a approaches the boundary of the **Adhesive Limit Surface** L_{cm} .

By optimizing this function, the planner identifies trajectories that are not only short but also "smooth" regarding the stress on the adhesive interface, effectively providing a

buffer against the viscoelastic uncertainties and creep observed during the material characterization phase.

6.2 The Adhesive Feasibility Field

The core contribution of this planning framework is the *is_force_feasible* module, which serves as a physics-based filter for the RRT^* expansion. This module evaluates whether the transition between a parent node $z_{nearest}$ and a candidate node z_{new} satisfies the mechatronic stability requirements of the adhesive interface.

6.2.1 Workspace Mapping Via Forward Kinematics

For the purpose of feasibility assessment, each segment of the RRT^* tree is modeled with a **Constant Velocity Profile**. This assumption simplifies the dynamic check by isolating the inertial loads to specific transition points and centripetal components.

The algorithm calculates the required linear acceleration (a_{req}) and angular acceleration (ω_{req}) necessary to move the object mass (m) between nodes within the specified time step. These kinematic inputs are then resolved into the gripper's local coordinate system:

- **Normal Component** (a_n) : Opposing the adhesive's cleavage resistance.
- **Tangential Component** (a_t) : Opposing the adhesive's shear resistance.

6.2.2 Reciprocal-Norm Failure Criterion

A unique challenge in adhesive manipulation occurs when the combination of linear and rotational loads places both ends of the bond (Edge A and Edge B) in a tensile state. To handle these "Biaxial Tension" scenarios, the algorithm applies a custom "**Biaxial Tension**" scenarios, the algorithm applies a custom **Reciprocal-Norm Failure Criterion** at every node check:

$$(6.3) \quad P_{crit,eff} = \sqrt{\frac{P_{crit,A}^2 \cdot P_{crit,B}^2}{P_{crit,A}^2 + P_{crit,B}^2}}$$

Where $P_{crit,A}$, $P_{crit,B}$ are the individual failure thresholds for each peeling front. This heuristic provides a conservative estimation of the joint's capacity by modeling the coupling between the two stress fields. It ensures that

6.2.3 Orthogonal Constraint Bottleneck

Final validation is enforced through a **Hard AND Bottleneck**. A candidate trajectory segment is added to the tree only if it passes both the linear and angular stability checks:

$$(6.4) \quad \text{Valid_if} \left(a_{req} \leq a_{limit} \right) \wedge \left(\ddot{\omega}_{req} \leq \ddot{\omega}_{limit} \right)$$

This "hard AND" logic is physically justified by the distinct failure modes at play.

While a_{limit} protects the interfacial bond from pure cleavage or shear slip, $\ddot{\omega}_{limit}$ ensures rotational stability against the tipping moments created by object offsets. Checking these independently ensures that the robot never performs a maneuver that is linearly feasible but rotationally unstable, providing a multi-layered defense against grasp failure.

6.2.4 Logic Design and Sub-process Flow

The sub-process for checking feasibility is detailed in **Fig 6.2**. This design ensures that the material parameters established in Chapter 5 are the direct drivers of the robot's motion constraints.

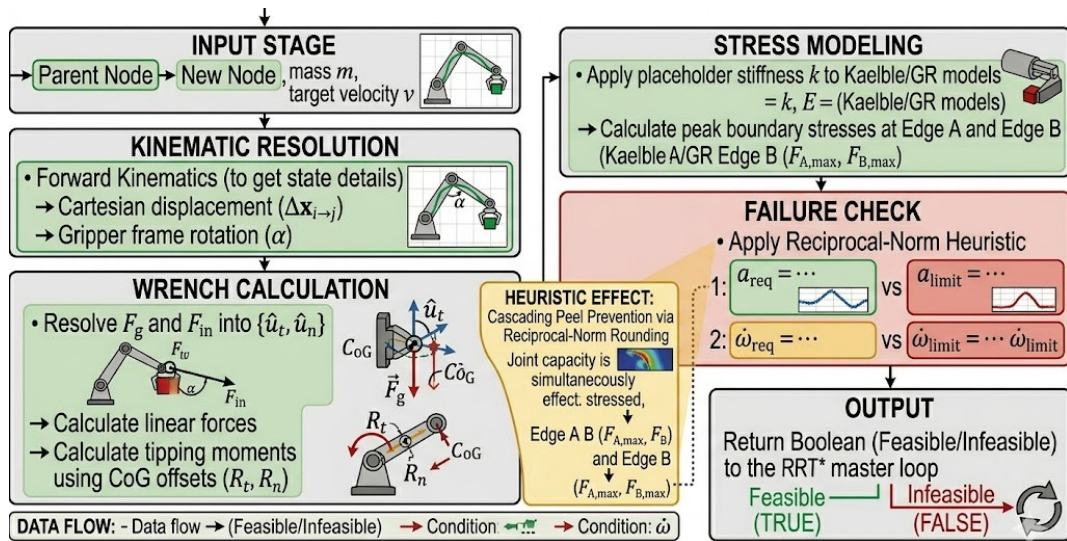


Fig 6.2 – Sub-process Flowchart for Checking Feasibility

6.2.5 The Energy Budget Criterion

Sections 6.2.2 and 6.2.3 define the feasibility field in terms of instantaneous force and moment limits. The implementation enforces a second, independent condition on every candidate segment: an energy balance between the work the adhesive bond can absorb before debonding and the work the manipulation demands. A segment satisfying Eq. 6.4 may still be rejected on these grounds, so the criterion is stated here for completeness. Two independent estimates of the energy required to separate the bond are computed, and the more conservative is adopted. Method 1 is the thermodynamic work of separation over the bonded contact area; Method 2 is the mean peel force acting over the contact length:

$$(6.5) \quad W_1 = \gamma \cdot A_{cont}$$

$$(6.6) \quad W_2 = 0.5 \cdot F_{tensile} \cdot l_{cont}$$

$$(6.7) \quad W_{lim} = \min(W_1, W_2)$$

Using $\gamma = 288.2 \text{ J/m}^2$ (uncoated cardboard, 5.3.4) and the gripper contact geometry of 5.1, these evaluate to $W_1 = 1.411 \text{ J}$, $W_2 = 0.509 \text{ J}$, so the peel-work estimate governs and $W_{lim} = 0.509 \text{ J}$.

The available budget is divided between translation and rotation by a fixed angular energy fraction f_{ang} :

$$(6.8) \quad W_{linear} = W_{lim} (1 - f_{ang}), W_{angular} = W_{lim} \cdot f_{ang}$$

For a segment traversing a distance d while the acceleration acts at angle θ_a to the vertical, the work done against gravity on the effective mass m_{eff} is W_g , and the energy remaining for useful motion is W_{net} . The segment is admissible only if this residual is positive:

$$(6.9) \quad W_g = m_{eff} \cdot g \cdot \sin(\theta_a) \cdot d$$

$$(6.10) \quad W_{net} = W_{linear} - W_g$$

$$(6.11) \quad \text{Validif} \rightarrow W_{net} > 0$$

When Eq. 6.11 fails, the bond cannot supply enough energy to move the object through the segment at all, irrespective of how gently the motion is executed. The condition

therefore represents a quasi-static limit that Eq. 6.4, which constrains only instantaneous accelerations, does not capture.

Because W_{lim} is fixed by the bond geometry and material while W_g scales linearly with mass, Eq. 6.12 becomes the binding constraint as the payload increases. This is the mechanism responsible for the rise in rejection rate reported in Table 6.2: from 18.5% of candidate segments at 0.10 kg to 27.5% at 1.00 kg. Setting $W_g = 0$ for a vertical translation over a 0.05 rad step gives a limiting mass of approximately 0.83 kg, above which purely vertical segments of that length become inadmissible, and the planner is forced to approach the goal through shallower, partially horizontal motions.

This coupling - a geometric planner whose admissible motion set contracts as the payload approaches the adhesive limit - is the practical expression of the force-aware behavior that distinguishes this planner from a conventional RRT*.

6.3 Optimization and Dynamic Cost Functions

While the feasibility field described in the previous section ensures that the robot avoids catastrophic detachment, it does not by itself select which of the many feasible paths between two configurations is best to execute. To reach a well-formed trajectory, the RRT* algorithm optimizes a cost function over the set of edges that have already passed the hard feasibility gate of section 6.2: the acceleration limits of Eq. 6.4 and the energy budget of Eq. 6.12. This is a deliberate separation of concerns. Whether a motion is admissible at all is decided categorically, by the hard gate; which admissible motion is preferred is decided by the soft cost described in this section. Adhesive safety is therefore enforced before the cost function is ever evaluated, not by it.

6.3.1 Multi-Objective Cost Formulation

The cost of a potential trajectory is not measured solely by its Euclidean length in joint space. Instead, a composite objective function J is defined over a path composed of n segments, combining three terms: the time required to execute the segment, its distance in configuration space, and its contribution to progress toward the goal:

$$(6.12) \quad J = \sum_i \left[w_{time} \cdot t_i + w_{dist} \cdot d_i + w_{advance} \cdot a_i \right]$$

Where:

t_i is the estimated execution time of segment i , computed from the velocity and acceleration limits derived in section 6.2 (so a segment that is only marginally force-feasible, and must therefore be executed more slowly, is already penalized through this term without a separate safety weighting);

d_i is the Euclidean distance of segment i in joint space;

a_i is a goal-advancement term that rewards edges which reduce the remaining distance to the goal, biasing the tree's growth and improving convergence speed;

$w_{time}, w_{dist}, w_{advance}$ are user-set weighting coefficients that trade off execution speed, path directness, and convergence rate.

Unlike a formulation that applies an explicit proximity-to-limit penalty within the soft cost, this planner does not reward a feasible edge for having additional margin inside the limit surface; all edges that pass the hard gate are, from the cost function's perspective, equally admissible, and are ranked only by time, distance, and progress. This is a real limitation, not a simplification made for exposition: a path that skims the boundary of the feasible region and a path that stays well inside it are treated identically by Eq. 6.6 if their time, distance, and progress costs are equal. Extending the cost function with an explicit margin-based term is identified as future work in Chapter 9.

6.3.2 Path Safety vs. Distance Trade-offs

Because adhesive safety is enforced by the hard gate rather than by a term in Eq. 6.12, the cost function does not itself reshape the configuration space into “high cost” and “low-cost” safety zones. What it does do is bias the search, among the already-safe candidates, toward motions that are faster and more direct — which in practice tends to favor smoother, less abrupt reorientations, since these require less execution time under the acceleration limits of section 6.2, even though nothing in Eq. 6.12 references the adhesive limit surface directly.

This distinction matters because of the viscoelastic uncertainty of the 3M™ 1900 tape observed during material characterization. Because the adhesive exhibits time-dependent creep, as seen in the shear test, where a 15.30 N load produced a total

displacement of 1.87 mm, maintaining a safety margin is important, and the current implementation provides it entirely through the hard feasibility gate of section 6.2 rather than through the cost function described here. The Reciprocal-Norm Failure Criterion introduced in that section is what distributes load consideration across both edges of the gripper; the cost function's role is limited to selecting an efficient path among the motions that criterion has already approved.

6.3.3 Algorithmic Rewriting and Asymptotic Convergence

The RRT* algorithm achieves asymptotic optimality through its unique rewiring step. When a new node is added to the tree, the algorithm searches its neighborhood to see if any existing nodes can be reached more cheaply through it, using the cost function of Eq. 6.12.

In the context of adhesive manipulation, this rewiring process shortens and smooths the tree using the cost function of Eq. 6.12: if a new node allows an existing one to be reached with lower combined time, distance, and progress cost, the algorithm restructures the tree to adopt that connection. Because every edge considered has already passed the hard feasibility gate, this process cannot introduce an adhesive-infeasible motion regardless of which edge the rewiring step prefers; what it converges toward is the fastest and most direct feasible trajectory, not necessarily the one with the greatest margin from the limit surface, for the reasons given in section 6.3.1.

6.4 Implementation and Pseudocode

The following pseudocode outlines the integrated planning loop, showing exactly where the material-dependent feasibility checks and the safety-weighted cost function intersect with the standard RRT^* expansion.

```
# Force-Aware RRT* Implementation
while iterations < max_iter:
    q_rand = sample_free_space()
    q_near = find_nearest_node(q_rand)
    q_new = steer(q_near, q_rand, delta)

    # Physics-based Feasibility Check
    if is_force_feasible(q_near, q_new, velocity):
        if is_collision_free(q_near, q_new):
            # Calculate cost including Safety Weighting
            cost_new = q_near.cost + dist(q_near, q_new) * (1 + gamma * safety_penalty)

            # Neighborhood Search and Rewire
            q_min = choose_best_parent(q_new, neighborhood, gamma)
            add_to_tree(q_new, q_min)
            rewire_neighbors(q_new, neighborhood, gamma)
```

Fig 6.3 – Pseudocode: Force Aware RRT* Implementation

This implementation treats the adhesive limit as a hard constraint for node creation and a soft penalty for path selection, providing dual-layered protection for the mechatronic system.

The following pseudocode details the complete algorithm, highlighting the integration of the safety penalty and the mechatronic rewiring process.

```
# Force-Aware RRT* Global Loop
Initialize tree T with q_start
for i from 1 to max_iterations:
    q_rand = Sample_Free_Space()
    q_near = Nearest_Neighbor(T, q_rand)
    q_new = Steer(q_near, q_rand, delta)

    # Logic Design Step: Physics-based Feasibility
    if is_force_feasible(q_near, q_new, mass, velocity):
        if is_collision_free(q_near, q_new):
            # Calculate cost including safety weighting
            s_penalty = calculate_safety_risk(q_near, q_new, Lcm)
            q_new.cost = q_near.cost + Dist(q_near, q_new) * (1 + gamma * s_penalty)

            # Neighborhood Optimization (Rewire)
            Neighborhood = Find_Near_Nodes(T, q_new, radius)
            q_best_parent = Choose_Parent(q_new, Neighborhood, gamma)
            Add_Node(T, q_new, q_best_parent)
            Rewire_Tree(T, q_new, Neighborhood, gamma)
```

Fig 6.4 – Pseudocode: Force Aware RRT*

By prioritizing paths with the lowest safety risk during the Choose_Parent and Rewire_Tree steps, the algorithm identifies trajectories that are not only efficient but also naturally stay within the center of the Adhesive Limit Surface (**Fig 5.4**).

Table 6.3 and Fig 6.6 summaries the outcome of a 50-scenario simulation bank (6.5.2); Fig 6.5 above shows one representative run in detail.

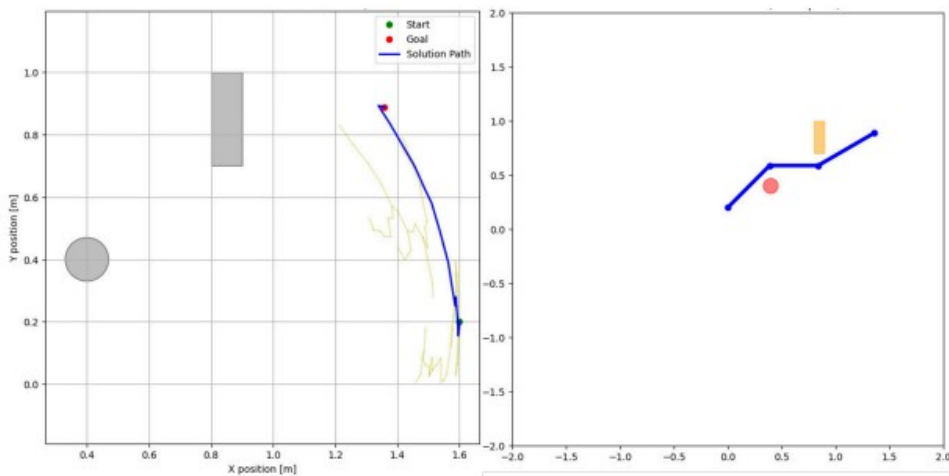


Fig 6.5 – End Effector's Path from Start to Goal (Left), and Robotic Arm's Finished Position (Right)

6.5 Simulation Results

6.5.1 Simulation Setup

The Force-Aware RRT* was evaluated over a bank of planning scenarios in which both the object mass and the obstacle layout were varied. Mass was treated as the primary parameter, because it directly determines how close the manipulation runs to the adhesive limit surface derived in 6.2.

Table 6.1 – Simulation configuration

Parameter	Value
Adhesive tensile capacity, F_{tensile}	10.19 N
Object mass range	0.60 – 1.00 kg
Object weight range	5.89 – 9.81 N
Load as fraction of adhesive capacity	58 – 96 %
Step size, Δ	0.05 rad
Goal tolerance	0.05 rad
Goal bias	0.20
Rewire radius	0.30 rad
Max iterations	900
Obstacle layouts	5 variations (position, size)

The mass range was chosen deliberately. At the 0.1 kg mass used during initial development the adhesive constraint was effectively inactive: the derived acceleration limits (101.9 m/s^2 , $15,617 \text{ rad/s}^2$) sat far above the manipulator's own joint limits (0.3 rad/s^2 after the safety factor S_a), so the feasibility gate never bound and the planner reduced to a conventional RRT*. Raising the object to 0.60–1.00 kg brings the load to 58–96% of static adhesive capacity, at which point the gate becomes the governing constraint over a substantial fraction of the configuration space.

Table 6.2 – Adhesive gate activity versus object mass

Object mass [kg]	Load [% of capacity]	Segments rejected by adhesive gate
0.10	10 %	18.5 %

0.60	58 %	19.0 %
0.80	77 %	22.0 %
0.90	87 %	23.0 %
1.00	96 %	27.5 %

Measured over 200 randomly oriented 0.05 rad segments per mass. The rejection rate rises monotonically with load, confirming that the feasibility field of §6.2 is actively shaping the search rather than passively admitting every motion.

6.5.2 Results

Fifty scenarios completed successfully across 60 attempts and are documented in full: each successful run is recorded with the arm at its start and goal configurations, the RRT* tree with its solution path in Cartesian space, and a video of the arm executing the planned trajectory.

Table 6.3 – Aggregate results, 50-scenario simulation bank

Metric	Value
Successful scenarios	50
Waypoints	3 – 38 (mean 20.7)
Tree nodes	12 – 96
Runtime	0 – 21 s
Goal error	0.00 rad, every run

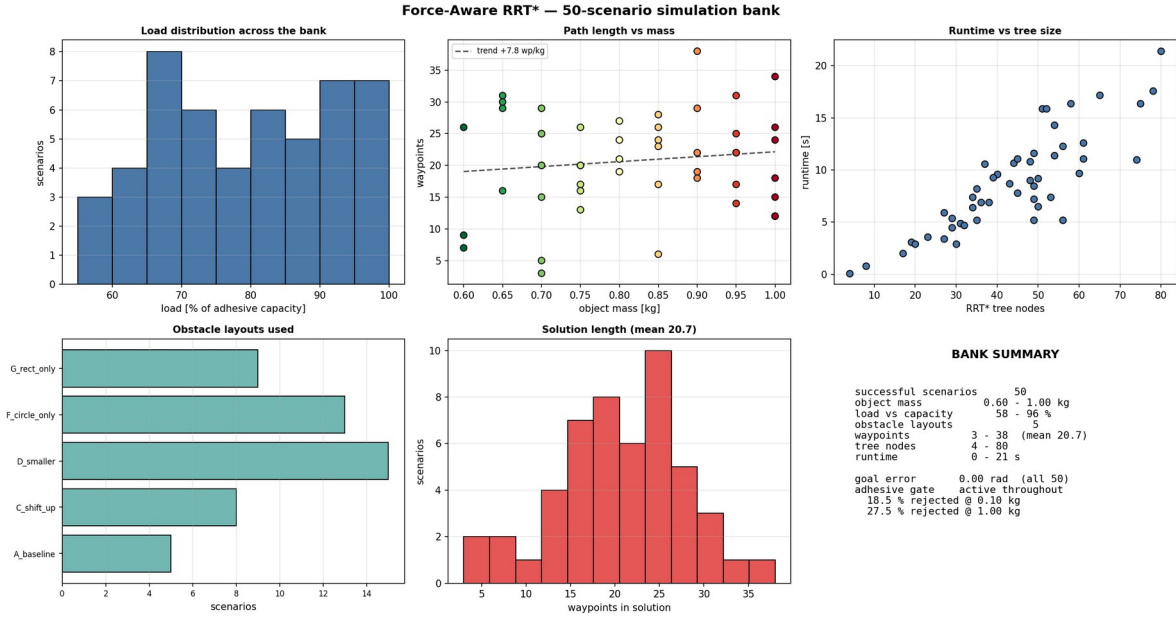


Fig 6.6 – Simulation bank summary: load distribution, path length vs. mass, runtime vs. tree size, obstacle layout usage, waypoint distribution, and aggregate statistics

6.5.3 Analysis

Terminal accuracy is exact. Every successful run terminated with a joint-space goal error of 0.00 rad. This is a structural property of the implementation rather than a statistical result: the goal configuration is appended to the tree as an exact copy once a validated edge reaches it, so the returned trajectory terminates precisely at the commanded configuration. The goal tolerance governs when a connection is attempted, not the accuracy of the result.

The adhesive constraint is genuinely governing. At 96% of capacity the feasibility gate rejects 27.5% of candidate expansions (Table 6.2). The planner is therefore not merely avoiding obstacles but actively excluding motions whose required acceleration would exceed the adhesive limit surface, the intended behavior of the two-layer structure described in 6.2 and 6.3.

Solution complexity does not scale simply with load. Waypoint counts range from 3 to 38 with no monotonic dependence on mass; the shortest solutions occurred at both low and high load, and obstacle geometry dominates path length far more than payload does. This indicates that within the tested range the adhesive constraint restricts how a motion may be executed more than whether a route exists.

6.5.4 Limitations

Obstacle layouts are variations on a small set of working geometries. More aggressive layouts, narrow gaps between tall obstacles, large central obstructions, three or more scattered obstacles, produced no solutions at all within the iteration budget. For a 3-DOF planar arm at the 0.05 rad step size that terminal precision requires, such layouts leave no traversable corridor that the sampling process locates in reasonable time.

Approximately one sixth of attempted scenarios failed and were excluded (10 of 60 attempts). The failures are attributable to search efficiency rather than to correctness or to the adhesive model: where a reasonably direct corridor exists, the planner converges within seconds, and where it does not, sampling fails to discover the required detour.

Corridor obstruction is a poor predictor of solvability. An attempt to pre-screen queries by counting blocked waypoints along the straight-line joint-space path proved unreliable, a query with only 1 of 26 waypoints blocked still failed across three random seeds at 1500 iterations. Scenario selection therefore verified solvability by attempting to plan, rather than by geometric screening.

These limitations motivate the bidirectional planning extension proposed for future work (9.5); growing trees simultaneously from start and goal is the standard remedy for precisely this narrow-corridor failure mode.

7. On-line Grasp Validation

The operational reliability of an adhesive gripper depends on its ability to autonomously verify the integrity of the thin-film bond in real-time. Unlike mechanical grippers, where motor torque or encoder feedback can indicate a secure grasp, adhesive systems require integrated surface sensing to infer the quality of the interfacial bond. This chapter details the development of a dual-layer validation system that utilizes a distributed photoresistor array and machine learning architectures to monitor the grasp from initial acquisition through dynamic transport. The comprehensive logic flow for this system is visualized in **Fig 7.1**.

The sensing, classification, and threshold components developed in this chapter are the building blocks of a larger validation algorithm; the full two-phase protocol that sequences them -- pre-motion assessment, continuous monitoring, and the graduated recovery actions triggered by a degrading grasp -- is presented as part of the integrated system in Chapter 8, once the trajectory-planning components it also depends on have been introduced.

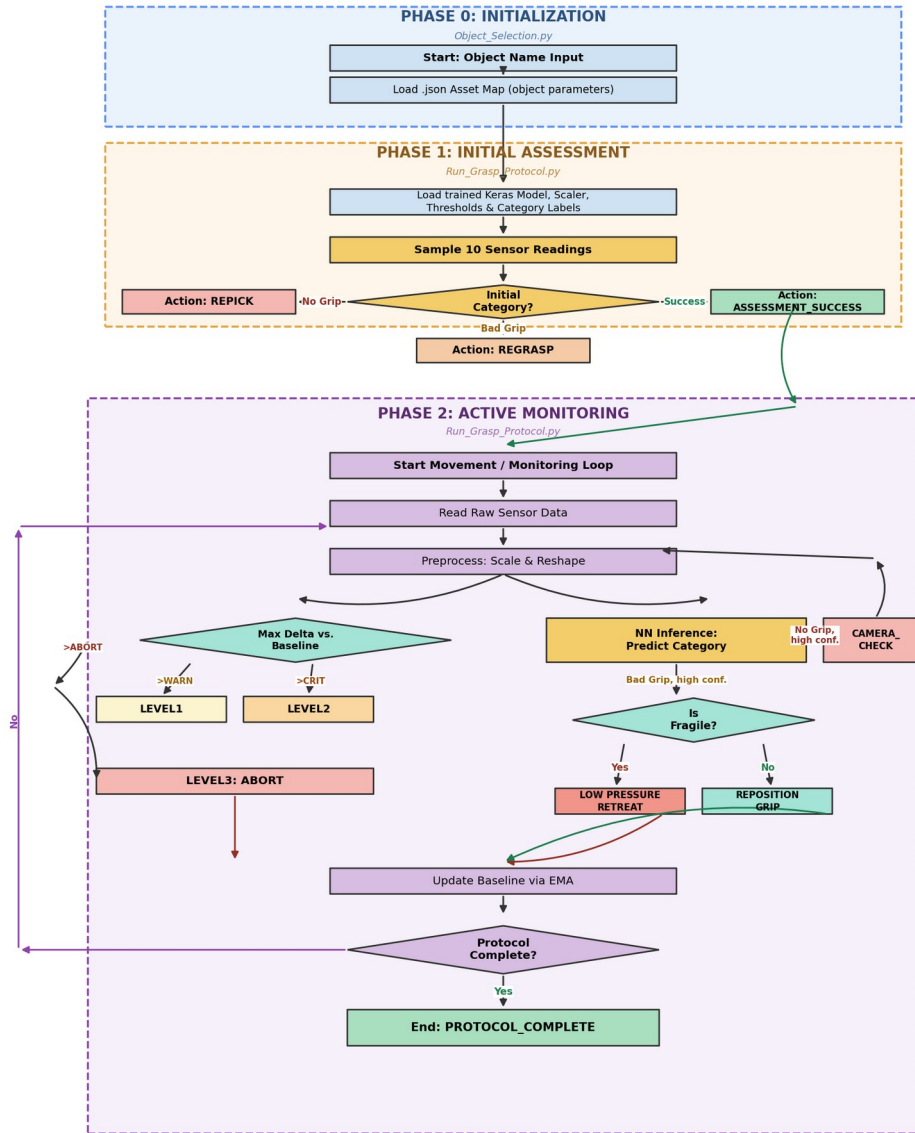


Fig 7.1 – Block Diagram for the Grasp Validation Algorithm

7.1 Sensor Technology Selection

The reliability of the validation system depends on the selection of a physical sensor capable of detecting micro-deformations and light-occlusion signatures at the adhesive-object interface in real-time. Two primary alternatives were considered: Infrared (IR) sensors and Light-Dependent Resistors (Photoresistors/LDRs), as shown in **Fig 7.2**.

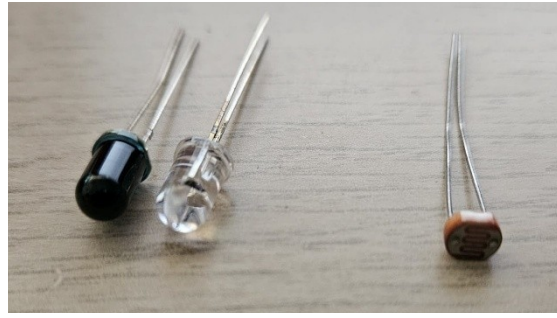


Fig 7.2 – IR Transmitter and Sensor (left) and LDR Sensor (Right)

7.1.1 POC and Comparative Study

To evaluate the performance of each technology, a specialized Proof-of-Concept (POC) Jig was created, **Fig 7.3**, mimicking the planned placement of the sensors within the final gripper design. The experimental setup utilized an Arduino microcontroller and a breadboard for rapid data acquisition, seen in **Fig 7.4**.

The comparative study involved applying a 3M™ 1900 adhesive strip over the sensors and measuring the change in electrical output between an "empty" state and a "loaded" state. To ensure robustness, the test was repeated across three objects spanning a range of surface finishes and geometries - a compliant sponge, a rigid sponge holder, and a glossy-surfaced earbud case - in three gripper orientations each (Up, Down, Side), against an empty-gripper baseline recorded in the same three orientations. A further arm repeated the full object set under direct illumination from a close lamp to test robustness to harsh lighting.



Fig 7.3 – IR POC Test (left) and LDR POC Test (Right)

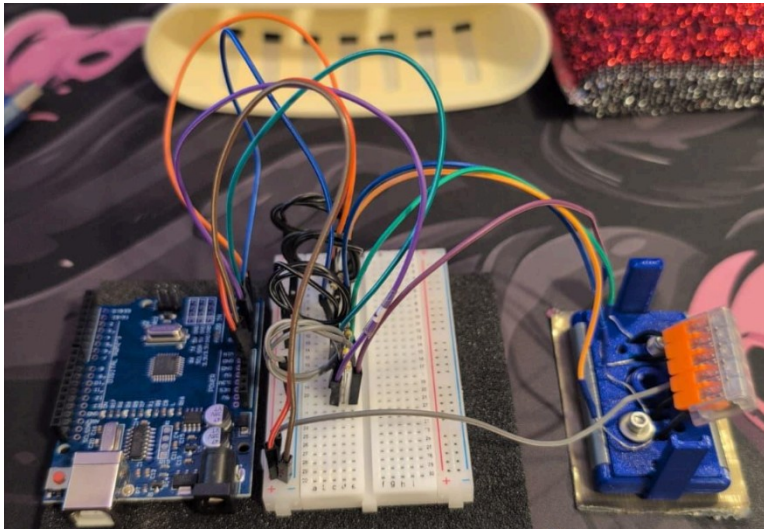


Fig 7.4 – POC Test Setup

7.1.2 Results and Sensor Selection

The experimental results indicated that LDR sensors offer significantly better performance for this specific application. The key findings are summarized below:

1. **Reflectivity Robustness:** IR sensors were found to be highly sensitive to the varied reflectivity of the target objects. While they performed well on smooth surfaces, porous or textured materials (like the cleaning sponge) produced inconsistent reflection signatures, leading to false-negative grasp detections.
2. **Environmental Robustness:** LDRs demonstrated superior stability under variable lighting. By measuring the occlusion of ambient light and the specific reflection signatures produced by the silver 3M™ 1900 tape during contact, the LDR array provided a more consistent "footprint" of the grasp.
3. **Frequency and Latency:** The simple resistive nature of the LDR allowed for high-frequency sampling (up to 500 Hz) via the microcontroller's analog inputs. This high sampling rate is critical for the temporal slippage detection algorithms detailed in later sections, where sub-millisecond signal oscillations must be captured to identify incipient slip. The full comparative results, including quantitative effect sizes and a harsh-lighting robustness arm, are presented in the following subsection.

These qualitative findings are supported by a quantitative comparison across three test objects (a sponge, a sponge holder, and an earbud case) in three gripper orientations each, against an empty-gripper baseline, using three criteria: the fraction of the 10-bit ADC range actually used, the effect size (Cohen's d) of each object condition against the

empty-gripper baseline, and the accuracy of a linear discriminant classifier (5-fold cross-validated) on object-presence detection.

Table 7.1 – ADC dynamic range utilization, IR vs. photoresistor

Modality	Min	Max	Span	% of ADC range
IR	853	1023	170	16.6 %
IR (excl. saturation events)	853	1001	148	14.5 %
Photoresistor	0	897	897	87.7 %

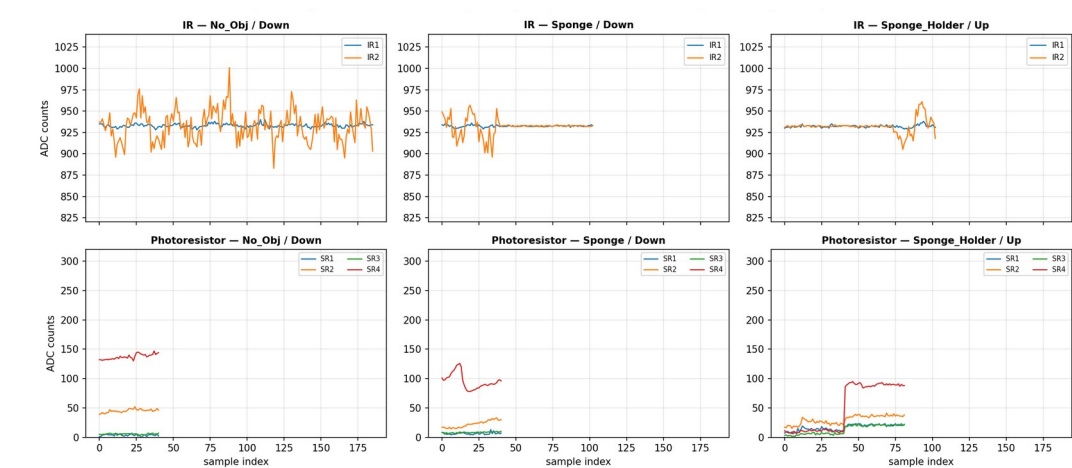


Fig 7.5 – Representative raw traces, IR (top) vs. photoresistor (bottom), fixed y-scale per row

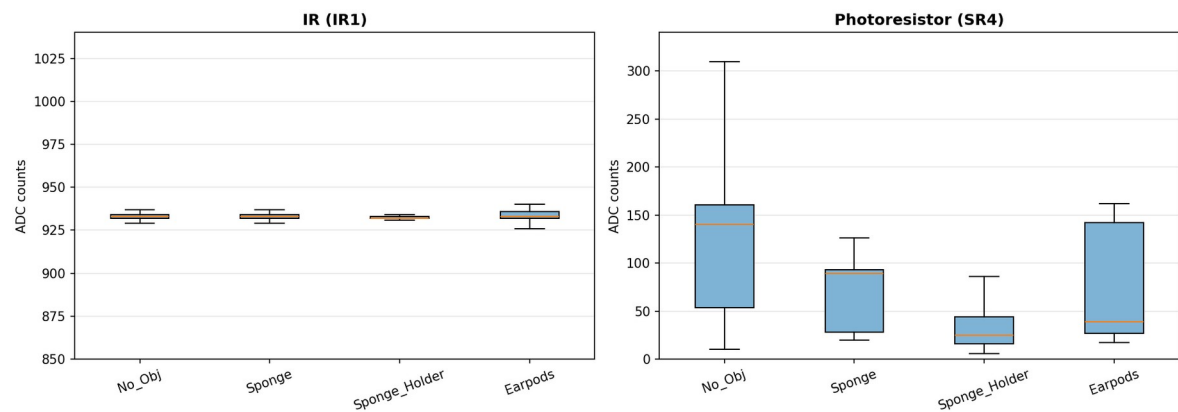


Fig 7.6 – Signal distribution by object, primary channel of each modality

The IR pair occupies a narrow band clustered near 933 counts across every condition, using under a sixth of the converter's range; the photoresistor array spans nearly the whole scale.

Table 7.2 – Effect size vs. empty-gripper baseline, by channel

Object	IR channel 1	IR channel 2	Photoresistor	Photoresistor
			ch. 2	ch. 4
Sponge	0.04	0.02	0.32	0.91
Sponge holder	0.16	0.06	0.05	1.70
Earbud case	0.36	0.12	0.28	0.37

By the conventional interpretation of Cohen's d , every IR effect size is negligible ($d < 0.2$) with the single exception of the earbud case ($d = 0.36$) - the only object with a glossy, specular surface, and a value inflated by transient saturation events discussed below. The photoresistor array achieves a large effect for the sponge ($d = 0.91$) and the sponge holder ($d = 1.70$).

Table 7.3 – Linear-discriminant classification accuracy (5-fold cross-validated)

Modality	Task	Accuracy	Majority baseline	Gain
IR	Object			
	identification (4-class)	0.318 ± 0.014	0.262	+0.056
IR	Object presence (2-class)	0.765 ± 0.001	0.765	+0.000
Photoresistor	Object			
	identification (4-class)	0.550 ± 0.020	0.357	+0.193
Photoresistor	Object presence (2-class)	0.842 ± 0.020	0.713	+0.129

The IR presence-detection result is decisive: at 0.765 ± 0.001 it matches the majority-class baseline to three decimal places, with near-zero cross-fold variance. The classifier has not learned to detect the object; it always answers, "object present", which is correct exactly as often as that class occurs. The IR channels carry no usable information about

object presence in this configuration. The photoresistor array exceeds its baseline by 12.9 points on the same task, and by 19.3 points on object identification.

The mechanism is interpretable. The IR channels rest at roughly 933 counts (about 91% of full scale) in every condition, the signature of a receiver operating near its rail: already close to saturation from the emitter's own reflected output and ambient IR, so additional reflected energy from a nearby object produces almost no further change. This is supported by the surface-dependence of the one measurable IR response (the glossy earbud case only) and by saturation transients recorded in the EarPods-Up trial, where both channels railed to 1023 for 5-8 consecutive samples before returning to baseline -- consistent with specular glints, and the reason the earbud case's effect size is inflated rather than genuinely informative. The photoresistor array instead exploits occlusion of ambient light rather than reflectance, so its response depends on object geometry and position rather than surface finish.

7.1.3 Robustness Under Direct Illumination

Because the photoresistor array senses occlusion of ambient light, it is a priori the modality more exposed to strong direct lighting. This was tested by repeating the object recordings with a lamp directed at the gripper from close range. No no-object baseline was recorded under this condition, so the analysis below uses three-class object identification throughout.

Table 7.4 – Signal level, nominal vs. direct illumination

Object	Nominal SR4 mean	Harsh SR4 mean	Increase
Sponge	71.4	97.6	×1.37
Sponge holder	34.8	114.9	×3.30
Earbud case	92.8	478.5	×5.16

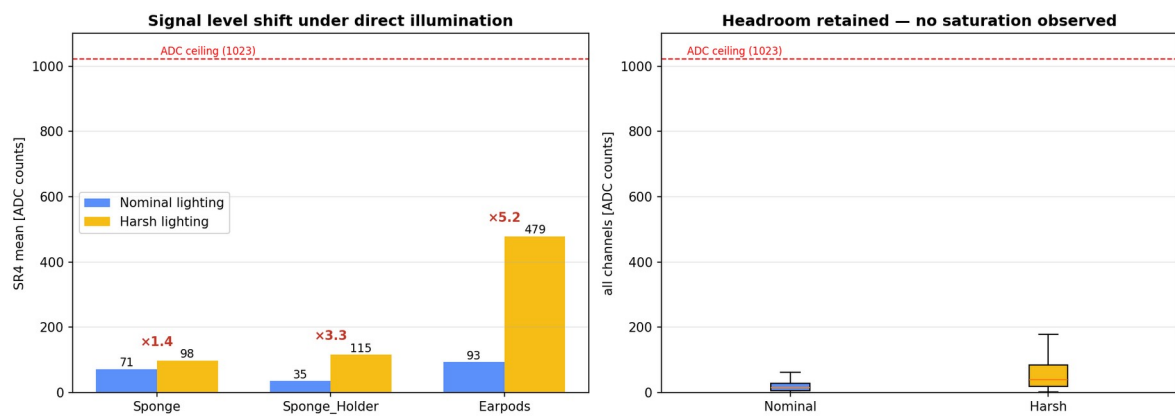


Fig 7.7 – Photoresistor response to direct lamp illumination — level shift and retained headroom

Direct illumination raises the signal level substantially and unevenly, but the array does not saturate: the maximum reading observed is 885 counts against the 1023 ceiling, roughly 13% headroom in reserve - the opposite of the IR failure mode, which sits near its rail even under nominal conditions. Discriminative power is not lost under harsh lighting either: within-condition classification reaches 0.600 ± 0.067 against a 0.400 baseline (a 20.0-point gain), comparable to the 10.6-point gain achieved under nominal lighting.

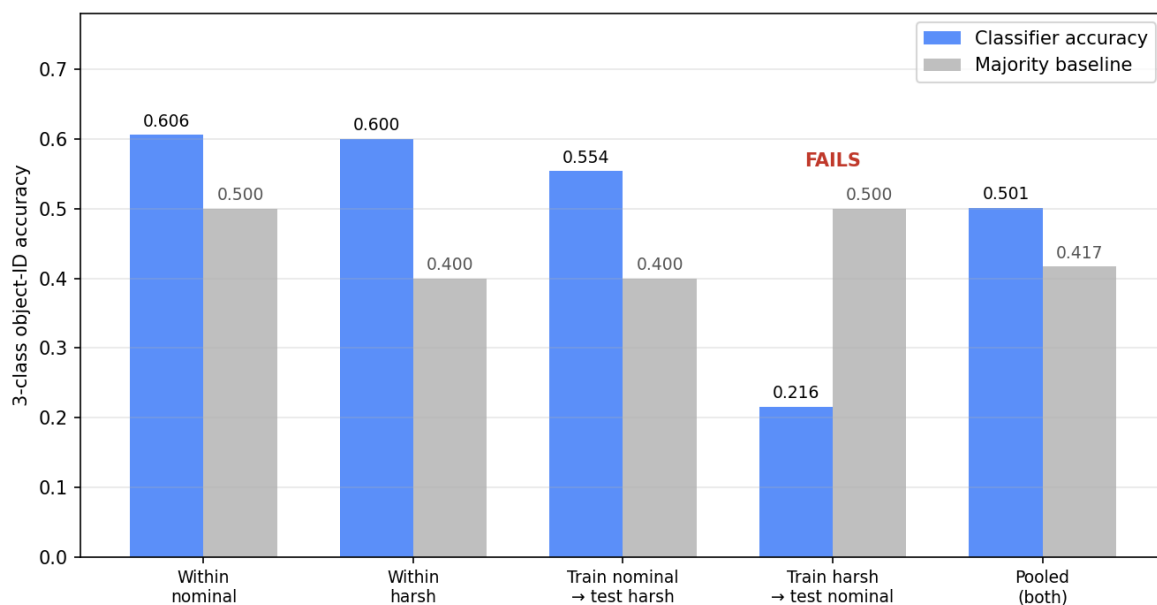


Fig 7.8 – Crosslighting generalization of the photoresistor array

Table 7.5 – Crosslighting transfer

Train → Test	Accuracy	Majority baseline	Outcome
Nominal → Harsh	0.554	0.400	Degraded
Harsh → Nominal	0.216	0.500	Fails — worse than guessing
Pooled (both conditions)	0.501 ± 0.012	0.417	Degraded

The operationally decisive question is not whether the array works under each lighting condition separately, but whether a model calibrated under one condition remains valid under the other. It does not: a classifier trained under direct illumination and applied under nominal lighting scores 0.216, below the 0.500 majority baseline - actively worse than always predicting the most common class. The failure is systematic: under nominal lighting the sponge holder produces the lowest SR4 level of the three objects, but under direct illumination it produces a higher level than the sponge, so a decision boundary fitted in one condition is inverted, not merely imprecise, in the other. Two candidate mitigations were tested and neither is sufficient alone -- illumination-normalized features (each channel as a fraction of the four-channel total) improve within-condition accuracy slightly but make transfer worse (0.283), since the lamp does not illuminate the four sensors uniformly; and pooled training across both conditions (0.501) underperforms either condition alone. A weak but usable mitigation is that the array can partially detect the lighting regime itself (0.700 accuracy against a 0.583 baseline), suggesting a practical architecture of detecting the illumination condition and applying the model calibrated for it.

The photoresistor array was selected as the sensing modality for the grasp validation system on three grounds: it uses 87.7% of the available ADC range against the IR pair's 16.6%; it achieves large effect sizes against the empty-gripper baseline ($d = 0.91$ and 1.70) where the IR pair achieves negligible ones ($d = 0.02$ - 0.16) for the same objects; and it exceeds the majority baseline for object-presence detection by 12.9 points, where the IR pair's accuracy matches its majority baseline exactly, indicating no usable presence information. The direct-illumination trials qualify but do not overturn this conclusion: the array retains headroom and discriminative power under a close lamp, but its absolute calibration does not transfer between lighting regimes, which is why the two-phase validation protocol of 7.3 re-establishes its baseline immediately before each manipulation rather than relying on a fixed threshold set at design time.

7.2 Comparative Analysis of Classification Methodologies

To identify the optimal architecture for real-time grasp validation, a comprehensive benchmark of six distinct methodologies was conducted. Each model was evaluated on a standardized dataset consisting of 4-channel LDR signatures sampled from over 50 sets of multiple grasping scenarios, including centered, off-center, and edge-only contacts.

7.2.1 Evaluated Architecture and Methodologies

The following models were implemented and tested to define the complexity required for accurate classification:

1. Single-Phase Neural Network (NN): A unified multi-layer perceptron (MLP) designed for end-to-end feature extraction.
2. Two-Phase NN: A hierarchical architecture that separates orientation detection from quality assessment.
3. Decision Tree (DT): A non-parametric supervised learning method for hierarchical classification.
4. Logistic Regression (LR): A statistical model used as a linear baseline.
5. Single-Layer Perceptron (SLP): A basic neural unit used to test linear separability.
6. Simple Thresholding: A rule-based system using static voltage limits.

```
Algorithm 1 Simple Thresholding (simple_threshold)  
Require: Test sensor matrix  $X_{test}$ , true grip labels  $y_{test}$ , grip category list  $C$   
Ensure: Best accuracy  $a^*$ , predictions  $\hat{y}^*$ , threshold sweep  $\{\theta_i\}$ , accuracy sweep  $\{a_i\}$   
1:  $s \leftarrow X_{test}[:, 0]$  ▷ Sensor 1 readings only  
2:  $\Theta \leftarrow 100$  values linearly spaced between  $\min(s)$  and  $\max(s)$   
3:  $a^* \leftarrow 0$ ;  $\theta^* \leftarrow \text{None}$ ;  $\hat{y}^* \leftarrow \text{None}$ ;  $\{a_i\} \leftarrow \emptyset$   
4: for each  $\theta \in \Theta$  do  
5:    $\hat{y} \leftarrow [C_{last} \text{ if } s_j > \theta \text{ else } C_{first}]_{j=1}^n$  ▷ Binary rule on a 4-class label set  
6:    $a \leftarrow \text{Accuracy}(y_{test}, \hat{y})$   
7:   append  $a$  to  $\{a_i\}$   
8:   if  $a > a^*$  then  
9:      $a^* \leftarrow a$ ;  $\theta^* \leftarrow \theta$ ;  $\hat{y}^* \leftarrow \hat{y}$   
10:  end if  
11: end for  
12: return  $a^*, \hat{y}^*, \Theta, \{a_i\}$ 
```

Fig 7.9 – Pseudocode: Simple Thresholding Grip Classification

Algorithm 2 Logistic Regression Baseline (`logistic_regression`)

Require: Scaled train/test sensor matrices X_{train}, X_{test} , grip labels y_{train}, y_{test} , categories C **Ensure:** Trained model M , accuracy a , predictions $\hat{y}$, learning-curve data

- 1: $M \leftarrow \text{LogisticRegression}(\text{max_iter} = 1000)$
 - 2: $M.\text{fit}(X_{train}, y_{train})$
 - 3: $\hat{y} \leftarrow M.\text{predict}(X_{test})$
 - 4: $a \leftarrow \text{Accuracy}(y_{test}, \hat{y})$
 - 5: Save M to disk with timestamp
 - 6: $(n_{sizes}, \text{train_scores}, \text{val_scores}) \leftarrow \text{LEARNINGCURVE}(M, X_{train}, y_{train}, cv=5)$
 - 7: **return** $M, a, \hat{y}, n_{sizes}, \text{train_scores}, \text{val_scores}$
-

Fig 7.10 – Pseudocode: Logistic Regression Grip Classification

Algorithm 3 Single-Layer Perceptron (`single_layer_perceptron`)

Require: Scaled train/test sensor matrices, grip labels, categories C **Ensure:** Trained model M , accuracy a , predictions $\hat{y}$, learning-curve data

- 1: $M \leftarrow \text{Perceptron}(\text{max_iter} = 1000)$
 - 2: $M.\text{fit}(X_{train}, y_{train})$
 - 3: $\hat{y} \leftarrow M.\text{predict}(X_{test})$
 - 4: $a \leftarrow \text{Accuracy}(y_{test}, \hat{y})$
 - 5: Save M to disk with timestamp
 - 6: $(n_{sizes}, \text{train_scores}, \text{val_scores}) \leftarrow \text{LEARNINGCURVE}(M, X_{train}, y_{train}, cv=5)$
 - 7: **return** $M, a, \hat{y}, n_{sizes}, \text{train_scores}, \text{val_scores}$
-

Fig 7.11 – Pseudocode: Single-Layer Perceptron Grip Classification

Algorithm 4 Decision Tree Classifier (`decision_tree`)

Require: Scaled train/test sensor matrices, grip labels, categories C **Ensure:** Trained model M , accuracy a , predictions $\hat{y}$, learning-curve data

- 1: $M \leftarrow \text{DecisionTreeClassifier}(\text{max_depth} = 10, \text{min_samples_split} = 20)$
 - 2: $M.\text{fit}(X_{train}, y_{train})$
 - 3: $\hat{y} \leftarrow M.\text{predict}(X_{test})$
 - 4: $a \leftarrow \text{Accuracy}(y_{test}, \hat{y})$
 - 5: Save M to disk with timestamp
 - 6: $(n_{sizes}, \text{train_scores}, \text{val_scores}) \leftarrow \text{LEARNINGCURVE}(M, X_{train}, y_{train}, cv=5)$
 - 7: **return** $M, a, \hat{y}, n_{sizes}, \text{train_scores}, \text{val_scores}$
-

Fig 7.12 – Pseudocode: Decision Tree Grip Classification

Algorithm 5 One-Phase Neural Network (`one_phase_nn`)

Require: Scaled train/test sensor matrices ($\in R^4$), grip labels, categories C ($|C|=4$)**Ensure:** Trained model M , accuracy a , predictions $\hat{y}$, training history H

- 1: $\text{input} \leftarrow \text{Input}(\text{shape} = (4,))$
 - 2: $h_1 \leftarrow \text{Dropout}(0.3)(\text{Dense}(64, \text{relu})(\text{input}))$
 - 3: $h_2 \leftarrow \text{Dropout}(0.3)(\text{Dense}(32, \text{relu})(h_1))$
 - 4: $\text{output} \leftarrow \text{Dense}(4, \text{softmax})(h_2)$ ▷ Single grip-quality head
 - 5: $M \leftarrow \text{Model}(\text{input}, \text{output})$
 - 6: $M.\text{compile}(\text{Adam}(lr=0.001), \text{sparse_categorical_crossentropy}, [\text{accuracy}])$
 - 7: $H \leftarrow M.\text{fit}(X_{train}, y_{train}, \text{val_data}=(X_{test}, y_{test}), \text{epochs}=100, \text{batch}=32)$
 - 8: $\hat{y} \leftarrow \arg \max_c M.\text{predict}(X_{test})$
 - 9: $a \leftarrow \text{Accuracy}(y_{test}, \hat{y})$
 - 10: Save M to disk with timestamp
 - 11: Plot confusion matrix of $(y_{test}, \hat{y})$
 - 12: **return** $M, a, \hat{y}, H$
-

Fig 7.13 – Pseudocode: One-Phase Neural Network Grip Classification

Algorithm 6 Two-Phase Neural Network Training (`GripValidationSystem.train_system_with_tuning`)

Require: Raw sensor dataset D , epochs E , test/validation split ratios
Ensure: Trained orientation model M_{orient} , trained grip model M_{grip} , evaluation results

- 1: $(X, y_{orient}, y_{grip}) \leftarrow \text{PREPAREDATA}(D)$ ▷ Scale sensors; label-encode orientation & grip
- 2: Split X, y_{orient}, y_{grip} into train/test (stratified on y_{grip}); sub-split train into train/val
- Phase 1 — Orientation Model*
- 3: $M_{orient} \leftarrow \text{TUNEORIENTATIONMODEL}(X_{train}, y_{orient,train}, \text{val_data})$ ▷ Keras-Tuner random search, 20 trials, early stopping
- 4: $H_{orient} \leftarrow M_{orient}.\text{fit}(X_{train}, y_{orient,train}, \text{epochs}=E, \text{early_stop})$
- 5: $p_{orient,train} \leftarrow M_{orient}.\text{predict}(X_{train}); p_{orient,test} \leftarrow M_{orient}.\text{predict}(X_{test})$
- Phase 2 — Grip-Quality Model (sensor features + orientation probabilities)*
- 6: $X_{train}^{comb} \leftarrow [X_{train} \parallel p_{orient,train}]; X_{test}^{comb} \leftarrow [X_{test} \parallel p_{orient,test}]$
- 7: $M_{grip} \leftarrow \text{TUNEGRIPMODEL}(X_{train}^{comb}, y_{grip,train}, \text{val_data})$ ▷ Separate Keras-Tuner search
- 8: $H_{grip} \leftarrow M_{grip}.\text{fit}(X_{train}^{comb}, y_{grip,train}, \text{epochs}=E, \text{early_stop})$
- Evaluation*
- 9: $\hat{y}_{orient} \leftarrow \arg \max_c M_{orient}.\text{predict}(X_{test})$
- 10: $\hat{y}_{grip} \leftarrow \arg \max_c M_{grip}.\text{predict}(X_{test}^{comb})$
- 11: **return** $M_{orient}, M_{grip}, H_{orient}, H_{grip}, \hat{y}_{orient}, \hat{y}_{grip}$

Fig 7.14 – Pseudocode: Two-Phase Neural Network Training (Orientation → Grip)

Algorithm 7 Grasp-Validation Method Comparison (`GraspingValidationSystem.run_complete_analysis`)

Require: Raw per-orientation Excel logs (No/Bad/Semi/Good grip $\times$ 3 orientations)
Ensure: Ranked comparison table of all six methods by grip-classification accuracy

- 1: $D \leftarrow \text{GENERATESYNTHETICDATA}$ ▷ Load, label, and concatenate all orientation/grip Excel logs into one DataFrame; cache as `all_sponge_data.csv`
- 2: $(X, y_{grip}, y_{orient}) \leftarrow$ sensor columns and mapped numeric labels from D
- 3: $(X_{train}, X_{test}, y_{grip,train}, y_{grip,test}, y_{orient,train}, y_{orient,test}) \leftarrow \text{TRAINTESTSPLIT}(X, y_{grip}, y_{orient}, 0.2, \text{stratify}=y_{grip})$
- 4: $X_{train}^{scaled}, X_{test}^{scaled} \leftarrow \text{STANDARDSCALER}(X_{train}, X_{test});$ save fitted scaler
- 5: $R \leftarrow \{\}$ ▷ Results dictionary
- 6: $R[\text{simple_threshold}] \leftarrow \text{SIMPLETHRESHOLD}(X_{test}, y_{grip,test}, C)$ ▷ Alg. 1
- 7: $R[\text{logistic_regression}] \leftarrow \text{LOGISTICREGRESSION}(X_{train}^{scaled}, X_{test}^{scaled}, y_{grip,train}, y_{grip,test}, C)$ ▷ Alg. 2
- 8: $R[\text{single_layer_perceptron}] \leftarrow \text{SLP}(X_{train}^{scaled}, X_{test}^{scaled}, y_{grip,train}, y_{grip,test}, C)$ ▷ Alg. 3
- 9: $R[\text{decision_tree}] \leftarrow \text{DECISIONTREE}(X_{train}^{scaled}, X_{test}^{scaled}, y_{grip,train}, y_{grip,test}, C)$ ▷ Alg. 4
- 10: $R[\text{one_phase_nn}] \leftarrow \text{ONEPHASENN}(X_{train}^{scaled}, X_{test}^{scaled}, y_{grip,train}, y_{grip,test}, C)$ ▷ Alg. 5
- 11: $R[\text{two_phase_nn}] \leftarrow \text{TWOPHASENN}(D, E=100)$ ▷ Alg. 6
- 12: $\text{table} \leftarrow$ sort R by grip accuracy, descending
- 13: **return** table

Fig 7.15 – Pseudocode: Grasp-Validation Method Comparison Orchestration

7.2.2 Analysis of Training and Learning Curve

The performance of these models was analyzed using learning curves (accuracy vs. training size) and training histories (accuracy/loss vs. epoch).

The Validation Accuracy results are presented in Table 7.6.

- **Linear Baseline Instability:** The Single-Layer Perceptron demonstrated extreme instability, with accuracy fluctuating between 0.48 and 0.61 (Fig 7.16). This erratic behavior confirms that the spatial LDR signatures are non-linearly separable and cannot be resolved by a simple hyperplane.
- **Convergence at Low Performance:** Logistic Regression reached an accuracy ceiling of approximately 70% (Fig 7.16), indicating high bias (underfitting) due to its inability to model the complex interaction between ambient light and tape reflection.

- **Two-Phase Model Bottlenecks:** Analysis of the Two-Phase NN components reveals that while the "Grip Quality Model" converges smoothly to ~88% accuracy, the "Orientation Model" plateaus at only ~60% (Fig 7.18). This discrepancy indicates that separating the tasks increases architectural complexity without providing a gain in predictive power, likely due to the highly coupled nature of orientation and contact area in LDR signatures.
- **Optimal Stability in Single-Phase NN:** The unified Single-Phase NN exhibited the most robust learning history. Both training and validation loss converged smoothly toward 0.45, while validation accuracy stabilized at the highest level of 86.84%, as shown in Fig 7.19.

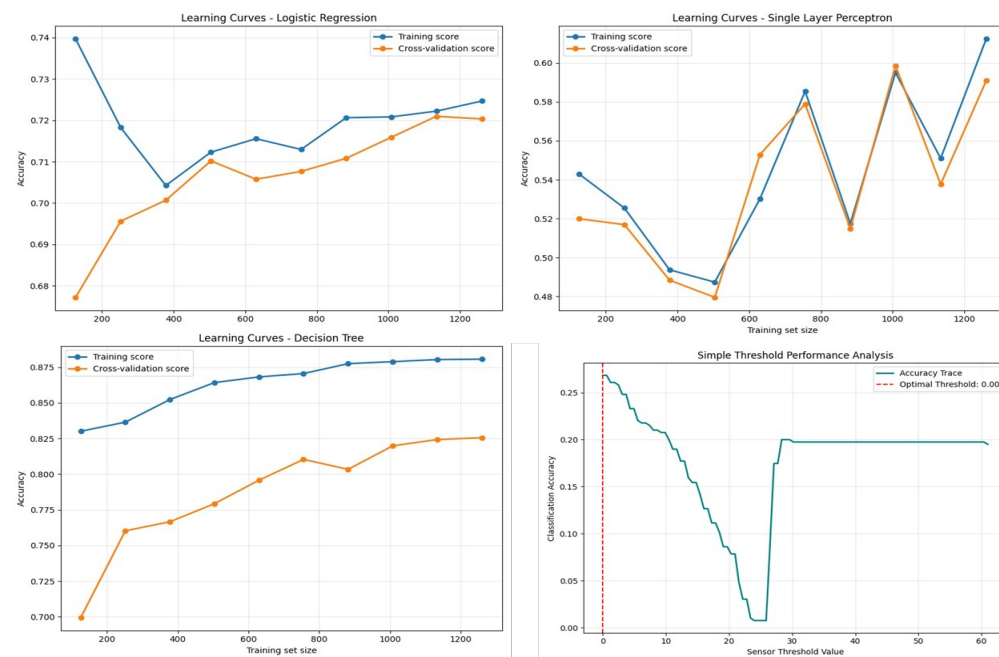


Fig 7.16 – Learning Curves for Logistic Regression (Top Left), SLM (Top Right), Decision Tree (Bottom Left), Simple Threshold (Bottom Right).

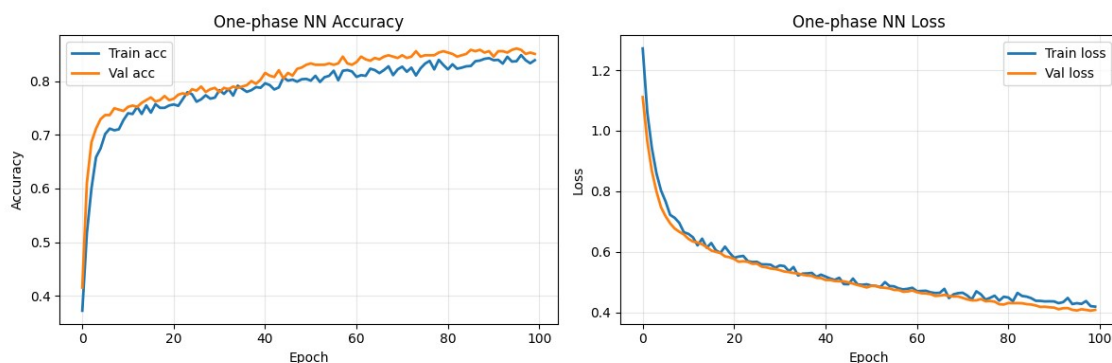


Fig 7.17 – Single Phase NN Accuracy Curve (Left) and Loss Curve (Right).

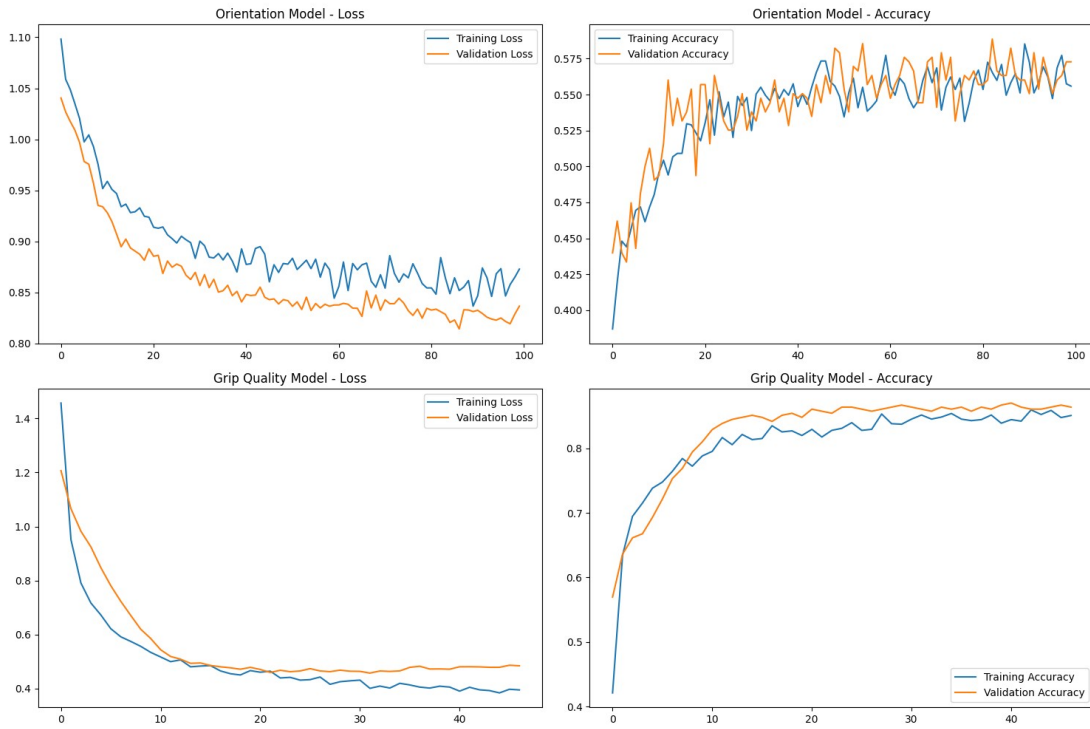


Fig 7.18 – Two Phase NN Curves -Orientation Model Loss (Top Left) and Accuracy (Top Right), Grip Quality Model Loss (Bottom Left) and Accuracy (Bottom Right).

7.2.3 Confusion Matrix Analysis

The classification performance of the selected Single-Phase NN was scrutinized using the multi-class confusion matrix provided in **Fig 7.19**.

Analysis of the matrix reveals that the model is extremely effective at identifying the "No Grip" state, achieving 100% recall (77/77). This is critical for preventing the robot from initiating transport without a secured agent. Furthermore, the model demonstrates a favorable "conservative bias" in its error distribution. Specifically, 18 instances of a "Good Grip" were categorized as "Semi-Good Grip," and 12 instances of "Semi-Good" were labeled as "Bad."

From a mechatronic safety perspective, these are acceptable errors; they lead the trajectory planner (Chapter 6) to apply a stricter acceleration limit a_{limit} , ensuring the safety of the bond at the cost of a minor reduction in throughput. The most significant risk, misclassifying a "Bad Grip" as "Good", occurred in only 13% of bad-grip

scenarios, which the real-time LSTM monitoring layer is designed to capture during the transport phase.

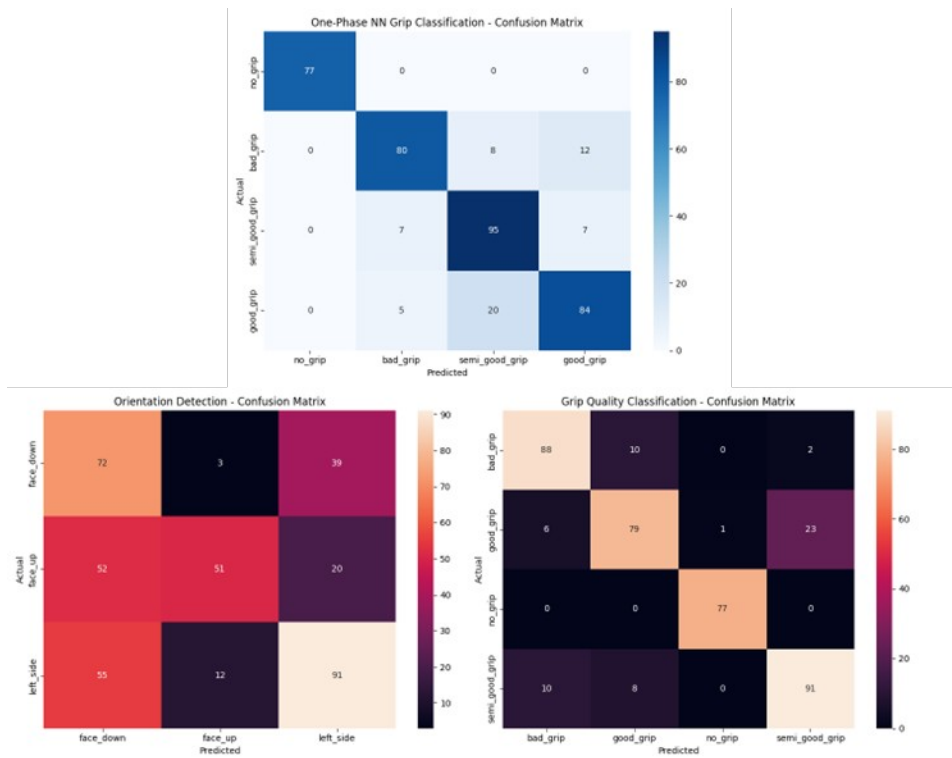


Fig 7.19 – Confusion Matrix for Single Phase NN (Top), and Double Phase NN Orientation Model (Bottom Left) and Grip Quality Model (Bottom Right)

7.2.4 Failure of Static Threshold

The Simple Thresholding method yielded the lowest performance, peaking at 26.84% accuracy. As visualized in the performance analysis, the accuracy trace remains flat across most threshold values, as seen in **Fig 7.4**, with a sharp drop as the threshold increases. This confirms that static voltage limits are entirely insufficient for the 3M™ 1900 interface, which requires the dynamic, spatial feature extraction provided by neural networks to handle variations in object material and ambient lighting.

7.2.5 Quantitative Performance Results

The table below summarizes the Validation Accuracy results produced by the comparison.

Table 7.6 – Comparison Results for all Methodologies

Methodology	Validation Accuracy [%]	Decision Rationale
-------------	-------------------------	--------------------

		Selected: Best balance of accuracy and generalization
Single-Phase NN	86.84	
Two-Phase NN	84.30	Limited by orientation detection bottlenecks
Decision Tree	84.30	Significant gap between training and validation accuracy
Logistic Regression	69.87	High bias; unable to model nonlinear signatures
Single Layer Perception	69.87	Highly unstable training behavior
Simple Threshold	26.84	Inadequate for dynamic surface signatures

The quantitative results underscore a critical design trade-off in the validation system. While the Two-Phase NN and Decision Tree are competitive, they exhibit higher variance in unseen conditions. The **Single-Phase NN** not only provides the highest accuracy but also the most stable transition from training to validation datasets. This robustness is vital for handled objects with varying surface textures and reflectivity, as the network effectively generalizes the underlying contact geometry from the raw 4-channel resistance vector. The categorical analysis from the confusion matrix further supports this choice, showing that the integrated architecture is superior at resolving the coupled relationship between object orientation and effective contact area. as the network effectively generalizes the underlying contact geometry from the raw 4-channel resistance vector.

7.2.6 Methodology Comparison Conclusions

The comparative study establishes that the nonlinear spatial features of the LDR array are best captured by a unified, multi-layer neural architecture. The Single-Phase NN offers a high validation accuracy of 86.84% while maintaining the low computational latency required for real-time integration into the robot's control loop. Most

importantly, its failure modes are biased toward safety, providing a robust "Go/No-Go" gatekeeper that enhances the overall reliability of the adhesive manipulation system.

7.3 NN-Based Grasp Quality Classification

The implementation of the Single-Phase Neural Network (NN) for grasp quality assessment serves as the primary decision engine of the manipulation cycle. This layer is responsible for translating the spatial light-occlusion signatures of the 4-channel LDR array into a set of discrete robotic behaviors. Unlike a conventional "Go/No-Go" sensor, which merely detects object presence, the neural architecture evaluates the quality of the bond to determine its mechanical capacity.

The classification layer functions as a post-acquisition diagnostic tool that dictates the robot's subsequent state machine. By analyzing the non-linear correlations in the sensor data, the network categorizes the established phase into one of four distinct states, each mapped to a specific mechatronic policy:

1. **Good Grip:** The object is centered with maximum contact area. The system authorizes the execution of the optimized RRT^* trajectory at nominal joint velocities.
2. **No Grip:** No bond has been established. The controller triggers an immediate "Try Again" sequence, resetting the gripper position for a new approach.
3. **Bad Grip:** A bond is detected, but the contact area is insufficient or highly unstable (e.g., edge-only contact). To prevent a dropped payload during motion, the system executes a "Regrip" maneuver to re-establish a more secure interface.
4. **Semi-Good Grip:** The bond is established but exhibits a geometric offset or in a noncomplete manner. The system continues the task but implements a "Move Slower" policy by reducing the acceleration limit (a_{limit}) to compensate for the increased tipping moments.

These four categories were assigned during a controlled data-collection protocol, in which the object was physically positioned at known, deliberately varied offsets and contact fractions relative to the gripper, centered and fully seated for Good Grip, offset by a fixed amount for Semi-Good Grip, reduced to edge-only contact for Bad Grip, and withdrawn entirely for No Grip, rather than being defined after the fact by thresholding

the raw sensor signal. The quantitative boundary between categories is therefore not a manually chosen threshold on any single sensor value; it is whatever separation the classifier of section 7.5 learns from the resulting labelled sensor patterns. How separable that boundary actually is in practice is reported empirically in the confusion-matrix and accuracy results of sections 7.2.3 and 7.2.5, rather than assumed here: the classifier distinguishes No Grip from the other three categories with perfect recall, while Bad Grip and Semi-Good Grip -- the two categories closest in physical contact fraction -- are correspondingly the pair most often confused with each other.

By providing this granular assessment, the validation system enables the robot to handle the stochastic uncertainty inherent in warehouse environments. This section details the engineering of the classification pipeline, including the preprocessing techniques that ensure signal invariance and the architectural design used to achieve high-accuracy state identification.

7.3.1 Data Processing and Feature Normalization

To ensure that the grasp classification layer is resilient to varying environmental conditions, the real-time algorithm implements a structured data conditioning pipeline. Raw analog signals transmitted from the distributed LDR array are transformed through three distinct stages, ingestion validation, statistical normalization, and temporal smoothing, before being processed by the neural architecture. This pipeline, visualized in **Fig 7.20**, ensures that the model identifies contact geometry rather than absolute lighting intensities.

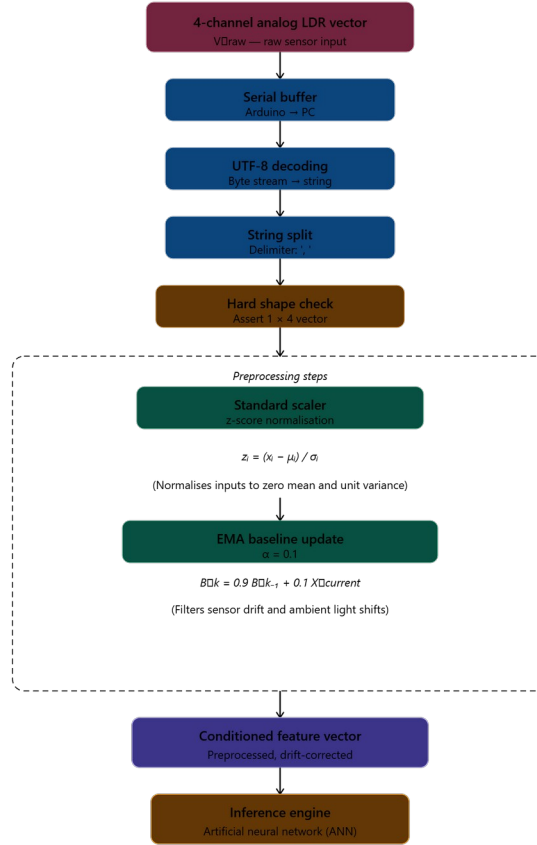


Fig 7.20 – Data Processing Pipeline

7.3.1.1 Ingestion and Shape Validation

The algorithm ingests data via a high-frequency serial buffer (up to 500 Hz). Each reading is pulled as a comma-separated UTF-8 string, which is subsequently decoded and converted into a NumPy array of floating-point values. To maintain data integrity, the system performs a "hard shape check" at every sample interval:

$$(7.1) \quad X_{raw} = [x_1, x_2, x_3, x_4]^T$$

Where X_{raw} is the input vector.

If the incoming vector does not contain exactly four channels corresponding to the physical LDR positions on the end-effector, the frame is rejected to prevent the propagation of erroneous state estimates.

7.3.1.2 Statistical Scaling and Feature Invariance

Because the resistance of an LDR varies significantly between different rooms or times of day, the raw values cannot be used directly for classification. The system utilizes a pre-trained StandardScaler to map the raw features into a normalized space with zero mean and unit variance (z-score normalization):

$$(7.2) \quad z_i = \frac{x_i - \mu_i}{\sigma_i}$$

Where μ_i, σ_i are the mean and standard deviation derived from the training dataset. This transformation ensures that the neural network's hidden layers focus on the spatial gradients of light occlusion (the relative difference between sensors) rather than absolute voltage offsets. This allows the "Good Grip" versus "Bad Grip" distinction to remain stable even as the robot moves through different shadows in the workspace.

7.3.1.3 Adaptive Baseline via Exponential Moving Average (EMA)

To account for sensor drift and gradual shifts in ambient light during the manipulation cycle, the system establishes a dynamic baseline. During the initial acquisition phase, the system averages ten stable reads to define the "Static Baseline." Once the robot enters the active transport phase, this baseline is updated using an Exponential Moving Average (EMA) with a smoothing factor of $\alpha = 0.1$:

$$(7.3) \quad B_k = (1 - \alpha) \cdot B_{k-1} + \alpha \cdot X_{current}$$

Where B_k, B_{k-1} are the updated and previous Baseline vector, $(1 - \alpha)$ is the Persistence Factor and $X_{current}$ is the current Raw Vector

This allows the controller to "track" changing environment conditions without losing the sensitivity required for the LSTM-based slippage detection discussed in later sections. The implementation of this preprocessing logic is detailed in **Pseudocode 6.1**.

7.3.2 Model Architecture

The core of the grasp validation system is a feed-forward Artificial Neural Network (ANN) configured as a Multilayer Perceptron (MLP). This architecture was selected

because it can efficiently approximate the complex, non-linear mapping between raw light-occlusion patterns and the mechanical stability of the adhesive bond. The network serves as a spatial feature extractor, resolving the positional data from the distributed LDR array into a single qualitative state.

7.3.2.1 Architectural Layer Sequence

The network is structured as a sequential pipeline of fully connected (Dense) layers, as illustrated in **Fig 7.17**.

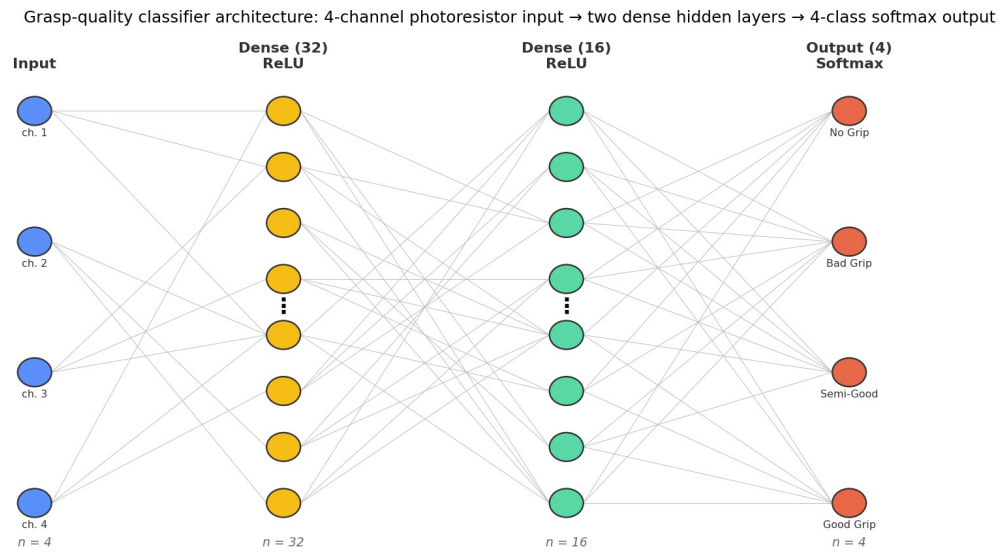


Fig 7.21 – Grasp-quality classifier architecture: 4-channel photoresistor input, two dense hidden layers (32, 16 units, ReLU), 4-class Softmax output

1. **Input Layer:** Consists of 4 neurons, each corresponding to a normalized channel from the photoresistor array (z_1, z_2, z_3, z_4).
2. **Hidden Layers:** The model utilizes two primary hidden layers with a high neuron density (e.g., 32 and 16 units) to resolve the spatial correlations between adjacent sensors. These layers use the **Rectified Linear Unit (ReLU)** activation function:

$$(7.4) \quad f(x) = \max(0, x)$$

ReLU was chosen for its ability to overcome the vanishing gradient problem and ensure rapid convergence, which is evident in the smooth loss curve reaching ≈ 0.45 after 100 epochs.

3. **Output Layer:** Consists of 4 neurons, each representing one of the discrete grasp states (Good, Semi-Good, Bad, No Grasp). This layer utilizes the **Softmax** activation function to produce a probability distribution:

$$(7.5) \quad P(y = j | x) = \frac{e^{z_j}}{\sum_{k=1}^4 e^{z_k}}$$

This probabilistic output allows the controller to establish a "confidence threshold." If the network predicts a "Good Grip" but with a low probability (e.g., < 0.70), the system can autonomously revert to a "Move Slower" policy as a safety precaution.

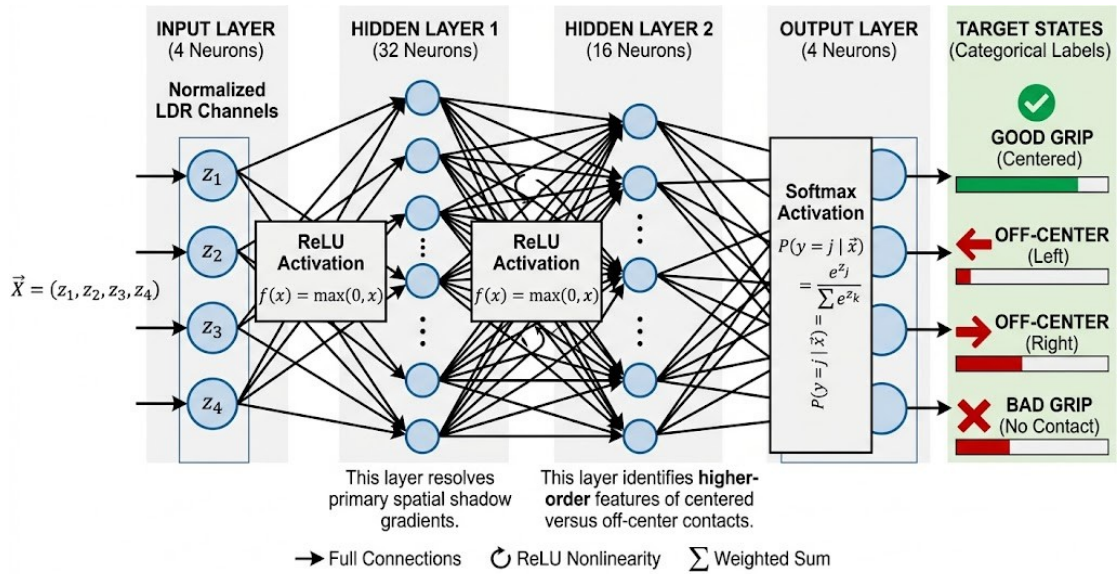


Fig 7.22 – Single Phase NN Architectural Topology

7.3.2.2 Training and Hyperparameter Configuration

The model was implemented using the Keras framework and trained using the **Categorical Cross-Entropy** loss function, which is ideal for multi-class classification tasks. The optimization was performed using the **Adam**

optimizer, selected for its adaptive learning rate properties which contributed to the high validation accuracy of **86.84%** observed in the experimental results. The high degree of convergence seen in the training history (**Figure 6.3**) confirms that the network has successfully generalized the underlying contact physics. By utilizing a "One-Phase" integrated approach rather than separating orientation and quality tasks, the model avoids the cumulative error seen in the "Orientation Model" bottleneck of the Two-Phase attempt, resulting in a more robust gatekeeper for the robot's motion planner.

7.3.3 Classification Logic and Planner Integration

The final stage of the grasp quality classification pipeline is the translation of the neural network's probabilistic output into a discrete robotic behavior. By bridging the gap between high-level perception and low-level motion control, the system ensures that the robot adapts its trajectory based on the actual physical state of the adhesive bond. This integration is managed through a specialized **Strategic Decision Engine**, visualized in the state machine of Figure 6.8.

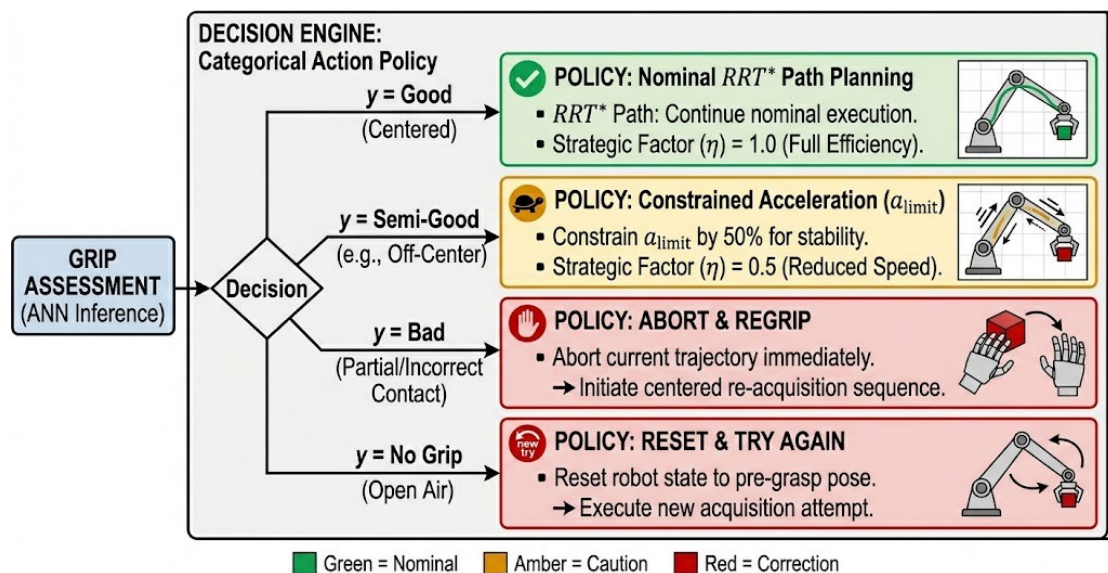


Fig 7.23 – Strategic Decision Engine

7.3.3.1 Strategic State-to-Action Mapping

The real-time algorithm maps the predicted class y from the Single-Phase NN to four distinct mechatronic policies. This logic ensures that the robot never attempts a maneuver that the bond cannot support.

1. **Good Grip (State 3): "Continue Policy"**

The system identifies a centered grasp with optimal contact area. The robot is authorized to execute the RRT^* path at nominal joint velocities, utilizing the full acceleration budget established by the 3M™ 1900 Limit Surface (L_{cm}).

2. **Semi-Good Grip (State 2): "Move Slower Policy"**

The network identifies a bond that is established but exhibits a geometric offset. While transport is permissible, the centering error increases the "tipping moments" described in Section 6.3.3. To maintain stability, the controller implements an **Acceleration Scaling Factor** ($\eta = 0.5$), effectively reducing the permissible a_{limit} by 50% to provide an additional safety buffer against edge peeling.

3. **Bad Grip (State 1): "Regrip Policy"**

A bond is established, but the signature indicates insufficient contact area or unstable edge engagement (e.g., as seen in the 12 semi-good-to-bad misclassifications in Fig 6.2). To prevent a failure during motion, the robot halts its current trajectory, detaches, and performs a "Regrip" manoeuvre to re-acquire the object with better centering.

4. **No Grip (State 0): "Try Again Policy"** The system detects a total establish failure (100% precision in the confusion matrix). The controller resets the end-effector to its pre-grasp configuration and triggers a new acquisition attempt.

7.3.3.2 Mathematical Integration with the Trajectory Planner

The integration with the force-aware planner is enforced by modifying the feasibility inequality. The effective acceleration limit (a_{eff}) for the current manipulation cycle is defined as:

$$(7.6) \quad a_{eff} = \eta(y) \cdot a_{limit}(E, G, h, a)$$

Where a_{limit} is the material-dependent boundary calculated via the Kaelble or GR models, and $\eta(y)$ is the **Strategic Scaling Factor** derived from the NN state y :

$$(7.7) \quad \eta(y) = \begin{cases} 1.0 & \text{if } y = \text{"Good"} \\ 0.5 & \text{if } y = \text{"Semi_Good"} \\ 0.0 & \text{if } y = \{\text{"Bad"}, \text{"No_Grip"}\} \end{cases}$$

This formulation ensures that the characterization results from Chapter 4, specifically the 150MPa Young's Modulus and 0.5MPa Shear Modulus, are dynamically modulated by the real-time quality score to ensure grasp survival.

7.3.3.3 Implementation of the Decision Loop

The real-time integration loop runs synchronously with the robot's state machine. Prior to the first joint movement, the algorithm calls the **initial_grasp_assessment** function, which averages the normalized LDR signatures to produce a high-confidence prediction. The implementation of this decision logic is detailed in **Fig 7.23**. By enforcing these policies, the validation system mitigates the risks associated with the viscoelastic uncertainty of the adhesive. If a "Semi-Good" grasp is identified, the system chooses safety over throughput, ensuring that the resultant force vector remains deep within the conservative center of the Limit Surface, far from the critical energy release rate G_c required for detachment.

7.4 Temporal Slippage Detection (LSTM)

While the grasp quality classification (Section 7.3) ensures a stable entry condition, the viscoelastic nature of the 3M™ 1900 adhesive introduces risks during the active transport phase. Even if the initial acceleration vector resides within the established L_{cm} , long-duration stresses can induce creep or progressive debonding. To mitigate these risks, a dynamic monitoring layer utilizes a Long Short-Term Memory (LSTM) network to identify temporal signatures of bond degradation in real-time.

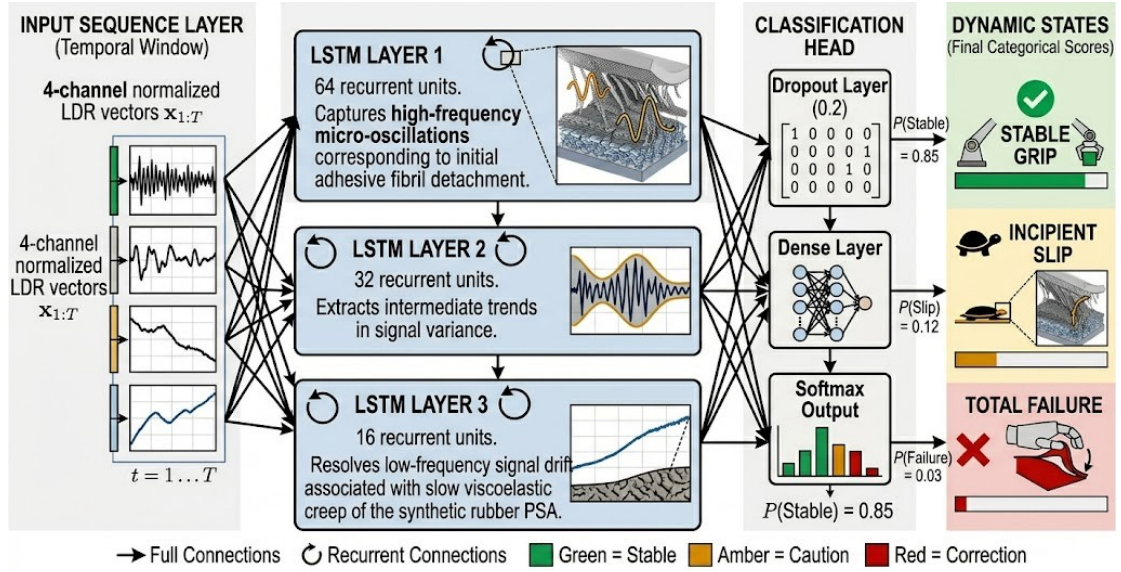


Fig 7.24 – LSTM Architectural Topology

7.4.1 Sequence Modeling for Incipient Slip

Failure in pressure-sensitive adhesives is typically a progressive process rather than a sudden fracture. As established in the shear tests (Section 5.1), PSAs subjected to continuous loads undergo large-scale plastic drawing and fibrillation. Before catastrophic detachment occurs, the adhesive fibrils detach and reattach at the microscopic level, creating unique temporal "signatures" in the LDR signal, such as high-frequency micro-oscillations and slow-frequency signal drift.

A standard feed-forward network is unable to capture these history-dependent patterns. The LSTM architecture is specifically chosen for its ability to maintain a "hidden state" that encodes the temporal evolution of the bond. This allows the system to distinguish between high-frequency electrical noise from the robotic actuators and the genuine "sawtooth" patterns associated with incipient shear-slip.

7.4.2 Recurrent Architecture Design

The integration of the sequence model into the robotic control architecture enables a proactive approach to grasp maintenance. By processing the 4-channel LDR stream through the stacked LSTM layers, the system identifies the onset of debonding before it results in a dropped payload. This subchapter details the logic used to transition from dynamic perception to physical robotic response.

7.4.2.1 Probabilistic Triggering and Confidence Thresholds

The monitoring loop operates at a control frequency of 500 Hz, synchronized with the sensor ingestion pipeline. At every time step 500 Hz, the LSTM produces a probability distribution p_k across the three dynamic states (Stable, Incipient Slip, Failure). To minimize false-positive emergency stops caused by transient electrical noise or severe robot vibrations, the controller enforces a **High-Confidence Trigger**:

$$(7.8) \quad \text{Trigger_Reaction} \quad \text{If} \quad P(\text{State}=1) > \tau_{\text{conf}}$$

Where θ_{conf} is the **Confidence Threshold**, set to 0.85 based on the pre-integration performance testing. This conservative threshold ensures that the robot only deviates from its optimized RRT^* path when the "sawtooth" signatures of adhesive fibrillation are clearly distinguishable from the background noise.

7.4.2.2 Integration with the Adaptive Baseline

To maintain sensitivity during long-duration transport tasks, the monitoring loop continuously references the **Exponential Moving Average (EMA)** baseline established in Section 7.4.1. The LSTM evaluates the **Current Deviation Vector** D_k :

$$(7.9) \quad D_k = X_{\text{current}} - B_k$$

Where B_k is the baseline updated with a smoothing factor of $\alpha=0.1$. If the LSTM identifies a State 1 (Incipient Slip) condition while D_k shows a persistent downward trend in occlusion, the system confirms that the object is shifting away from the gripper center. This synergy between the recursive EMA filter and the recurrent neural memory allows the system to ignore slow environmental lighting changes while remaining highly responsive to rapid adhesive creep.

7.4.2.3 Strategic Recovery Policies

Upon a confirmed detection of **Incipient Slip**, the state machine interrupts the active trajectory and implements one of two mechatronic recovery strategies designed to "heal" the bond:

1. **Dynamic Deceleration:** The joint velocities are immediately capped at 20% of their planned RRT^* values. This reduction in joint speed decreases the centripetal acceleration ($a_c = v^2 \cdot k$) and the associated tangential wrench, allowing the viscoelastic adhesive to stabilize and stop the sliding progression.
2. **Wrist Reorientation (Normal Force Compensation):** The controller calculates the direction of the inertial load vector. The wrist joint (q_3) is then reoriented to align the gripper face perpendicular to this vector. This reorientation shifts the load from a shear/peel state into a compressive state. As establish in Section 5.1.4, PSAs exhibit higher strength in compression; this "healing manoeuvre" effectively re-wets the adhesive interface, restoring the bond's integrity before the trajectory resumes.

The logical flow for this real-time monitoring and recovery loop is detailed in **Fig 7.25**, and the implementation of the decision logic is provided in Pseudocode 6.4.

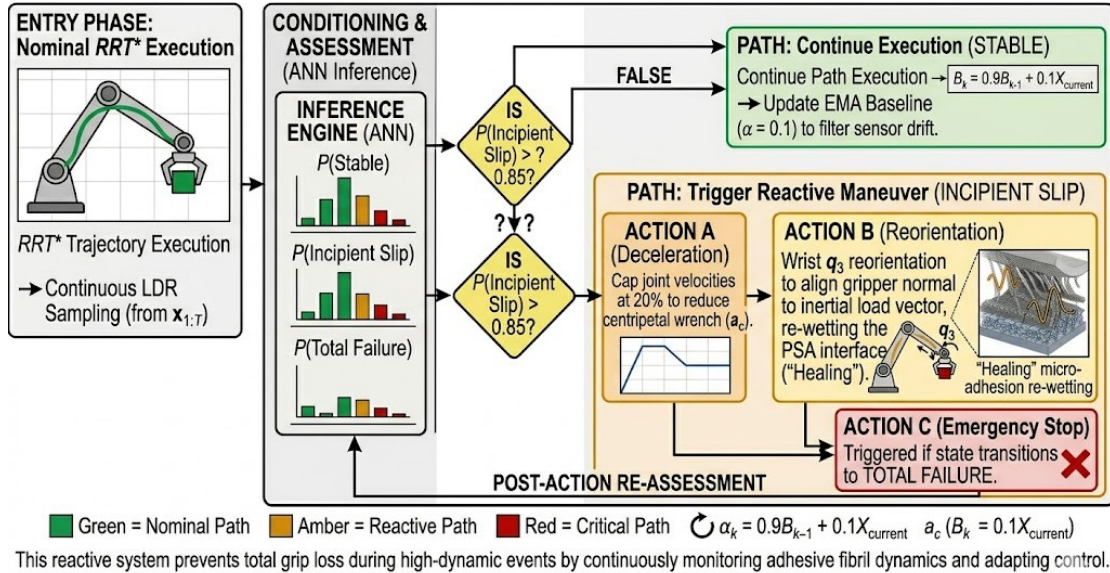


Fig 7.25 – Logical Flow of the Dynamic Monitoring and Recovery Loop

7.4.3 Dynamic State Monitoring and Classification Logic

The monitoring loop operates at the same sampling frequency as the sensor ingestion pipeline (up to 500 Hz). The system classifies the transport phase into three continuous states:

- **Stable Grip (State 0):** Characterized by near-constant sensor values with a low standard deviation. The robot continues its RRT^* trajectory without interruption.
- **Incipient Slip (State 1):** Detected when the LSTM identifies a localized trend of increasing signal variance or a persistent drift from the established EMA baseline $(\alpha = 0.1)$. This state indicates that the bond is approaching the critical energy release rate G_c and failure is imminent.
- **Total Failure (State 2):** Identified by a rapid return of all 4 LDR channels to their baseline (ambient light) values, signifying a complete loss of contact.

7.4.4 Reactive Control and Recovery Strategies

The primary value of the LSTM layer is its ability to trigger **Reactive Maneuvers** when Incipient Slip is identified. Because the system "remembers" the failure progression, it can close the loop before the object is dropped.

Upon detection of State 1, the controller interrupts the current trajectory and implements one of two mechatronic recovery strategies:

1. **Acceleration Reduction:** The joint velocities are immediately capped at a lower fraction of their nominal value to reduce the inertial wrench on the bond.
2. **Wrist Reorientation (Healing Manoeuvre):** The robot reorients the end-effector to shift the resultant force vector from a tensile/shear direction into a compressive direction. This re-application of mechanical pressure helps "heal" the bond by re-wetting the PSA interface before total separation occurs.
3. The implementation of this high-frequency monitoring loop is detailed in **Fig 7.26**.

Algorithm 1 Standalone Real-Time Grip Inference (`run_realtime_inference`)

Require: Trained Keras model path, fitted scaler, serial port, baud rate, grip category list C

Ensure: Continuous console stream of predicted grip category, confidence, and per-prediction latency

```

1:  $M \leftarrow \text{LOADKERASMODEL}(\text{MODEL\_PATH})$ 
2: if  $M$  failed to load then
3:   return
4: end if
5:  $\text{ser} \leftarrow \text{OPENSERIAL}(\text{SERIAL\_PORT}, \text{BAUD\_RATE})$ ; wait 2s for port init
6:  $t_{\text{start}} \leftarrow \text{Now}$ ;  $n \leftarrow 0$ ;  $\{\Delta t_i\} \leftarrow \emptyset$ 
7: while true do
8:   if serial buffer has data then
9:      $t_0 \leftarrow \text{Now}$ 
10:     $\text{line} \leftarrow \text{READSERIALLINE}(\text{ser})$ 
11:    if line is empty then
12:      continue
13:    end if
14:     $\mathbf{r} \leftarrow$  parse line as 4 comma-separated floats
15:    if  $|\mathbf{r}| \neq 4$  then
16:      continue ▷ Skip malformed reading
17:    end if
18:     $\mathbf{x} \leftarrow \text{scaler.transform}(\mathbf{r}.\text{reshape}(1, 4))$ 
19:     $\mathbf{p} \leftarrow M.\text{predict}(\mathbf{x})$ ;  $\hat{c} \leftarrow \arg \max \mathbf{p}$ 
20:     $t_1 \leftarrow \text{Now}$ ; append  $(t_1 - t_0)$  to  $\{\Delta t_i\}$ ;  $n \leftarrow n + 1$ 
21:    print  $n$ ,  $\mathbf{r}$ ,  $C[\hat{c}]$ ,  $\mathbf{p}[\hat{c}]$ ,  $(t_1 - t_0)$ , running mean of  $\{\Delta t_i\}$ 
22:  end if
23: end while
  On exit (KeyboardInterrupt or serial error):
24: print total runtime,  $n$ , mean/min/max of  $\{\Delta t_i\}$ , throughput  $n/\text{runtime}$ 
25: Close serial port

```

Fig 7.26 – Pseudocode: Standalone Real-Time Grip Inference

Algorithm 2 Dynamic Grasp Validation and Graduated Recovery Protocol (`start_grasp_validation`)

Require: Object name (selects an asset-map config: model, scaler, categories, fragility flag, slippage thresholds, quality thresholds)

Ensure: Final protocol action $\in \{\text{REPICK}, \text{REGRASP}, \text{LEVEL3}, \text{CAMERA.CHECK}, \text{PROTOCOL.COMPLETE}, \text{MANUAL.HALT}, \text{RUNTIME.ERROR}\}$

Step 0 – Configuration

```

1:  $\text{profile} \leftarrow \text{LOADCONFIGURATIONPROFILE}(\text{object\_name})$  ▷ Reads {object}.asset_map.txt; loads Keras model +
    $\text{joblib scaler}$ ; validates paths exist
2:  $(M, \text{Sc}, C, \text{IsFragile}, \text{SlipThr}, \text{QualThr}) \leftarrow \text{profile}$ 
3:  $\text{ser} \leftarrow \text{OPENSERIAL}(\text{SERIAL\_PORT}, \text{BAUD\_RATE})$ 

```

Phase 1 – Initial Grasp Assessment (pre-movement)

```

4:  $\text{reads} \leftarrow \emptyset$ 
5: while  $|\text{reads}| < 10$  do ▷ Collect 10 stable sensor samples
6:    $(\text{cat}, \text{conf}, \mathbf{r}) \leftarrow \text{READANDPREDICT}(M, \text{Sc}, \text{ser}, C)$ 
7:   if  $\mathbf{r} \neq \text{None}$  then
8:     append  $\mathbf{r}$  to  $\text{reads}$ 
9:   end if
10:  sleep 50ms
11: end while
12:  $\mathbf{b} \leftarrow \text{mean}(\text{reads})$  ▷ Baseline reading
13: if  $\text{cat} = C[0]$  then ▷ No-grip
14:    $\text{SENDCOMMAND}(\text{ser}, \text{REPICK})$ 
15:   return  $\text{REPICK}$ 
16: else if  $\text{cat} = C[1]$  then ▷ Bad-grip
17:    $\text{SENDCOMMAND}(\text{ser}, \text{REGRASP})$ 
18:   return  $\text{REGRASP}$ 
19: end if

```

Phase 2 – Active Monitoring (during movement)

```

20:  $\text{cycles} \leftarrow 0$ 
21: while true do
22:    $(\text{cat}, \text{conf}, \mathbf{r}_{\text{cur}}) \leftarrow \text{READANDPREDICT}(M, \text{Sc}, \text{ser}, C)$ 
23:   if  $\mathbf{r}_{\text{cur}} = \text{None}$  then
24:     sleep 10ms; continue
25:   end if
26:    $\delta \leftarrow \max \left( \frac{|\mathbf{r}_{\text{cur}} - \mathbf{b}|}{\mathbf{b} + \epsilon} \right)$  ▷ Max relative sensor deviation from baseline

```

Graduated slippage response:

```

27:   if  $\delta > \text{SlipThr.ABORT}$  then
28:      $\text{SENDCOMMAND}(\text{ser}, \text{LEVEL3})$ 
29:     return  $\text{LEVEL3}$  ▷ Fatal slip 2014 immediate abort
30:   else if  $\delta > \text{SlipThr.CRITICAL}$  then ▷ Non-fatal; monitoring continues
31:      $\text{SENDCOMMAND}(\text{ser}, \text{LEVEL2})$ 
32:   else if  $\delta > \text{SlipThr.WARNING}$  then ▷ Non-fatal; monitoring continues
33:      $\text{SENDCOMMAND}(\text{ser}, \text{LEVEL1})$ 
34:   end if

```

NN grip-quality response:

```

35:   if  $\text{cat} = C[0]$  and  $\text{conf} \geq \text{QualThr.NO\_GRIP\_CRITICAL.CONF}$  then
36:      $\text{SENDCOMMAND}(\text{ser}, \text{CAMERA\_CHECK})$ 
37:     return  $\text{CAMERA\_CHECK}$  ▷ Object presumed lost
38:   else if  $\text{cat} = C[1]$  and  $\text{conf} \geq \text{QualThr.BAD\_GRIP\_WARNING.CONF}$  then
39:      $a \leftarrow \text{LOW\_PRESSURE\_RETREAT}$  if  $\text{IsFragile}$  else  $\text{REPOSITION\_GRIP}$ 
40:      $\text{SENDCOMMAND}(\text{ser}, a)$  ▷ Non-fatal; monitoring continues
41:   end if
42:    $\mathbf{b} \leftarrow 0.9 \mathbf{b} + 0.1 \mathbf{r}_{\text{cur}}$  ▷ Exponential moving-average baseline update
43:    $\text{cycles} \leftarrow \text{cycles} + 1$ 
44:   if  $\text{cycles} > 5000$  then ▷ Fail-safe timeout (~50s)
45:     return  $\text{PROTOCOL\_COMPLETE}$ 
46:   end if
47:   sleep 10ms
48: end while

```

Exception handling (wraps the entire protocol):

```

49: on KeyboardInterrupt: return  $\text{MANUAL\_HALT}$  2
50: on other exception: return  $\text{RUNTIME\_ERROR}$ 

```

Fig 7.27 – Pseudocode: Dynamic Grasp Validation and Graduated Recovery Protocol

7.4.5 Physical Hardware Validation

The architecture described above was designed around a four-channel photoresistor array sampled at up to 500 Hz. The physical rig available for validation differs from this design on both counts: two of the four photoresistor channels read a constant 0.0 with zero variance across every recording, and the Arduino streams at a baud rate that limits sampling to roughly 2 Hz rather than 500 Hz. The wrist-reorientation recovery and confidence-threshold triggering described in the preceding sections were not implemented on this hardware. What was tested is therefore a deliberately simplified version of the same core question the LSTM design is built around: whether a short window of consecutive sensor frames classifies grip state more reliably than a single frame, using the two sensor channels that are actually functional on this gripper. Ten recordings of 180 frames (~90 s) each were captured directly from the rig, an empty gripper twice, roughly forty minutes apart, and four objects (a red sponge, a soap holder, a glitter scrubber, and a yellow sponge) gripped once well and once poorly, and are summarized in Table 7.7. Fig 7.28 shows the physical setup: the KUKA arm and mounted gripper, the gripper's internal adhesive interface and wiring, and the four test objects.

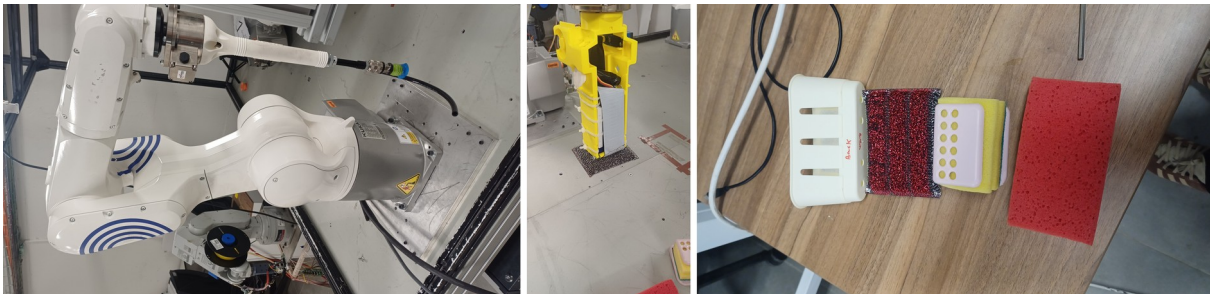


Fig 7.28 – Physical validation setup: KUKA arm and mounted gripper (left), gripper interior showing the adhesive pad, wiring, and servo (center), and the four test objects -- soap holder, glitter scrubber, yellow sponge, red sponge (right)

Table 7.7 – The ten recordings captured from the physical rig

sensor_2 range	sensor_2 mean±std	sensor_1 mean±std	Class	Run
14-16	15.22 ± 0.76	4.88 ± 0.37	no_grip	empty_1
19-21	20.62 ± 0.52	5.71 ± 0.47	no_grip	empty_2
4-7	5.43 ± 0.55	0.51 ± 0.50	good_grip	redsponge_good
6-8	6.86 ± 0.39	0.84 ± 0.38	good_grip	holder_good
6-7	6.71 ± 0.45	0.73 ± 0.45	good_grip	scrubber_good
9-10	9.92 ± 0.28	1.92 ± 0.27	good_grip	yellow_good
17-62	36.53 ± 8.68	11.44 ± 2.17	bad_grip	redsponge_bad
5-26	9.22 ± 4.66	1.26 ± 0.93	bad_grip	holder_bad
7-178	76.48 ± 54.61	12.38 ± 10.71	bad_grip	scrubber_bad
9-30	11.89 ± 3.16	2.48 ± 0.67	bad_grip	yellow_bad

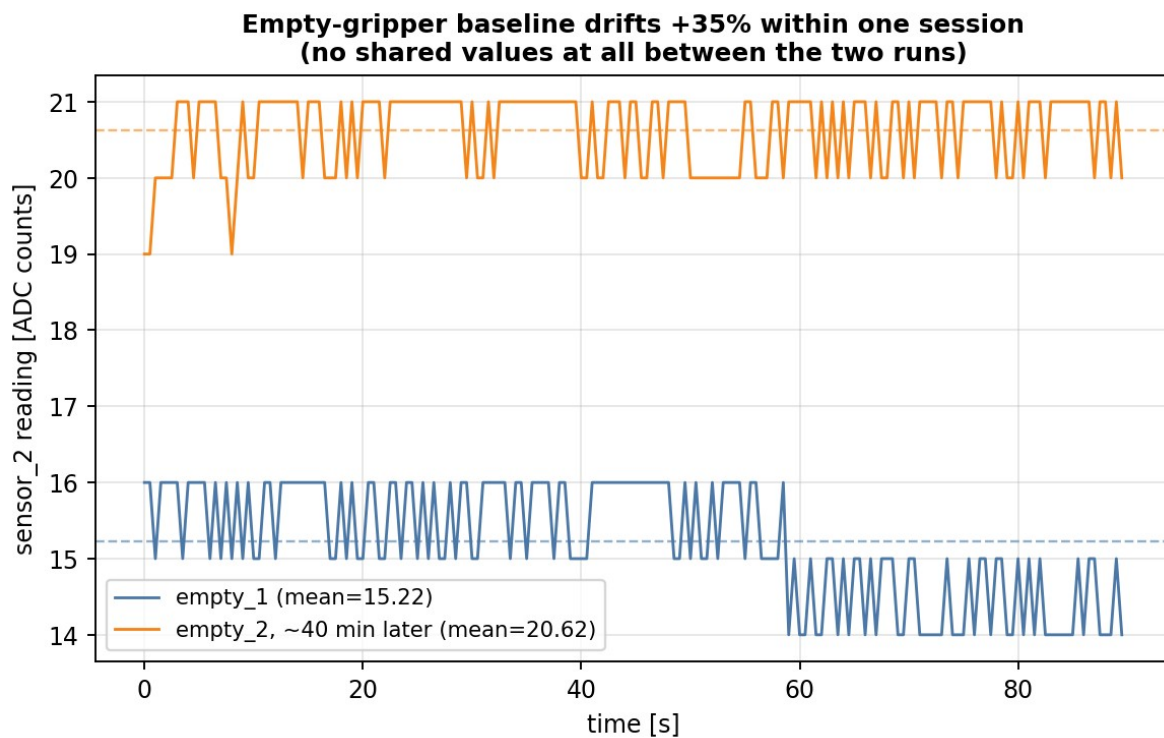


Fig 7.29 – Empty-gripper baseline recorded twice, roughly 40 minutes apart, on the same rig

The two empty-gripper recordings illustrate why the two-phase protocol of section 7.3 re-establishes its baseline immediately before each manipulation rather than relying on a value fixed at design time. sensor_2's mean rises from 15.22 to 20.62, a 35% drift, and the two runs share no common values at all, every frame of empty_1 falls between 14 and 16, every frame of empty_2 between 19 and 21. A threshold calibrated at the start of this session would already be wrong by its end.

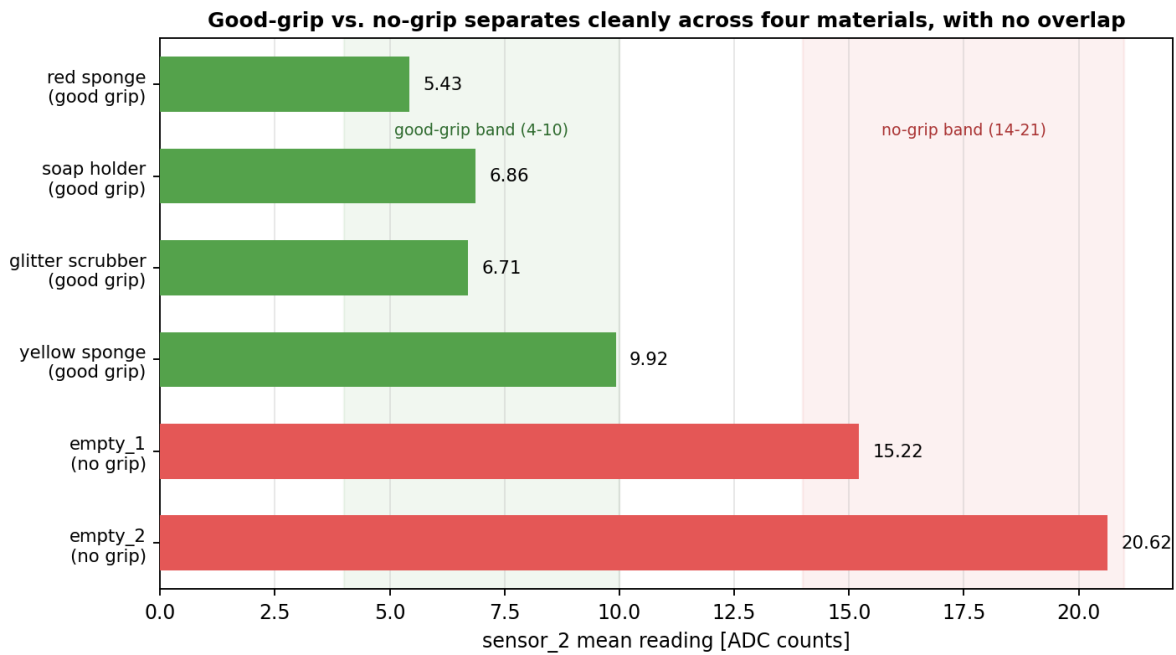


Fig 7.30 – Good-grip and no-grip sensor_2 means across four objects and two empty-gripper runs

Across the four gripped objects, the good-grip and no-grip conditions separate completely on sensor_2, with no overlap at all: every good grip falls between 4 and 10, both empty runs between 14 and 21. The four objects span a compliant sponge, a rigid slotted-plastic holder, a metallic scrubbing pad, and a composite sponge, yet their good-grip means cluster within a factor of 1.8 of each other (5.43 to 9.92). This is consistent with the sensor responding primarily to the gripper's own contact geometry -- how much of the photoresistor's field of view the object occludes at full contact -- rather than to the object's material.

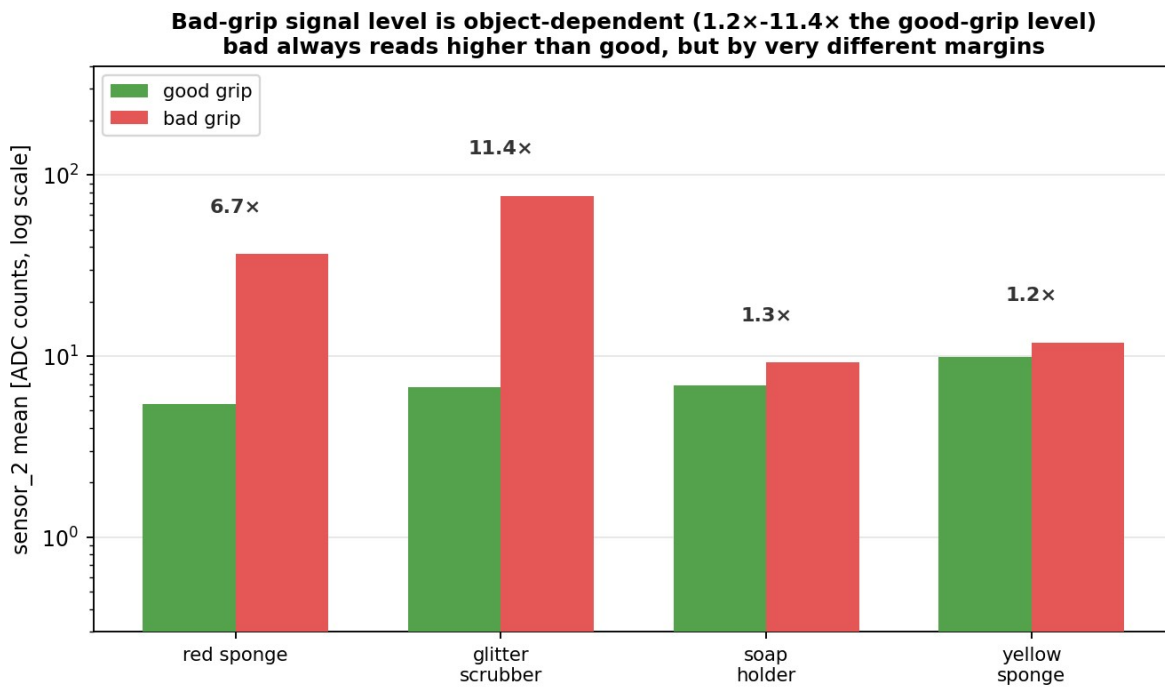


Fig 7.31 – Good-grip vs. bad-grip sensor_2 mean, per object, log scale

The bad-grip condition is less consistent. In every one of the four objects the bad-grip mean reads higher than the corresponding good-grip mean, consistent with a partial or edge-only contact reflecting more light back to the sensor than a full seal blocks. But the margin between good and bad varies by close to an order of magnitude across objects: 11.4× for the glitter scrubber, 6.7× for the red sponge, and only 1.3× and 1.2× for the soap holder and yellow sponge respectively, where the bad-grip recording is closer to a marginal grip than a clearly failed one. A single sensor-value threshold for “bad grip” would need to be calibrated per object to be useful; this independently supports the per-object asset-map thresholds of section 7.3 rather than a single global cutoff.

The central comparison, whether the sequence model generalizes to objects it has not seen, which is what the design in this section depends on, was tested by training both a single-frame classifier and the 15-frame LSTM on the red sponge and soap holder only, then evaluating both, unmodified, on the glitter scrubber and yellow sponge. Table 7.8 and Fig 7.32 report the result: the single-frame model reaches 56.6% accuracy and the sequence model 75.7%, a 19-point gain from the same two sensor channels and the same underlying data, differing only in whether the classifier sees one frame or fifteen.

Table 7.8 – Per-class accuracy, unseen objects (trained on red sponge + soap holder, tested on glitter scrubber + yellow sponge)

LSTM (15-frame window)	Dense (single-frame)	Class
56 / 56 correct	0 / 180 correct	no_grip
112 / 112 correct	360 / 360 correct	good_grip
44 / 112 correct	149 / 360 correct	bad_grip
75.7%	56.6%	Overall accuracy

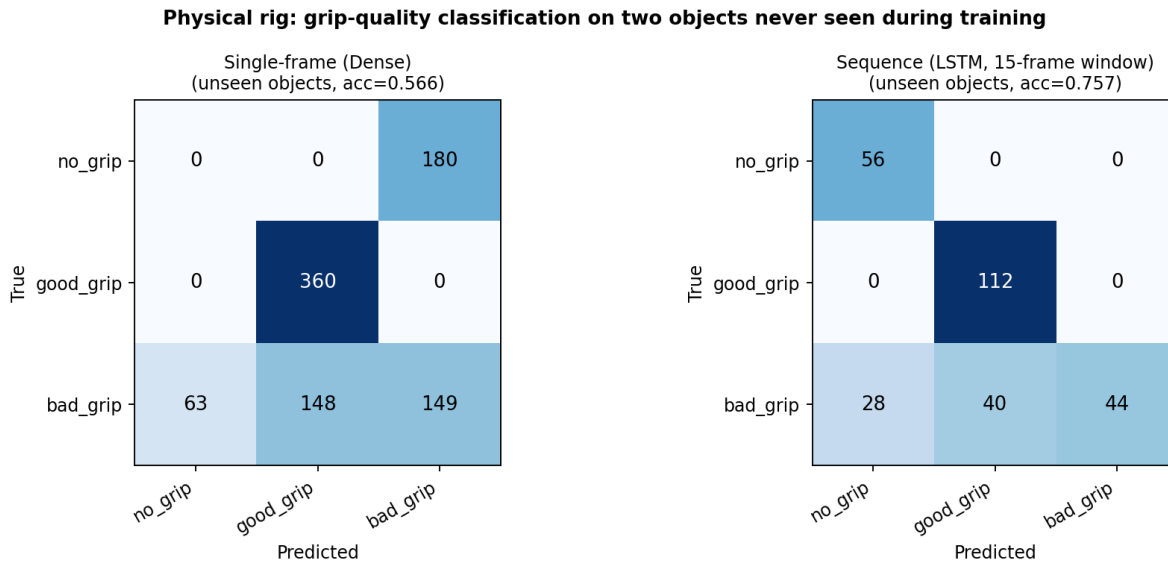


Fig 7.32 – Confusion matrices, single-frame vs. sequence model, unseen objects

The gap is concentrated entirely in the no_grip class. The single-frame model misclassified every one of the 180 no_grip test frames, calling all of them bad_grip. This follows directly from the baseline drift result above: the model learned a fixed empty-gripper value from empty_1 during training (mean 15.2) and, at test time, met empty_2's drifted baseline (mean 20.6), a value it had never seen labelled no_grip and therefore assigned to whichever class its decision boundary placed it in. The sequence model scored 56 out of 56 on the identical frames. It cannot use the absolute sensor level to make this decision either, since it also never saw the drifted baseline during training; what it evidently uses instead is the shape of a 15-frame window, and an empty gripper is flat and low-variance regardless of what level it sits at. This is the specific mechanism the architecture of this section was designed to exploit, and this is the first direct evidence that it does so on physical hardware, even in this reduced two-channel form.

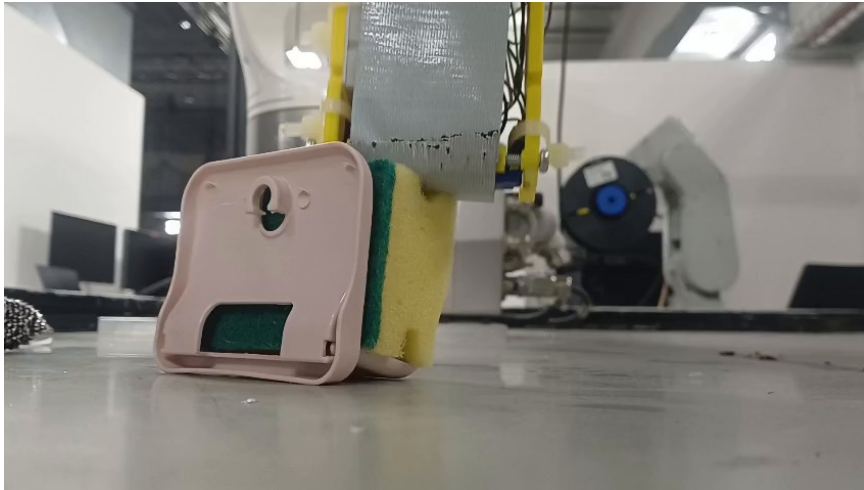


Fig 7.33 – Frame from a recorded trial: the gripper approaching the soap-holder and sponge assembly during a live capture

Several limitations qualify these results. Half of the designed sensor array is non-functional on this rig, so every result above uses two channels rather than the four the architecture assumes. Each condition was captured once; these are point estimates from a pilot test, not a repeated-measures study, and no measure of run-to-run variation is available. The holder_bad recording drifted from 5 to 26 with a mean (9.22) that overlaps yellow_good's (9.92), suggesting it may be closer to a marginal grip than a clearly failed one, which likely depresses the reported bad_grip recall. That recall -- 39% for the LSTM, chance-level for the single-frame model -- indicates that distinguishing bad from good grip is not yet reliably learnable from two channels, even though distinguishing gripped from not-gripped is. These results characterise the rig in its current, degraded state; they validate the sequence-modelling principle behind this section's design, but they do not validate the four-channel, 500 Hz architecture described above it, which remains untested.

8. The Full System

8.1 Overview

The preceding chapters developed three components in isolation: a characterization of the adhesive's mechanical limits (Chapter 5), a force-constrained trajectory planner that respects those limits (Chapter 6), and a sensor-based grasp validation system that monitors the bond during motion (Chapter 7). This chapter describes how they combine into a single manipulation system, and states precisely which parts of that combination have been demonstrated, and which remain designed but untested.

The organizing principle is a separation of enforcement in time. The adhesive bond can fail in two distinct ways, and each requires a different remedy: motion that demands more force than the bond can supply is predictable in advance and is handled offline, by the planner rejecting it before execution; unexpected degradation of the bond - poor initial seating, creep, disturbance - is only detectable at runtime, and is handled online, by the monitor detecting it and triggering recovery. A planner alone has no knowledge of how well the adhesive seated on a particular object; a monitor alone can only react after the bond has already been overloaded. The two are complementary rather than redundant.

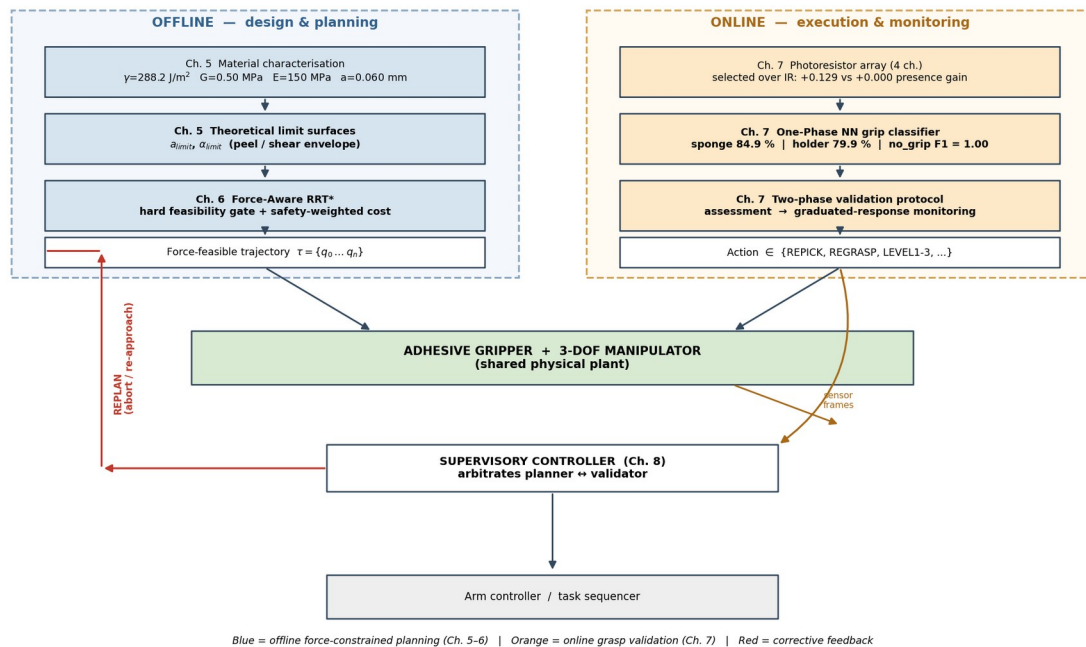


Fig 8.1 – Full system architecture

8.2 The Shared Physical Basis

The most important structural property of the system is that the offline constraint and the online alarm thresholds are not independent heuristics. Both derive from the same measured adhesive properties, established in Chapter 5.

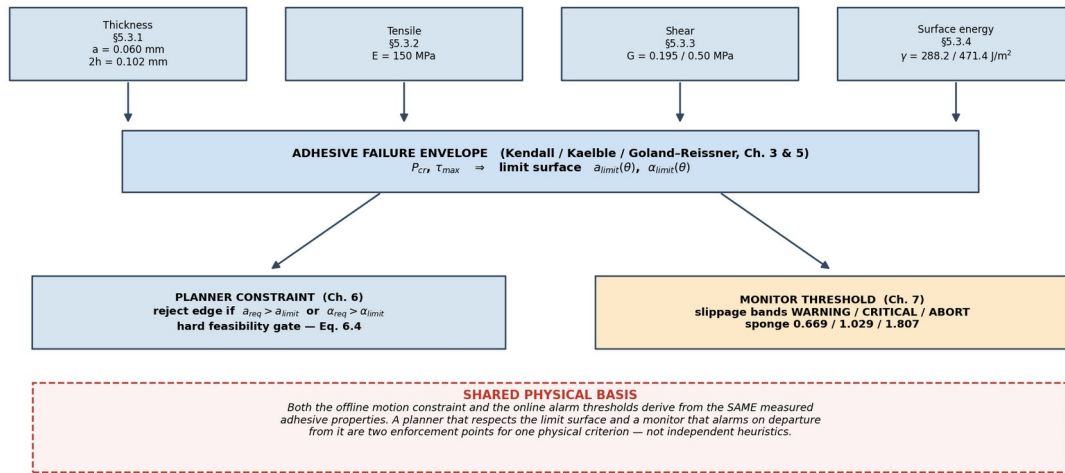


Fig 8.2 – Propagation of measured material properties into runtime parameters

Table 8.1 – Measured adhesive parameters (3M™ Value Duct Tape 1900)

Property	Symbol	Value	Source
Adhesive layer thickness	a	0.060 mm	5.3.1
Backing thickness	$2h$	0.102 mm	5.3.1
Backing elastic modulus	E	150 MPa	5.3.2
Shear modulus (static)	G	0.195 MPa	5.3.3
Shear modulus (adopted, dynamic)	G	0.50 MPa	5.3.3
Surface energy, uncoated cardboard	γ	288.2 J/m ²	5.3.4
Surface energy, coated cardboard	γ	471.4 J/m ²	5.3.4

These feed the peel and shear models of Chapter 3 to produce a limit surface: the envelope of linear and angular accelerations the bond can sustain in each orientation. That surface then serves two consumers. The planner (8.3) uses it as a hard rejection criterion: any candidate trajectory edge requiring an acceleration beyond the limit surface is discarded outright. The monitor (8.4) uses it as the physical justification for its graduated slippage bands: a departure from the established sensor baseline is interpreted as the bond moving toward that same envelope. A planner that respects the limit surface and a monitor that alarms on departure from it are therefore two enforcement points for one physical criterion, not two independent heuristics. Caveat: this coupling is architectural. In the present implementation the slippage thresholds were derived empirically from recorded sensor data rather than analytically from the limit surface. Establishing the quantitative mapping between fraction of limit surface consumed and expected sensor deviation is identified as future work.

8.3 Offline Stage: Force-Constrained Planning

Given a start and goal configuration and an obstacle set, the Force-Aware RRT* of Chapter 6 produces a trajectory that is collision-free, kinematically admissible and force-feasible for the adhesive bond. Its two-layer structure maps directly onto the two roles identified above: a hard constraint layer (collision checking, joint limits, and the adhesive feasibility gate of Eq. 6.4 and Eq. 6.12) that guarantees feasibility with respect to the modelled limit surface, and a soft cost layer that, among feasible edges, favors those with the greatest margin inside the envelope. The planner runs once per task, offline, before any motion begins.

Unresolved discrepancy: the cost function stated in Eq. 6.2,

$$J(\sigma) = \left(\frac{1}{L} \right) \int [1 + \lambda R(a(s))] ds$$

, describes a continuous risk-weighted path integral,

but the cost function implemented, Eq. 6.12, $J = \sum_i [w_{time} \cdot t_i + w_{dist} \cdot d_i + w_{advance} \cdot a_i]$, is a discrete sum with no explicit risk term $R(a)$ at all. The results reported in §6.5 use the implemented form. Reconciling the stated integral formulation with the implemented discrete form is required before the planner's cost formulation is quoted as definitive.

8.4 Online Stage: Grasp Validation

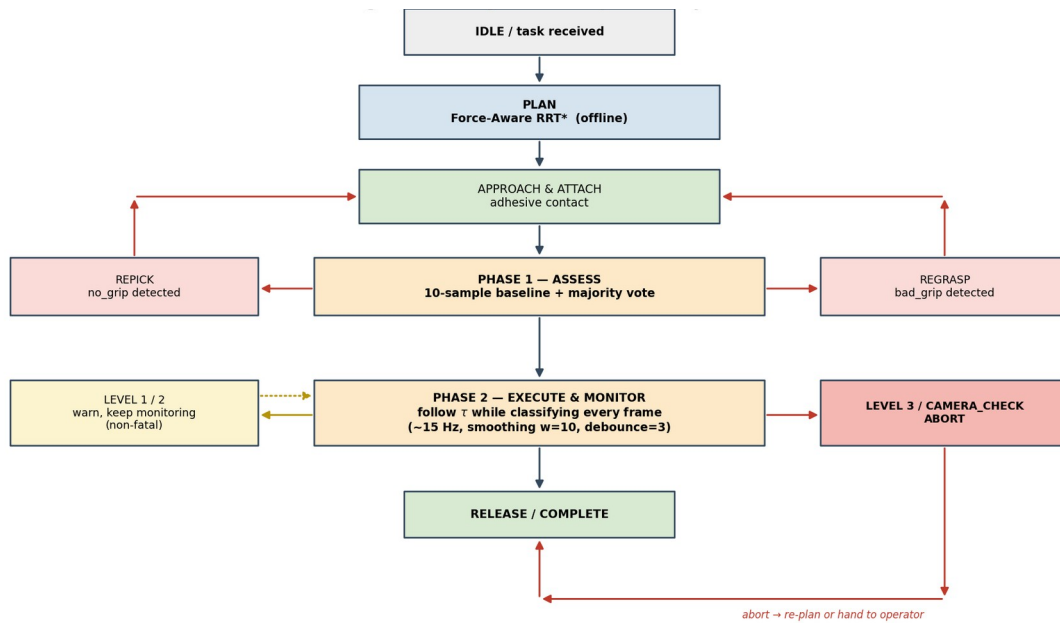
While the trajectory executes, the validation system of Chapter 7 runs continuously, in two phases. Phase 1 (pre-motion assessment) collects ten sensor frames with the object gripped but stationary; these establish the baseline vector, and a majority vote over their classifications decides whether the grasp is fit to move at all. A `no_grip` verdict returns `REPICK`; a `bad_grip` verdict returns `REGRASP`. Phase 2 (execution monitoring) classifies every subsequent frame and compares it against the baseline along two independent paths - a slippage path using the graduated bands, and a quality path using the neural classifier - both debounced over several consecutive frames, since the classifier is 84.9% (sponge) / 79.9% (holder) accurate per frame and acting on single frames produced spurious recovery commands during healthy grasps in simulation.

The Phase 1 baseline capture is significant beyond its obvious role. Because it is taken immediately before each manipulation, under whatever ambient lighting happens to prevail at that moment, all subsequent monitoring is relative to current conditions - which matters because 7.4.6 showed that absolute photoresistor calibration does not transfer between lighting regimes (a model trained under direct illumination scored 0.216 against a 0.500 baseline when applied under nominal lighting). Baseline-relative sensing is the system's structural defense against exactly that failure.

Table 8.2 – Runtime parameters per object

Parameter	Sponge	Holder	Origin
Slippage WARNING	0.669	0.630	p90 of healthy-grip deviation
Slippage CRITICAL	1.029	0.951	p99 of healthy-grip deviation
Slippage ABORT	1.807	1.396	max healthy-grip deviation
NO_GRIP confidence floor	0.90	0.90	no_grip F1 = 1.00 → high bar safe
BAD_GRIP confidence floor	0.60	0.60	weakest class → lower bar
Smoothing window W	10	10	cuts noise-floor p99 from 3.10 → 1.03
Consecutive hits N	3	3	suppresses per-frame

8.5 The Integrated Manipulation Cycle



Blue = planning Green = physical action Orange = validation Red = recovery Yellow = non-fatal warning

Fig 8.3 – Integrated manipulation cycle

Algorithm 1 Integrated Manipulation Cycle (supervisory controller)

Require: Object identity o , start configuration q_{start} , goal configuration q_{goal} , obstacle set $\mathcal{O}$, measured adhesive properties $\mathcal{M} = \{\gamma, G, E, a, 2h\}$

Ensure: Terminal status $\in \{\text{SUCCESS}, \text{ABORTED}, \text{PLANNING-FAILED}, \text{MAX-RETRIES}\}$

```
— Offline stage (Ch. 5-6) —
1:  $(a_{limit}, \alpha_{limit}) \leftarrow \text{LIMITSURFACE}(\mathcal{M}, \text{gripper geometry, object mass})$  ▷ Eq. 6.4
2:  $\text{profile} \leftarrow \text{LOADCONFIGURATIONPROFILE}(o)$  ▷ model, scaler, slippage & quality thresholds
3:  $\text{retries} \leftarrow 0$ 

4: while  $\text{retries} \leq R_{max}$  do
5:    $\tau \leftarrow \text{FORCEAWAREERTSTAR}(q_{start}, q_{goal}, \mathcal{O}, a_{limit}, \alpha_{limit})$  ▷ Alg. 6.x
6:   if  $\tau = \text{FAILURE}$  then
7:     return PLANNING-FAILED
8:   end if

— Physical attachment —
9:   EXECUTEAPPROACH( $\tau_0$ ); ATTACHADHESIVE

— Phase 1: pre-motion assessment (Ch. 7) —
10:   $(\mathbf{b}, s) \leftarrow \text{INITIALGRASPASSESSMENT}(\text{profile})$  ▷ 10-sample baseline, majority vote
11:  if  $s = \text{REPICK}$  or  $s = \text{REGRASP}$  then
12:    EXECUTERECOVERY( $s$ );  $\text{retries} \leftarrow \text{retries} + 1$ 
13:    continue ▷ re-attach, then re-plan from the new pose
14:  end if

— Phase 2: execute  $\tau$  under continuous validation —
15:   $k \leftarrow 0$ 
16:  while  $k < |\tau|$  do
17:    COMMANDJOINTTARGET( $\tau_k$ )
18:     $\text{action} \leftarrow \text{MONITORSTEP}(\text{profile}, \mathbf{b})$  ▷ one frame: classify + slippage band
19:    if  $\text{action} \in \{\text{LEVEL3}, \text{CAMERA\_CHECK}\}$  then
20:      HALTMOTION
21:       $\text{retries} \leftarrow \text{retries} + 1$ 
22:      goto RECOVER ▷ fatal: abort this attempt
23:    else if  $\text{action} \in \{\text{LEVEL1}, \text{LEVEL2}\}$  then
24:       $\tau \leftarrow \text{DERATETRAJECTORY}(\tau, k)$  ▷ scale remaining  $a_{req}$  toward  $a_{limit}$ ; motion continues
25:    else if  $\text{action} \in \{\text{REPOSITION\_GRIP}, \text{LOW\_PRESSURE\_RETREAT}\}$  then
26:      EXECUTEINLINECORRECTION( $\text{action}$ ) ▷ non-fatal; monitoring continues
27:    end if
28:     $\mathbf{b} \leftarrow \lambda \mathbf{b} + (1 - \lambda) \mathbf{r}_{smoothed}$  ▷ EMA baseline update,  $\lambda = 0.9$ 
29:     $k \leftarrow k + 1$ 
30:  end while
31:  RELEASEOBJECT
32:  return SUCCESS

33:  RECOVER:
34:  EXECUTERECOVERY( $\text{action}$ )
35:   $q_{start} \leftarrow \text{CURRENTCONFIGURATION}$  ▷ re-plan from wherever the arm actually stopped
36: end while
37: return MAX-RETRIES
```

Fig 8.4 – Integrated supervisory controller (pseudocode)

Algorithm 2 MonitorStep – single validation frame

```

1: function MONITORSTEP(profile, b)
2:   ( $c, p, \mathbf{r}$ )  $\leftarrow$  READANDPREDICT(model, scaler, serial)
3:   if  $\mathbf{r} = \text{NONE}$  then
4:     return NONE
5:   end if
6:    $\mathbf{r}_{sm} \leftarrow$  mean of last  $W$  raw frames  $\triangleright W = 10$ ; suppresses the noise floor
7:    $\delta \leftarrow \max(|\mathbf{r}_{sm} - \mathbf{b}| / (\mathbf{b} + \epsilon))$ 
8:   Graduated slippage response (debounced over  $N$  frames):
9:   if  $\delta > \Theta_{ABORT}$  for  $N$  consecutive frames then
10:    return LEVEL3
11:  else if  $\delta > \Theta_{CRIT}$  for  $N$  consecutive frames then
12:    return LEVEL2
13:  else if  $\delta > \Theta_{WARN}$  for  $N$  consecutive frames then
14:    return LEVEL1
15:  end if
16:  Neural-network grip-quality response (same debounce):
17:  if  $c = \text{NO\_GRIP}$  and  $p \geq \Theta_{noGrip}$  for  $N$  frames then
18:    return CAMERA_CHECK
19:  else if  $c = \text{BAD\_GRIP}$  and  $p \geq \Theta_{badGrip}$  for  $N$  frames then
20:    return LOW_PRESSURE_RETREAT if fragile else REPOSITION_GRIP
21:  end if
22:  return NOMINAL
23: end function

```

Fig 8.5 – MonitorStep: single validation frame (pseudocode)

The supervisory controller arbitrates between planner and validator. The coupling is deliberately asymmetric, reflecting the severity of each condition.

Table 8.3 – Action coupling between validation and planning

Validator output	Severity	System response	Re-plan?
REPICK	Pre-motion, no grip	Release, re-approach	Yes – from new pose
REGRASP	Pre-motion, poor grip	Detach, re-seat adhesive	Yes – from new pose
LEVEL1 (warning)	In-motion, minor	De-rate remaining trajectory	No
LEVEL2 (critical)	In-motion, significant	De-rate more aggressively	No
LEVEL3 (abort)	In-motion, fatal	Halt immediately	Yes – from halt pose
CAMERA_CHECK	Object presumed lost	Halt, external verification	Yes
REPOSITION_GRIP	Degrading quality	Inline correction	No
LOW_PRESSURE_RETREAT	Degrading, fragile object	Gentle inline correction	No

LEVEL1 and LEVEL2 deliberately do not stop motion: they de-rate the remaining trajectory, scaling commanded accelerations further inside the limit surface, and monitoring continues. Only LEVEL3 and CAMERA_CHECK are terminal. This exists because stopping a manipulation is itself costly, and the adhesive's viscoelastic behavior (5.3.3) means a marginal bond may well survive a gentler continuation.

Open design question (carried from §7): there is no automatic escalation from repeated LEVEL2 warnings to LEVEL3. A bond that sits persistently in the CRITICAL band generates warnings indefinitely without ever triggering an abort. Whether repeated LEVEL2 should escalate, and after how many occurrences, is unresolved.

8.6 Timescale Separation

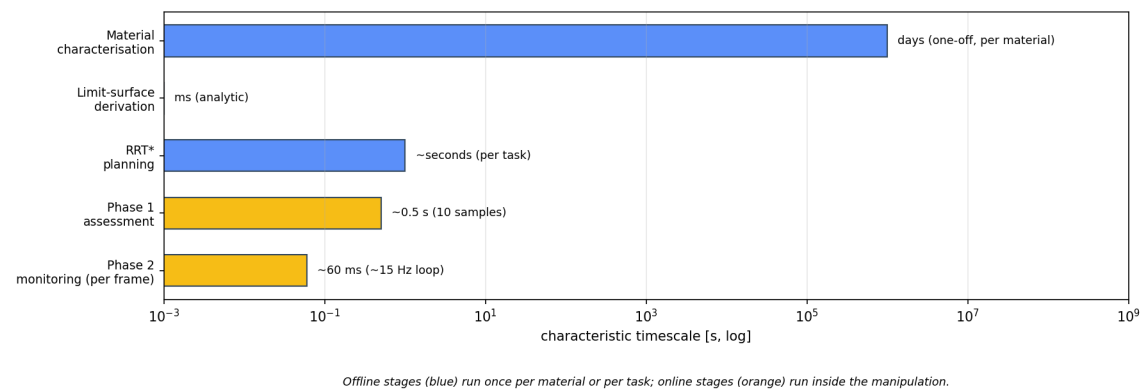


Fig 8.6 – Timescale separation across the system

Table 8.4 – Characteristic timescales

Stage	Timescale	Frequency
Material characterization	days	once per adhesive/substrate pair
Limit-surface derivation	milliseconds (analytic)	once per object/gripper
RRT* planning	~seconds	once per task
Phase 1 assessment	~0.5 s (10 frames)	once per manipulation
Phase 2 monitoring	~60 ms per frame (~15 Hz)	continuous during motion

The measured inference latency of ~60 ms per frame is the binding real-time constraint. At ~15 Hz, with a debounce of N = 3 frames, the minimum latency between a genuine

fault and a fatal command is approximately 200 ms. For the safety-derated velocities of Chapter 6 this is acceptable, but it is a hard floor on reaction time (9.3.3).

8.7 Implementation and Validation Status

The subsystems differ substantially in maturity, and presenting the system as a uniformly validated whole would misrepresent the work. Table 8.5 states the position directly.

Table 8.5 – Validation coverage by subsystem

Subsystem	Status	Evidence
Material characterization (Ch. 5)	Experimentally validated	Thickness, tensile, shear, surface-energy tests; γ cross-checked against literature
Sensor selection (Ch. 7)	Experimentally validated	IR vs photoresistor, 3 objects $\times$ 3 orientations, + harsh-lighting arm 84.9% / 79.9% on held-out test sets; models verified to reload from disk
Grip classifier (Ch. 7)	Experimentally validated	Exercised end-to-end against recorded data via mock serial; REPICK/monitoring paths confirmed
Validation protocol logic (Ch. 7)	Validated in simulation	50 documented successful simulation runs (6.5); ~1/3 of attempted scenarios still fail on detour-heavy queries
Force-Aware RRT* (Ch. 6)	Executable, partially validated	No end-to-end hardware trial with planner and validator coupled
Full closed loop (Ch. 8)	Designed, not demonstrated	

Two points deserve emphasis. First, the grasp validation half of the system is substantially further advanced than the planning half: the classifier is trained, saved and verified; the protocol has been exercised end-to-end against recorded sensor data; the thresholds are derived from measurements rather than assumed. The planner, by contrast, only recently became executable at all (6.6), and roughly a third of attempted planning scenarios still fail on detour-heavy queries. Second, the closed loop has not been demonstrated: every interface in Fig 8.1 is specified, and the supervisory controller is written against those interfaces, but planner and validator have not been executed together against real hardware. The architecture of this chapter should therefore be read as a design that the component results support, not as a validated system-level result.

8.8 Summary

The system couples offline force-constrained planning to online grasp validation through a shared physical basis: the adhesive limit surface derived from measured material properties. The planner prevents motions that are known in advance to overload the bond; the validator detects degradation that no model could have predicted and returns graduated recovery actions whose severity determines whether the trajectory is de-rated, or aborted and re-planned.

The component results supporting this architecture are established to differing degrees. The material characterization, the sensor selection and the grip classifier rest on experimental measurement. The validation protocol has been verified in simulation against recorded data. The planner is now executable and has produced 50 documented successful trajectories, though a meaningful fraction of scenarios still fails. The closed loop itself has not been demonstrated, and Chapter 9 treats the hardware integration as priority work.

Practically, this architecture is designed to slot into a specific point in Pickcommerce AI Robotics' PickoBot system, the autonomous pick-and-pack platform this thesis's gripper hardware was developed for and validated on (Chapter 4). PickoBot's Physical AI Engine already performs per-item gripper selection across four modalities -- vacuum, fingers, magnetics, and Pickcommerce's patented adhesive end-effector -- choosing which to use for a given SKU before a pick begins. The system developed here

is intended to become the control layer for that fourth case: once PickoBot's gripper-selection stage commits to the adhesive modality for a given item, the Force-Aware RRT* planner would consume the object's measured mass and pose from PickoBot's existing computer-vision pipeline to certify a feasible trajectory, and the grasp-validation system would report each pick's outcome -- REPICK, REGRASP, or a graduated recovery action -- back to PickoBot's own supervisory controller in place of the simplified stand-in used throughout this chapter. This places the integration boundary at two existing PickoBot interfaces, object pose/mass output and pick-outcome input, rather than requiring changes to its vision or packing-optimization layers. No such integration has been implemented or tested; it is stated here as the intended next step, consistent with the closed-loop hardware validation identified as .priority work in Chapter 9

9. Future Work

Four directions for future work follow directly from the limitations identified in this thesis. First, the grasp validation system of Chapter 7 was fitted onto an existing gripper design, with its four-channel photoresistor array placed empirically rather than as a first-class input to the gripper's mechanical design; a natural next step is to design the gripper around the sensing requirements the validation system depends on, which would likely also change what the Chapter 7 classifier needs to learn. Second, the force-to-acceleration derivation of section 5.2, though corrected extensively during this work (Chapter 6 documents thirteen implementation defects), has not been experimentally validated: the predicted a_{limit} and α_{limit} have not been checked against the acceleration at which the adhesive bond actually fails on a real gripper, which a direct test, accelerating a gripped object toward and beyond the predicted limits while recording the true point of failure, would establish. Third, the Force-Aware RRT* planner itself requires two further steps: extending the planner and its underlying limit-surface derivation from two to three dimensions, since the current formulation is planar and cannot yet be applied to general pick-and-place motion; and testing the planner on the physical manipulator and gripper described in Chapter 5, since its fifty documented successful trajectories (section 6.5) were generated and evaluated entirely in simulation. Finally, and most consequentially, the full system integrating the offline planner with the online validator has not been demonstrated in closed loop on physical hardware, as Chapter 8 states explicitly; closing this gap, ideally through a staged protocol moving from nominal completion, to induced mid-motion slippage, to initial-grasp failure, is the only experiment that would test the central claim of the integrated design -- that a force-feasible plan combined with a validator able to detect and respond to unexpected departures from it produces a more reliable manipulation system than either component alone.

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תקציר

תפסניות קצה דביקות מציעות חלופה רב-תכליתית למלקחיים מכניים, תפסניות ואקום ומגנטיים עבור איסוף והנחה רובוטי, אך בשונה ממהדק מכני, כושר האחיזה של קשר דביק הוא פונקציה דינמית של הכוחות שהתנועה של הזרוע עצמו מפעילה עליו. עבודה זו מפתחת ומעריכה מערכת מניפולציה הבנויה סביב אילון זה, המשלבת אפיון חומרי אמפירי, תכנון מסלול מודע-כוח, ואימות אחיזה בזמן אמת.

התכונות המכניות של הדבק המשמש לאורך עבודה זו (סרט 3M™ Value Duct Tape 1900) נמדדו ישירות ולא נלקחו מגיליון נתונים: עובי, מודול מתיחה (150MPa), מודול גזירה, ואנרגיית פנים על גבי משטחי קרטון לא מצופים (288.2 J/m^2) ומצופים (). ערכים נמדדים אלו מועברים כפרמטרים למתכנן Force-Aware RRT* (Rapidly-exploring Random Tree) (Star), הדוחה תנועות מועמדות שהאצה או הוצאת האנרגיה הנדרשות מהן יחרגו מהגבול המאופיין של הדבק. המתכנן הוערך על מאגר של חמישים מסלולים מוצלחים מודמים כאשר המטען הועלה ל-58-96% מקיבולת המתיחה הסטטית של הדבק, שבו האילון מודע-הכוח דוחה בפועל כרבע מהתנועות המועמדות ומוכח כמכוון את החיפוש.

מכיוון שתוכנית טובה היא רק כמו ההנחות שעליהן היא נבנתה, עבודה זו מפתחת גם מערכת אימות אחיזה המנתרת את הקשר במהלך הביצוע במקום להסתמך על התוכנית בלבד. שתי אופני חישה מועמדים, החזרת אינפרה-אדום וחסימת פוטו-נגד, הושאו כמותית על פני שלושה עצמים ובתנאי תאורה רגילים ותאורה ישירה כאחד; מערך הפוטו-נגדים נבחר על בסיס גודר-אפקט גדול משמעותי יותר לעומת בסיס של מלקחיים ריקים (Cohen's d עד 1.70, לעומת ערכים זניחים עבור אינפרה-אדום) ושימוש ב-87.7% מטווח הדינמי הזמין של החיישן לעומת 16.6% עבור אינפרה-אדום. מסווג רשת עצבית שאומן על מערך חיישנים זה מבחין בין ארבע קטגוריות איכות אחיזה בדיוק בדיקה של 84.9% ו-79.9% על שני עצמי בדיקה, ומזהה איבוד מוחלט של אחיזה בהיזכרות מושלמת בשניהם. פרוטוקול דו-שלבי הבנוי על מסווג זה מעריך את איכות האחיזה לפני תחילת התנועה ומיישם תגובה מדורגת, מהפחתת קצב המסלול ועד עצירה מיידית, להידרדרות המתגלה במהלך הביצוע, כאשר הספים נגזרים מסטטיסטיקה של נתוני החיישן המתועדים ולא מונחים מראש.

רכיבים אלו משולבים לכדי ארכיטקטורת מערכת אחת שבה המתכנן ה-offline והמנתר המקוון חולקים בסיס פיזיקלי אחד: אותו משטח גבול דביק נמדד המגביל את התנועה המתוכננת מבסס גם את ספי ההתראה של המנתר. המתכנן והמנתר אופיינו ואומתו כל אחד בנפרד, המתכנן בסימולציה, המנתר מול נתוני חיישן מתועדים, אך פעולתם המשולבת על גבי חומרה פיזית טרם הודגמה, ועבודה זו מציינת מגבלה זו במפורש במקום לרמוז על תוצאה שלא

הושגה. השאלות הפתוחות הנותרות, יחד עם צעדי ההמשך הנדרשים לסגירתן, מפורטים
בפרק המסכם.



אוניברסיטת בן גוריון בנגב

הפקולטה למדעי ההנדסה

המחלקה להנדסת מכטרוניקה

תכנון, עיצוב ושיפור של תפסנית דביקה

חיבור זה מהווה חלק מהדרישות לקבלת תואר מגיסטר בהנדסה

מאת: יואב מן מנהיים

מנחה: פרופ' אמיר שפירא

.....תאריך:

.....תאריך:

.....חתימת המחבר:

.....חתימת המנחה:

.....חתימת יו"ר ועדת תואר שני מחלקתית:

.....תאריך:



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30/09/26